

NeuralCert: certified computational discovery of extremal mathematical constructions

Abstract

Neural networks are becoming popular in solving mathematical problems, but stochastic models do not provide mathematical exactness by themselves. This study introduces a discovery-to-certification framework in which high-dimensional variational trial functions are learned in a compact separable representation, spectrally diagnosed and pruned, and then certified exactly through multimodular evaluation. Exact certification makes the numerical proofs fully explicit and independently verifiable. This framework can be run on a standard personal computer.

Across three extremal problems, we show that neural optimization can contribute to rigorous mathematics in three distinct ways: by discovering improved constructions, by exposing empirical invariants that lead to proofs, and by revealing optimization barriers whose geometry motivates new analytic or numerical representations.

More broadly, these results suggest a path toward AI-assisted mathematics in which flexible computational discovery and exact certification become complementary components of a single rigorous workflow.

Machine learning can contribute to mathematical discovery by identifying latent relations between mathematical objects, guiding search over vast combinatorial or algebraic spaces, and generating candidate constructions or algorithms. Davies *et al.* used learned predictors and attribution methods to expose latent mathematical structure and guide conjecture formation [1]. AlphaTensor showed that reinforcement learning can search large spaces of tensor decompositions and recover exact matrix-multiplication algorithms, including previously unknown constructions [2]. FunSearch coupled neural generation to an executable evaluator, demonstrating that iterative proposal, evaluation and selection can produce new mathematical constructions [3]. Also, neural function approximators can transform a mixed discrete–continuous geometric problem into a differentiable optimization problem [4].

Despite these advances, a methodological gap persists for problems in which the unknown object is a function or extremizer in a large or infinite-dimensional space. Such problems are commonly reduced computationally by choosing a finite ansatz,

$$f(x) = \sum_{j=1}^m a_j \phi_j(x), \quad (1)$$

and optimizing over the coefficients a_j . While this makes the problem tractable, it also imposes an a priori structural restriction,

$$f^* \approx \text{span}\{\phi_1, \dots, \phi_m\}, \quad (2)$$

so that improvements in optimization accuracy cannot remove approximation error induced by an inadequate representation. However, neural parameterizations offer a complementary strategy: instead of fixing the shape of a candidate solution in advance, they provide flexible differentiable families in which large-scale functional search can be conducted. For example, in geometric colouring a mixed discrete–continuous search space was replaced by a probabilistic differentiable representation amenable to gradient optimization [4].

Here we introduce NeuralCert, a discovery-to-certification framework that deliberately separates the representation used for computational search from the representation used for mathematical proof. Flexible neural parameterizations are used to learn one-dimensional factors within a separable trial family, without fixing the polynomial dictionary subsequently used for exact certification. Discovered candidates are subsequently compressed into explicit mathematical representations from which rigorous certificates can be constructed and independently verified. Here, certification means recording a

claimed bound together with the exact data needed to recompute it independently, so that the inequality can be confirmed without trusting, or re-running, any part of the discovery pipeline. Across three extremal problems, we show that this separation enables neural optimization to contribute to rigorous mathematics through three distinct modes: discovering improved constructions, exposing structure that leads to analytic results, and identifying optimization barriers that motivate alternative representations.

Since a floating-point neural candidate does not constitute a proof, the discovery candidate is reduced to an independently checkable object [2, 3] such that verification establishes the mathematical claim. This idea builds on previous work where neural search produced tensor decompositions which could be checked algebraically [2], and candidate programs judged by an explicit executable evaluator [3]. These examples suggest that discovery and verification impose fundamentally different computational requirements. Discovery benefits from (over)parameterization, approximation and exploration, whereas verification benefits from low-dimensional structure, exactness and controlled numerical error.

Our central methodological hypothesis is not that neural optimization should replace mathematical proof, but that the representation best suited for discovery need not be the representation best suited for proof. A flexible computational model can search a substantially larger functional space than is convenient for exact symbolic treatment, after which the useful structure can be transferred into an explicit mathematical object. Conversely, repeated failure or instability of such a search can reveal invariants or unsuitable coordinates and thereby guide analytic reformulation [1]. NeuralCert is designed around this asymmetry: discovery is deliberately flexible and expendable, whereas every reported mathematical claim must be supported by a representation that can be reconstructed and verified independently.

NeuralCert overview

NeuralCert is a Python framework for neural discovery of candidate functions followed by independent certification and verification. The architecture is deliberately separated into three components; discovery, certification and verification, which do not share numerical evaluation routines. Discovery uses flexible neural parameterizations to explore problem-specific function spaces and exports candidate functions and metadata, typically as NumPy `.npz` files. Certification is independent of the neural model: no network parameters, optimizer state or training data are required. Instead, the candidate is reconstructed in a problem-specific mathematical representation and converted into an explicit certificate. Verification is implemented as a standalone minimal code path that ingests only this certificate, typically as `.json`, and independently reconstructs the inequalities underlying the claimed result. Thus, neural optimization serves only as a proposal mechanism; final mathematical claims depend exclusively on independently reconstructed and verifiable proof objects.

Discovery and certification of sieve constants improve prime gap bounds

The Maynard–Tao sieve reduces bounded gaps between primes to lower bounds on a variational constant M_k , which we approached through two independent computational routes. The first uses neural search over separable functional families, followed by reconstruction in a polynomial representation and rigorous Rayleigh–Ritz certification. At small k , this approach recovers known extremal behaviour ($M_5 \geq 2.0071443298$ and $M_{20} \geq 3.12755795$); at $k = 25$ it improves the Polymath8b reference (certified 3.3221511 versus 3.3221426). Thus, high-quality extremizers can be discovered without prescribing the polynomial ansatz subsequently used for certification (Supplementary Tables 7 and 8). Neural discovery further benefits from overparameterization: a redundant channel representation supplies extra variational degrees of freedom during optimization, after which unnecessary channels are removed by backward elimination with re-optimization at each step, subject to a cumulative relative Rayleigh-quotient loss of at most $\tau_{\text{prune}} = 10^{-7}$.

At large k , an early neural run near $k = 3600$ produced a numerically stable Rayleigh value above 8 that remained below the Cauchy–Schwarz ceiling and was stable under grid refinement. Exact reconstruction nevertheless failed: the optimizer had collapsed onto a sharply localized single channel whose discovery score was inflated by an inconsistent quadrature frame. The discovered profile approaches

$1/(\log k \cdot t)$, a form the polynomial channels cannot represent across the increasingly flat Rayleigh landscape (Supplementary Methods 5.18). The artifact survived the available heuristic gates because its value did not cross the upper bound; only independent reconstruction of the candidate for certification exposed the inconsistency, which motivated NeuralCert’s scalar cross-checks and support-aware thresholds.

To access much larger k , we developed a second computational route using the rational profile $g(t) = 1/(c + (k - 1)t)$, already employed in Polymath8b (Theorem 6.7). Our contribution is a direct, rigorously certified Fourier-domain evaluation of the associated product trial function on the simplex. This makes it possible to optimize and certify the construction at a computational cost nearly independent of k , obtain improved explicit prime-gap bounds, and analyse its asymptotic deficit through a stable-law limit. This yields

$$M_{3655} \geq 8.000064869, \quad M_{208910} \geq 12.00000199, \quad M_{11655069} \geq 16.0000003924, \quad M_{644589002} \geq 20.0000009143,$$

for 3, 4, 5 and 6 primes in a tuple, together with the uniform bound $M_k \geq \log k - 0.307$ for $100 \leq k \leq 1.88 \times 10^9$. These results require only the Bombieri–Vinogradov theorem, assuming neither the Elliott–Halberstam conjecture nor Deligne-type distributional input.

Combining these thresholds with explicit admissible tuples yields

$$H_2 \leq 33\,118, \quad H_3 \leq 2\,718\,108, \quad H_4 \leq 214\,099\,720, \quad H_5 \leq 14\,541\,349\,288,$$

where the tuple for $k = 3655$ is taken from the Engelsma–Sutherland database [5], the tuple for $k = 208910$ was constructed by a hybrid shifted-Schinzel/greedy sieve, and the two largest use the first k primes exceeding k ; all four are checked by a standalone verifier that shares no code with the construction (Supplementary Methods). Measured against the corresponding records established without the Elliott–Halberstam conjecture and without Deligne’s theorems [polymath’wiki, 6], these improve the published bounds by factors of 14.3, 11.9, 9.5 and 8.6 respectively. They also improve on the stronger records that do invoke Deligne-type input [7], by factors of 12.0, 9.0, 6.5 and 5.3 (Figure 1 and Supplementary Data). This distinction matters because, as noted in [8], the previously best-known upper bounds H_m for $m \geq 2$ relies on equidistribution estimates of Zhang type; the bounds above are obtained without them.

An asymptotic analysis of the single-channel rational family identifies its limiting convolution profile with a spectrally positive 1-stable law and proves $M_k \geq \log k - C_* - o(1)$, where $C_* = \inf_{a \in \mathbb{R}} C(a)$; numerical evaluation suggests $C_* \approx 0.3343$ (Supplementary Methods). This explains why the rank-one structure remains asymptotically competitive, attaining the optimal $\log k$ growth rate with only a constant-order deficit, without implying global optimality of rank one. Consequently

$$0 \leq \liminf_{k \rightarrow \infty} (\log k - M_k) \leq C_*,$$

where the lower endpoint follows from the classical bound $M_k \leq \frac{k}{k-1} \log k$. The finite-range constant 0.307 is smaller than C_* because, within this rational family, the finite-range deficit is numerically consistent with $C_* - 1.047/\log k$; the directly evaluated deficit remains below 0.29 throughout the certified range.

Despite sharing no numerical evaluation, the two computational pipelines agree throughout their overlapping range. While the finite- k results remain below Bogaert’s independently computed (Krylov subspace) values for $k \neq 25$, $k \leq 30$, indicating that the Krylov method is powerful for modest k , NeuralCert improves $31 \leq k \leq 100$. Every reported finite- k inequality is accompanied by a machine-checkable certificate that can be independently verified.

We also implemented the ε -enlarged variant proposed in Polymath8b. At $k = 49$, optimization within our certified trial family saturates at 3.98867. The certified value is a lower bound, so this does not establish $M_{49} \leq 4$; it shows that enlarging the sieve weight alone, at Bombieri–Vinogradov level, did not reach the threshold in our search. Concurrent work is consistent with this reading: $H_1 \leq 240$ has since been established by combining the Bombieri–Vinogradov support with equidistribution estimates for smooth moduli [8], and subsequent refinements of that construction give $H_1 \leq 212$ [9] and $H_1 \leq 186$ [10]; all three use distributional input beyond Bombieri–Vinogradov. The additional ingredient is therefore on the arithmetic side rather than the variational one, which is where our saturation result locates it. We report the first certified ε -values at $k = 49, 52$ and above $k = 54$.

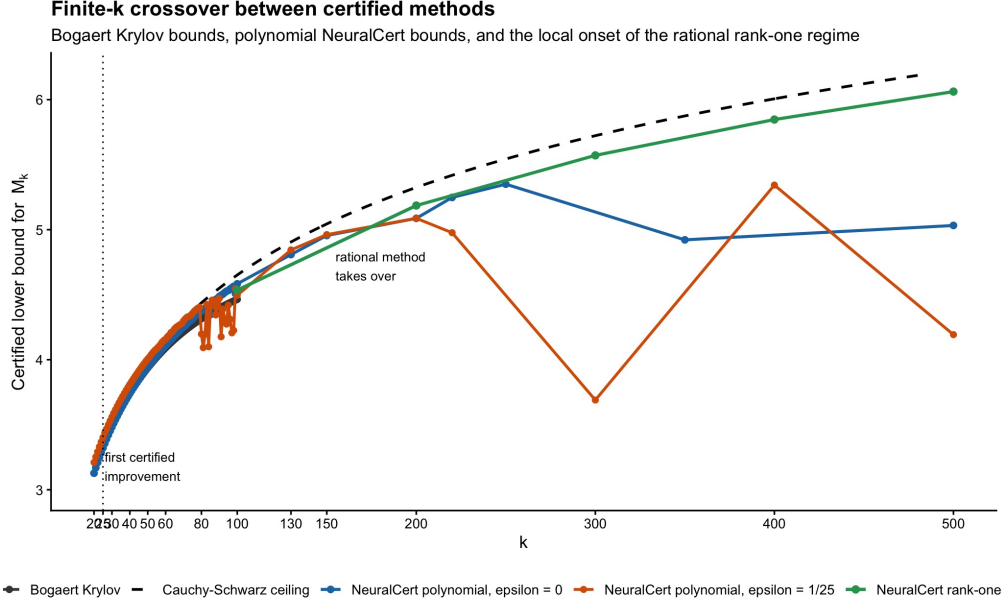


Figure 1: **Certified lower bounds for the Maynard variational constant across computational regimes.** Certified values obtained with Bogaert’s Krylov method are compared with NeuralCert polynomial constructions for the vanilla problem ($\epsilon = 0$) and enlarged support ($\epsilon = 1/25$), together with the certified rank-one rational family. The polynomial construction first exceeds the Krylov bounds at $k = 25$, but loses efficiency as k increases, whereas the rank-one rational representation becomes competitive and provides the continuation to larger k . The Cauchy–Schwarz bound is shown as a reference. Lines connecting discrete certified values are guides to the eye and do not represent interpolated certificates. Beyond $k = 500$, the rank-one rational family is the only representation that remained reliably optimizable and certifiable in the regime studied here.

Higher-order Delsarte discovery exposes a tensorization obstruction

Higher-order Delsarte hierarchies strengthen the classical LP for linear codes and show substantial finite-length gains, but asymptotic progress requires an explicit dual family rather than additional numerical optimization [11, 12]. We therefore treated the higher-order dual as a discovery problem, initially asking whether a low-complexity correction to the lifted level-one solution could improve the MRRW bound. Extending the symmetry-reduced hierarchy to $r = 3$ confirmed that higher levels contain genuine information, including the level-3 bound $V_3(12, 5) \leq 16$, against the level-2 value $V_2(12, 5) = 24.260255\dots$ located numerically. However, comparison with the explicit lift showed that the source of the improvement changes with hierarchy level: at $r = 2$ it can reside entirely in the partial-Fourier constraints, whereas at $r = 3$ the full-transform rows already improve on the lift. A single learned correction was therefore not a stable asymptotic target.

We redirected discovery to the spectral construction of [12] where the blocklength dependence can be removed analytically. For a normalized configuration G on $\mathbb{F}_2^\ell$, the relevant objective is $J_\ell(G) = H(G)/\ell$. Although we initially intended to parameterize G neurally, symmetry and analytic reduction collapsed the problem faster than a generic network became useful: for $\ell \leq 4$ the largest search space has only 15 free probability coordinates and no remaining dependence on n . We therefore optimized the reduced configuration directly. Across 28 instances and 13,600 randomized local starts, no configuration improved MRRW; instead, the recovered optima repeatedly matched the quasirandom tensor-product family to numerical precision.

This repeated null result exposed a structural obstruction. Defining the translation affinity

$$\Phi(G, v) = \sum_u \sqrt{G(u)G(u+v)},$$

the finite spectral walk condition implies $\Phi(G, v) \geq \varepsilon$ on a spanning set of shifts (Methods, hypotheses (A1)–(A2)). For

$$h(\varepsilon) = \mathbf{H}\left(\frac{1 - \sqrt{1 - \varepsilon^2}}{2}\right),$$

convexity of h , conditioning and Jensen’s inequality give

$$\mathbf{H}(G) \geq \sum_{i=1}^{\ell} h(\Phi(G, e_i)) \geq \ell h(\varepsilon).$$

Hence, with $\delta = (1 - \varepsilon)/2$,

$$J_{\ell}(G) \geq h(\varepsilon) = \mathbf{H}\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right),$$

exactly the first MRRW expression. The tensor-product configuration $\text{Bernoulli}(p_{\varepsilon})^{\otimes \ell}$ saturates the entropy–affinity inequality, identifying the object repeatedly recovered computationally. Hence, within the fixed single-orbit spectral ansatz, optimizing the configuration G cannot improve MRRW Theorem 18 and Corollary 19).

The null result is informative because it is highly structured rather than merely negative. Across 13,600 local optimizations, the recovered configurations repeatedly collapse onto the same tensor-product family to numerical precision (Supplementary Methods 7.5). This suggested that the observed optimum reflected an invariant of the reduced problem instead of optimization failure. The entropy–affinity argument supports this interpretation: the tensor-product configuration saturates the inequality and the numerically recovered object is an extremizer of the constrained configuration problem.

Furthermore, within the single-orbit spectral construction, additional optimization of G cannot improve the first MRRW expression; any improvement must instead exploit degrees of freedom not controlled by the obstruction, such as alternative sign functions, support on multiple configuration orbits, the partial-Fourier dual families or more general dual feasible solutions (Supplementary Methods 7.6 and 7.7). Thus, the computation eliminates an apparently natural search direction while identifying the components in which non-trivial asymptotic freedom remains.

Convex discovery improves sign-uncertainty bounds

For the +1 Bourgain–Clozel–Kahane sign-uncertainty problem, radial polynomial–Gaussian Fourier eigenfunctions are the canonical ansatz. Cohn, Dong and Gonçalves [13] showed that sublinear-degree members of this family encounter an asymptotic ceiling, motivating our initial search for more flexible Fourier-eigenfunction structures. We therefore treated the choice of envelope, scales and contact locations as a discovery problem, with learned proposals followed by independent high-precision evaluation.

The flexible search improved rapidly but stalled in dimension $d = 1$. Gaussian-mixture constructions reached $\rho = 0.578532$, above the published $A_+(1) \leq 0.572990$ bound. Neural optimization did not remove the plateau: apparently improving trajectories systematically entered coefficient systems with condition numbers 10^{17} – 10^{18} , where no reliable digits could be guaranteed from the floating-point solve. Multiprecision reevaluation rejected these candidates. Multistart runs also converged to distinct plateaus rather than a common value, consistent with changes in root multiplicity and contact structure in the last-sign-change functional. We therefore treated the failure as a diagnostic of the search representation rather than evidence that the function class itself had been exhausted.

The decisive simplification was to return to the Fourier-eigenfunction constraint. With $u = \pi|x|^2$, every radial +1 polynomial–Gaussian eigenfunction in the truncated Laguerre space has the form

$$f(x) = P(u)e^{-u}, \quad P \in \text{span} \left\{ L_0^{(d/2-1)}(2u), L_2^{(d/2-1)}(2u), \dots, L_{2N}^{(d/2-1)}(2u) \right\}.$$

The Fourier constraint is linear in the Laguerre coefficients and $e^{-u} > 0$. For fixed u_0 , admissibility therefore reduces to

$$P(0) = 0, \quad P(u) \geq 0 \quad (u \geq u_0).$$

To exclude the zero polynomial and fix the otherwise arbitrary scale, we impose the linear normalization

$$\int_{u_0}^{\infty} P(u)e^{-u} du = 1.$$

Thus the search can be reformulated as a convex semi-infinite feasibility problem in the coefficients, with bisection in u_0 , rather than a non-convex optimization over forced contacts. An adaptive exchange LP locates the active tangencies; these are then rationalized, used to reconstruct an exact Laguerre polynomial over $\mathbb{Q}$, and verified on the complete ray by exact factorization and a Sturm chain over $\mathbb{Z}$. The numerical search determines the quality of the construction found, whereas admissibility of each quoted bound follows independently from the rational certificate. No global optimality claim is made.

In $d = 1$ the degree ladder first reproduces the published configuration and then improves it, with approximate last-sign-change radii

$$\rho_{22} \approx 0.572989678, \quad \rho_{30} \approx 0.572706699, \quad \rho_{38} \approx 0.572588699.$$

The degree-22 search numerically resolves the same five-contact configuration as the published construction to its stated precision; the new improvement occurs at higher degree. The degree-38 candidate is certified by an explicit rational polynomial, giving $A_+(1) \leq 0.572588700$. The same pipeline gives $A_+(2) \leq 0.756206237$, which is approximately 7.6×10^{-7} below the published 0.756207 value and is therefore treated as a certified reproduction with a marginal gain rather than a substantive improvement (see Figure 2).

Higher-dimensional runs approach, rather than cross, the known polynomial–Gaussian asymptotic ceiling; their interpretation is limited by the numerical degree ceiling of the floating-point discovery stage (Methods). Thus, the new small-dimensional bounds do not evade the Cohn–Dong–Gonçalves obstruction. Instead, they show that the finite polynomial–Gaussian family had not been numerically exhausted.

Discussion

Can modern neural optimization rediscover and improve extremal constructions in hard analytic and combinatorial mathematics? Across the three problems studied here, the answer takes several distinct forms. In the Maynard–Tao sieve, neural optimization identifies stronger extremizers, but at large k the neural representation ultimately gives way to the rational family $g(t) = (c + (k - 1)t)^{-1}$. The large- k results illustrate how computational progress can arise from a new evaluation and certification strategy for an established mathematical family. In the higher-order Delsarte hierarchy, computational failure is itself informative: the search collapses onto a tensor-product family, and the resulting invariant yields an obstruction to improving the first MRRW expression through the entropy-based rate estimate under the spanning-affinity and finite-walk hypotheses considered here. The result leaves open constructions that bypass these hypotheses or use alternative spectral supports or dual feasible solutions. In sign uncertainty, neural optimization encounters a plateau; independent verification reveals severe ill-conditioning, and the failure exposes a convex reformulation that produces new certified bounds. In each case, the discovery representation is provisional: its purpose is to expose mathematical structure rather than to become part of the final proof [1, 14, 15]. Computation therefore contributes not only by finding better objects, but also by identifying obstructions and more appropriate representations. The numerical degree ceiling encountered in the sign uncertainty problem is a case in point: asymptotic theory indicates that the strongest bounds lie in the high-degree regime, yet that regime is precisely where the discovery representation loses numerical resolution, so progress required changing the representation rather than pushing the degree further.

Separating heuristic proposal generation from mathematically controlled inference also underlies neuro-symbolic systems such as AlphaGeometry, in which learned construction proposals are coupled to a symbolic deduction engine [16]. NeuralCert applies a related separation not to theorem search, but to continuous variational discovery: the learned component proposes mathematical objects, whereas the final inequality depends only on the reconstructed certificate. The computational model need not itself be interpretable, exact or even retained; it can use floating-point arithmetic, stochastic optimization, approximate quadrature and overparameterized models. Discovery output is instead regarded as an exploratory and untrusted proposal that must cross into a mathematically explicit representation

Objective landscape and trajectory search

The last-sign-change objective $\rho(\tau_1, \tau_2)$ on a slice through the certified record configuration ($d = 1$, degree 38, $m = 9$, $\rho = 0.5725887$).

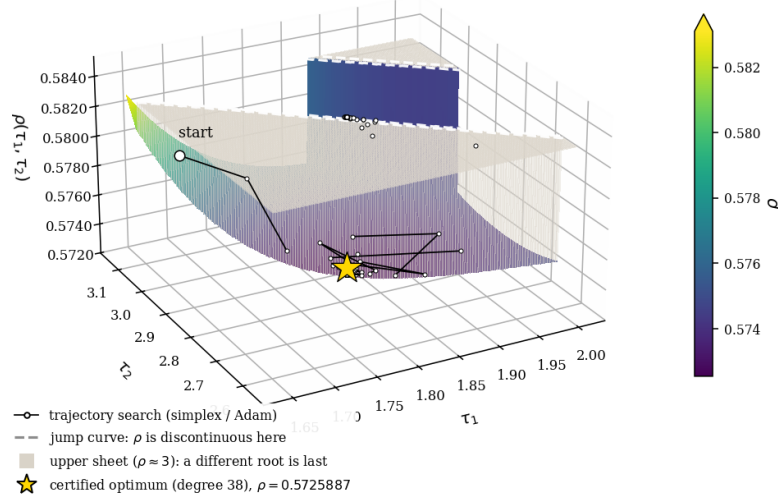


Figure 2: **Objective landscape and trajectory search.** The last-sign-change functional $\rho(\tau_1, \tau_2)$ on a slice through the certified record configuration ($d = 1$, degree 38, $m = 9$ contacts, $\rho = 0.5725887$), obtained by varying the two innermost contacts with the remaining seven held at τ^* . The surface is drawn from the raw grid without smoothing or interpolation, and is evaluated in the damped Laguerre basis $\psi_n(u) = L_n^{(\alpha)}(2u)e^{-u}$: at this degree the underlying polynomial reaches $\sim 10^{69}$ on the grid, so a monomial evaluation loses every digit at the outer tangencies and splits them into spurious sign changes. The landscape has two sheets, a narrow valley (colour) and an upper sheet at $\rho \approx 3$ (translucent), with no intermediate values — crossing the dashed curve, a different root of P becomes the last sign change and ρ leaps. Black markers show a simplex/Adam trajectory, its polyline broken wherever the objective jumps. The observed jumps illustrate the sensitivity of direct optimization to changes in the last-sign-change root. The convex reformulation avoids differentiating this discontinuous objective by using bisection in $u_0 = \pi\rho^2$ and solving a coefficient-feasibility problem at each step. The dashed curve is drawn as a high-gradient contour and indicates where ρ jumps rather than resolving the discontinuity set exactly.

before it can support a claim. Certification and verification then operate on that representation independently of the optimizer. This design follows certificate-based principles in verified computation [17]. The failed large- k Maynard candidate demonstrated the value of this design. A candidate near $k = 3600$ remained stable under grid refinement and below the ceiling, yet independent reconstruction revealed a scale inconsistency with inflation of the objective. The optimizer had exploited the numerical error, generating a plausible extremizer that failed certification. Conversely, the later rational rank-one construction was accepted because its Rayleigh quotient can be independently reconstructed and certified. Hence, structural similarity discovered numerically becomes mathematically relevant only when it survives the certification boundary, in line with the certificate discipline of computer-assisted proof [18]. The methodological contribution is therefore the transition between search representations and independently verifiable mathematical objects, with neural optimization serving as one component of that process.

The same separation has appeared concurrently on this very problem. Recently, the bound $H_1 \leq 246$ was improved to 240 [8], then to 212 [9] and 186 [10], the last two accompanied by Lean certificates and, in one case, a proof attributed to an automated system. The present work automates different objects. Instead of a proof generated through a proof assistant at a single small dimension ($k = 49, 45$ and 40 respectively), we discovered a trial function numerically and then removed from the argument entirely: what is certified is the Rayleigh quotient of one fixed, exactly specified function, at dimensions

up to $k \approx 6 \times 10^8$ and uniformly to $k = 1.88 \times 10^9$. Formal verification is the stronger guarantee, and extending certificates into that setting is a potent direction. Conversely, the variational constant is shared infrastructure: earlier work requires $k \approx 3.5 \times 10^4$ to reach $m = 2$ because the available lower bound on M_k falls well short of $\log k$, and combining the sharper bound $M_k \geq \log k - 0.307$ with the improved equidistribution estimates used in [8–10] would be expected to improve H_m for $m \geq 2$ beyond either ingredient alone.

The framework also has clear limitations. NeuralCert is not a general mechanism for converting numerical optimization into proof. Certification requires that a discovery can be transferred to an explicit mathematical representation, and this transfer may itself become a bottleneck. A failed reconstruction is not hard evidence that the discovered function is mathematically uninteresting; it only establishes that the proposed claim has not crossed the certification boundary. Likewise, certificates for variational constructions establish the performance of explicit admissible trial functions, not their global optimality. A further limitation concerns the standard of verification itself. The certificates here are rigorous numerical objects; interval enclosures and exact modular arithmetic, not machine-checked proofs, and a Lean or Coq development of the reduction and error bounds would be a strictly stronger guarantee. That the same problem has recently attracted both approaches suggests the combination is within reach. More generally, flexible neural (over)parameterizations are particularly useful when their additional representational freedom assists discovery. Once lower-dimensional structure, convexity or an explicit analytic family has been identified, retaining a neural representation offers limited advantage; the mathematically simpler representation should replace it.

Methods

1 Discovery and certification of sieve constants improve prime gap bounds

1.1 Maynard–Tao variational formulation

The Maynard–Tao sieve reduces bounded gaps between primes to the construction of admissible trial functions with large variational quotient [6, 19]. For the separable constructions considered here, the relevant denominator and numerator functionals can be expressed through pairwise one-dimensional convolution integrals. We write the trial function as

$$F_{\theta,c}(t_1, \dots, t_k) = \sum_{j=1}^m c_j \prod_{i=1}^k g_{\theta,j}(t_i), \quad (3)$$

where $g_{\theta,j}$ are one-dimensional channels and $c \in \mathbb{R}^m$ is a linear mixing vector. For each pair,

$$h_{j\ell}(x) = g_{\theta,j}(x)g_{\theta,\ell}(x), \quad (4)$$

and the k -dimensional functionals reduce to quadratic forms

$$I_{k,\varepsilon}(F_{\theta,c}) = c^\top A(\theta)c, \quad J_{k,\varepsilon}^{(1)}(F_{\theta,c}) = c^\top B(\theta)c, \quad (5)$$

whose entries depend on one-dimensional integrals involving $h_{j\ell}^{*(k-1)}$. At fixed channels the objective is therefore the generalized Rayleigh quotient

$$R(\theta) = k \lambda_{\max}(B(\theta), A(\theta)) = k \max_{c \neq 0} \frac{c^\top B(\theta)c}{c^\top A(\theta)c}. \quad (6)$$

The nonlinear optimizer consequently searches only over the channel manifold; the optimal linear combination is re-solved as a generalized Ritz vector at each objective evaluation [20, 21]. The precise normalization of M_k , its ε -enlarged analogue, and the conversion of thresholds in M_k to bounds on H_m are given in Supplementary Methods, Section 5.1.

1.2 Neural discovery of separable trial functions

The discovery stage is used only to identify strong explicit trial functions and does not establish a rigorous inequality. To expose the increasingly sharp boundary structure at large k , each channel is parameterized as

$$g_{\theta,j}(x) = r_{\theta,j}(x)e^{-\rho_j x}, \quad \rho_j > 0, \quad (7)$$

where the residual functions are produced by a shared multilayer perceptron with channel-specific outputs. The network receives (x, e^{-kx}) as input, explicitly exposing the natural $O(k^{-1})$ boundary-layer scale. In the positive-channel solver, $r_{\theta,j}(x) = \exp(f_{\theta,j}(x, e^{-kx}))$, allowing the convolution recursion to be performed without cancellation in logarithmic coordinates.

The leading generalized Ritz vector is profiled out rather than learned jointly with the channel parameters. If

$$B(\theta)v_\theta = \lambda_\theta A(\theta)v_\theta, \quad v_\theta^\top A(\theta)v_\theta = 1, \quad (8)$$

we freeze v_θ and differentiate

$$q_{v_\theta}(\theta) = k \frac{v_\theta^\top B(\theta)v_\theta}{v_\theta^\top A(\theta)v_\theta}. \quad (9)$$

Under the usual nondegeneracy condition, the Hellmann–Feynman/envelope identity [22, 23] gives the derivative of the profiled leading eigenvalue without differentiating through the eigendecomposition. Adam [24] is used for exploration and a block L-BFGS minorize–maximize procedure for polishing [25]. Channel-rank and grid-resolution continuation are used during search. Network architecture, optimization schedules, finite-difference gradient controls and continuation rules are specified in Supplementary Methods, Sections 5.4–5.10.

1.3 Stable evaluation of high-order convolution powers

Direct recursion on h^{*p} becomes unstable as p grows because algebraic concentration, exponential decay and global amplitude must otherwise be represented simultaneously. For each channel pair we write

$$h_{j\ell}(x) = r_{j\ell}(x)e^{-a_{j\ell}x}, \quad a_{j\ell} = \rho_j + \rho_\ell, \quad (10)$$

and factor its convolution powers as

$$\nu_p^{(j\ell)}(s) = s^{p-1} e^{-a_{j\ell}s} e^{\sigma_p^{(j\ell)}} \psi_p^{(j\ell)}(s). \quad (11)$$

The algebraic and exponential terms are handled analytically, the scalar σ_p carries the global amplitude, and only the residual shape ψ_p is represented numerically. Substitution into the convolution identity and the change of variables $x = su$ yield a beta-weighted recursion,

$$\psi_{p+q}(s) = B(p, q) \int_0^1 \psi_p(su) \psi_q(s(1-u)) \frac{u^{p-1}(1-u)^{q-1}}{B(p, q)} du, \quad (12)$$

which is evaluated with Gauss–Jacobi quadrature [26]. The concentration associated with increasing p and q is thereby moved into a known quadrature weight rather than left as unresolved structure in the represented residual.

For positive channels we store $\xi_p = \log \psi_p$ and evaluate the positive quadrature sum by log-sum-exp, preventing underflow across repeated convolutions. The required power $h^{*(k-1)}$ is assembled through a binary addition chain, reducing convolution depth from $O(k)$ to $O(\log k)$. Representation, convolution, primitive-integration and outer-integration grids are kept distinct. Full recursion formulas, grid definitions and quadrature construction are given in Supplementary Methods, Sections 5.5–5.7.

1.4 Structure-preserving Rayleigh–Ritz optimization

The denominator matrix A is mathematically a Gram matrix and must satisfy $A \succeq 0$. Entrywise accurate discretization does not automatically preserve this property. We therefore use positive two-point interpolation in the convolution recursion and a shared positive-weight outer integration mesh

across all channel pairs. Whenever the inner quadrature nodes are common to all pairs, the discrete denominator has the form

$$A_{j\ell}^{(N)} = \sum_{\alpha} w_{\alpha} \Phi_{\alpha j} \Phi_{\alpha \ell}, \quad w_{\alpha} > 0, \quad (13)$$

and therefore retains the positive-semidefinite Gram structure up to floating-point round-off. In the saturated regime, where the inner rule for the channel primitives becomes pair-dependent, this common representation is preserved only up to inner-quadrature error; numerical adequacy is then assessed by the positive-semidefiniteness, two-grid and refinement diagnostics of Section 1.5, which are safeguards rather than rigorous bounds on that error. No bound reported here depends on this discrete structure: the certified values are produced by the independent exact certifier.

Because channel norms may span hundreds of orders of magnitude, the Gram pencil is diagonally equilibrated by the exact congruence transformation

$$A \mapsto D^{-1/2} A D^{-1/2}, \quad B \mapsto D^{-1/2} B D^{-1/2}, \quad D = \text{diag}(A_{11}, \dots, A_{mm}). \quad (14)$$

Numerically unresolved directions are removed by rank-revealing spectral whitening; no ridge term is added to the variational denominator. The interpolation, shared-mesh construction, equilibration and rank-revealing Ritz solve are detailed in Supplementary Methods, Sections 5.8–5.9.

1.5 Validation of neural candidates

Validation is embedded into model selection because a flexible optimizer can exploit structured discretization error. Candidate states are therefore required to reproduce the independently recomputed Rayleigh quotient, preserve numerical positive semidefiniteness of the denominator, retain sufficient effective rank, remain stable under a more conservative rank threshold, respect applicable rigorous upper bounds for both individual channels and the full quotient (rejecting numerical states that spuriously exceed the Cauchy–Schwarz ceiling [6]), and reproduce on a finer representation grid. Closed-form controls based on $g(x) = 1$ and $g(x) = x$ exercise the convolution and integration paths before optimization. Final candidates are re-evaluated on multiple grids; extrapolation is used only diagnostically and never as a certified lower bound. Exact validation criteria, controls and refinement procedures are provided in Supplementary Methods, Sections 5.11–5.12.

1.6 Exact certification of neural candidates

A discovered candidate is projected onto rational polynomial channels,

$$F(t_1, \dots, t_k) = \sum_{j=1}^m c_j \prod_{i=1}^k q_j \left(\frac{t_i}{1 + \varepsilon} \right), \quad (15)$$

and the mixing vector is transferred from the preconditioned discovery frame to the corresponding exact channel frame. For fixed rational channels and rational coefficients the associated Maynard quotient is rational; certification therefore reduces to exact evaluation of the two quadratic forms.

Rather than constructing and reconstructing every exact Gram-matrix entry, the certifier accumulates the pair contributions to $c^{\top} A c$ and $c^{\top} B c$ modulo machine-word primes. For each prime p it computes only two aggregate residues,

$$S_A(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_{\ell} A_{j\ell}^{\text{int}} \pmod{p}, \quad S_B(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_{\ell} B_{j\ell}^{\text{int}} \pmod{p}. \quad (16)$$

Only the aggregate integers S_A and S_B are reconstructed, with analytically known positive denominators D_A, D_B :

$$c^{\top} A c = \frac{S_A}{D_A}, \quad c^{\top} B c = \frac{S_B}{D_B}. \quad (17)$$

Rigorous a priori integer bounds $|S_A| \leq \mathcal{B}_A$ and $|S_B| \leq \mathcal{B}_B$ determine the reconstruction budget. If $M_P = \prod_{p \in P} p$ and

$$M_P > 2 \max(\mathcal{B}_A, \mathcal{B}_B), \quad (18)$$

the centered Chinese-remainder representative is unique. The final quotient is then formed exactly from the reconstructed integers and known denominators. Thus the neural optimizer, numerical quadrature and grid extrapolation no longer enter the proof once the rational trial function has been fixed.

Before exact evaluation, the candidate is compressed by loss-controlled backward elimination: channels are removed only while the re-optimized Rayleigh quotient remains within a prescribed relative loss budget. The exact backend certifies the explicit reduced function, so pruning affects certificate strength and computational cost but not validity. Detailed denominator formulas, coefficient-extraction identities, rational representation transfer, pruning rules, integer bounds and deterministic CRT reconstruction are given in Supplementary Methods, Sections 5.13–5.17.

1.7 Large- k rational construction

At large k , neural optimization is used as a structural discovery instrument rather than as the source of a certified numerical value. The neural candidate is compressed into clustered rational families

$$g_{\text{rat}}(t) = \sum_{a=1}^r \sum_{j=1}^{p_{\max}} u_{a,j} (c_a + nt)^{-j}, \quad n = k - 1, \quad (19)$$

where $c_a > 0$ are pole locations and the higher powers represent confluent directions. For fixed c_a , the coefficients $u_{a,j}$ are eliminated by variable projection, so only the nonlinear pole locations are optimized. The confluent basis has a direct geometric interpretation because

$$\frac{\partial^q}{\partial c^q} \frac{1}{c + nt} = (-1)^q q! (c + nt)^{-(q+1)}. \quad (20)$$

The quality of a distilled representation is not determined by pointwise fitting error alone. Each candidate is rebuilt in an independent deterministic evaluator, the corresponding Gram pencil is re-computed, and the generalized Rayleigh problem is solved afresh. Candidates are rejected if they fail probability-mass conservation, analytic moment checks, Fourier-inversion stability, Gram-rank diagnostics, cross-cluster conditioning, or the rigorous ceiling

$$R_k \leq \frac{k}{k-1} \log k. \quad (21)$$

Surviving pole locations are then locally refined under the deterministic Rayleigh objective.

In the large- k regime, additional rational clusters can reduce the pointwise approximation error while failing these distributional checks. The stable deterministic representation collapses to the first-order family

$$g(t) = \frac{1}{c + nt}, \quad n = k - 1, \quad (22)$$

which is subsequently optimized independently of the neural model. This distillation step therefore uses neural search to identify structure, while the numerical value submitted for certification is obtained from a separate rational representation. The clustered fit, deterministic Fourier evaluator, failure diagnostics and local refinement are detailed in Supplementary Methods, Section 5.18.

1.8 Certified large- k evaluation in ball arithmetic

For the first-order family in Eq. (22), certification can be reduced to a one-dimensional probabilistic Fourier calculation. Let $w = g^2$, $m_0 = \int_0^1 w(t) dt = [c(c+n)]^{-1}$, and let S be the sum of $n = k - 1$ independent draws from the probability density w/m_0 on $[0, 1]$. With

$$G(\rho) = \int_0^\rho g(t) dt, \quad H(\rho) = \int_0^\rho w(t) dt, \quad (23)$$

the separable Maynard quotient reduces exactly to

$$R = \frac{kN}{D}, \quad N = \mathbb{E}[G(1-S)^2 \mathbf{1}_{S < 1}], \quad D = \mathbb{E}[H(1-S) \mathbf{1}_{S < 1}]. \quad (24)$$

Writing

$$\varphi(\theta) = \left(\frac{\widehat{w}(\theta)}{m_0} \right)^n, \quad \Psi_X(\theta) = \int_0^1 X(\rho) e^{i\theta\rho} d\rho, \quad (25)$$

for $X \in \{G^2, H\}$, Fourier inversion gives

$$I_X = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi(\theta) e^{-i\theta} \Psi_X(\theta) d\theta. \quad (26)$$

The certifier evaluates a truncated trapezoidal form of Eq. (26) using Arb ball arithmetic [27]. The difference from the exact integral is bounded by three separately controlled contributions: periodic aliasing, omitted Fourier mass, and rigorous numerical enclosure of the transforms and auxiliary integrals. Aliasing is one-sided because the corresponding convolution density is nonnegative and compactly supported; its residual mass is bounded by certified moment inequalities. The omitted Fourier range is divided into a finite band controlled by Arb wide-ball suprema and an analytic tail obtained from integration-by-parts decay.

All error budgets remain in ball arithmetic until the final quotient is assembled. If N_{ball} and D_{ball} are the certified truncated-sum enclosures and E_N, E_D the rigorous aliasing and truncation budgets, the exported lower bound is

$$R_{\text{low}} = \text{lb} \left(\frac{k \text{ lb}(N_{\text{ball}} - E_N)}{\text{ub}(D_{\text{ball}} + E_D)} \right), \quad (27)$$

with all conversions explicitly rounded outward. The stored claim is rounded downward, so changes in valid Arb subdivision or enclosure radii cannot strengthen the reported inequality. The complete reduction, Fourier lemmas, moment bounds, tail estimates, special-function evaluation and directed rounding procedure are given in Supplementary Methods, Sections 5.20–5.23.

1.9 Finite-range certification and stable-law asymptotics

The first-order rational family also permits a certified bound over a continuous range of integer dimensions. Let $R_k^{(1)}$ denote the Rayleigh quotient of $g_c(t) = (c + (k-1)t)^{-1}$ after selection of c . To avoid optimizing c independently at every integer k , we use the standardized simplex headroom

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)} \quad (28)$$

as a scale coordinate. Its moments are available in closed form, and the large- k optimizers empirically concentrate near a fixed value of η . A log-uniform ladder beginning at $k = 100$ was therefore generated by solving $\eta(c, k) = -0.4746$ and certifying the resulting fixed rank-one trial at each rung. For consecutive certified values $k_i < k_{i+1}$, monotonicity of M_k gives, for every integer $k \in [k_i, k_{i+1}]$,

$$M_k \geq R_{\text{cert}}^{(1)}(k_i) \geq \log k - \left[\log k_{i+1} - R_{\text{cert}}^{(1)}(k_i) \right]. \quad (29)$$

The maximum interval deficit over the complete ladder is 0.306754936, yielding the uniform certified statement

$$M_k \geq \log k - 0.307, \quad 100 \leq k \leq 1.88 \times 10^9. \quad (30)$$

The ladder construction, scale surrogate and interval-by-interval calculation are given in Supplementary Methods, Sections 5.24–5.25.

The asymptotic behavior of the same rational family can be analysed independently of the numerical discovery pipeline. Put $n = k - 1$ and choose the fixed-shift scaling

$$c_k^{-1} = \log n + a, \quad a \in \mathbb{R}. \quad (31)$$

After rescaling the squared rational channel, the associated triangular array has Lévy density converging to $x^{-2} dx$. The centered boundary variable converges in distribution to

$$X_a = (a + 1) - \Lambda, \quad (32)$$

where Λ is the spectrally positive 1-stable random variable with characteristic function

$$\mathbb{E}e^{is\Lambda} = \exp \left\{ \int_0^\infty (e^{isx} - 1 - isx\mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2} \right\}. \quad (33)$$

Uniform density and logarithmic-integrability estimates permit passage from weak convergence to the logarithmic observables in the Rayleigh quotient. The resulting asymptotic defect is

$$R(F_{k,c_k}) = \log k - \mathcal{C}(a) + o(1), \quad \mathcal{C}(a) = a - 2\mathbb{E}[\log X_a \mid X_a > 0]. \quad (34)$$

Thus, with

$$\mathcal{C}_* = \inf_{a \in \mathbb{R}} \mathcal{C}(a), \quad (35)$$

the Maynard constant satisfies

$$M_k \geq \log k - \mathcal{C}_* - o(1). \quad (36)$$

Numerical evaluation of the limiting expression suggests $\mathcal{C}^* \approx 0.3343$. This is a lower-bound construction obtained from fixed-shift scalings and is not asserted here to be the asymptotic optimum over all possible sequences c_k . The stable-limit proof, logarithmic uniform-integrability argument and numerical evaluation of $\mathcal{C}_*$ are detailed in Supplementary Methods, Section 5.26.

1.10 ε -enlarged sieve calculations

We additionally evaluate the enlarged-domain variant of the Maynard variational problem. For each computed pair (k, ε) , the same discovery-to-certification separation is retained: a candidate is optimized numerically, transferred to the explicit representation used by the ε -certifier, and the resulting lower bound is recomputed independently. To compare the enlarged and vanilla searches we define

$$\Delta_{\text{cert}}(k, \varepsilon) = L_{k,\varepsilon} - L_{k,0}, \quad (37)$$

where $L_{k,\varepsilon}$ and $L_{k,0}$ are the certified lower bounds obtained by the two pipelines. Across the reliable sweep, Δ_{cert} decreases with k and changes sign for the tested values of ε . Linear interpolation of the reliable sign-bracketing pairs gives the crossover estimates reported in the Results (Supplementary Table 9).

Because Eq. (37) compares two lower bounds rather than the unknown exact variational constants, a negative value does not prove $M_{k,\varepsilon} < M_k$. It establishes only that the certified vanilla construction found by our pipeline outperforms the certified enlarged-domain construction at that k . We therefore treat the finite crossover as an empirical property of the certified construction ladder and state its extension to the exact variational constants as a conjecture. In particular, the observed crossover locations motivate

$$\varepsilon k^*(\varepsilon) \longrightarrow \infty \quad (\varepsilon \rightarrow 0), \quad (38)$$

but the available data do not determine a unique finer scaling law. The sign brackets, excluded optimization failures, interpolation procedure and scope of the conjecture are given in Supplementary Methods, Section 5.27.

1.11 Independent certificate verification

Large- k certificates are self-contained records containing the exact rational trial parameter, Fourier discretization parameters, error summaries and the claimed lower bound. The standalone verifier does not trust stored intermediate numerical bounds. It reconstructs the analytic constants, regenerates the far-field partition, recomputes wide-ball suprema, certified moments, Fourier nodes, trapezoidal sums, aliasing and truncation errors, and assembles a new R_{low} in ball arithmetic. The claim is accepted only when this independently recomputed lower bound supports the stored inequality.

The certificate proves the Rayleigh quotient of one fixed admissible trial function; it does not establish optimality of its parameter, re-prove the external Maynard–Tao sieve theorem, or certify the diameter of the admissible k -tuple used to convert a certified M_k value to a prime-gap bound. Monte–Carlo and small- k comparisons are falsification tests rather than proof components. Certificate schema and trust boundary are specified in Supplementary Methods, Section 5.31.

2 Higher-order Delsarte discovery and certification

Symmetry-reduced hierarchy. For hierarchy level r , blocklength n and minimum distance d we construct the symmetry-reduced higher-order Delsarte LP over type vectors

$$I_{r,n} = \left\{ \alpha \in \mathbb{N}^{2^r} : \sum_{u \in \mathbb{F}_2^r} \alpha_u = n \right\},$$

where a type records the multiplicity of each column $u \in \mathbb{F}_2^r$ in an $r \times n$ binary matrix. For linear codes the program is invariant under $\text{GL}(r, 2)$, so variables are indexed by group orbits. Full Fourier constraints are represented by multivariate Krawtchouk coefficients $K_\alpha(\beta)$, and the Loyfer–Linial strengthening adds the corresponding partial transforms [11]. We implemented these transforms recursively and extended the symmetry reduction to $r = 3$, where sorting-based orbit representatives are no longer valid; construction, orbit enumeration and validation anchors are given in Supplementary Methods 7.1.

Exact dual certification. Floating-point optimization is used only to locate competitive dual solutions. Candidates are reconstructed at high precision, rounded to dyadic rationals and checked against every dual inequality in exact arithmetic. Residual rounding violations are repaired using

$$\sum_{\beta \in I_{r,n}} \binom{n}{\beta} K_\alpha(\beta) = 2^{rn} \mathbf{1}_{\{\alpha = n\varepsilon_0\}},$$

which supplies an exactly feasible repair direction with a rational cost. Finite- n bounds therefore rest on exact dual feasibility and weak duality alone, not on the numerical LP tolerance, and are upper bounds on the level- r optimum (Supplementary Methods 7.2).

Reduction to an asymptotic configuration search

We next considered the spectral dual construction of Coregliano et al., in which the complementary spectral function is supported on a single configuration orbit [12]. A normalized configuration is a probability distribution G on $\mathbb{F}_2^\ell$. Because a level- ℓ dual bounds $|C|^\ell$, the asymptotically relevant rate objective is

$$J_\ell(G) = \frac{H(G)}{\ell},$$

with $H(\cdot)$ the Shannon entropy in bits; this normalization, and two discrepancies in the cited preprint that it exposes, are recorded in Supplementary Methods 7.4. For a non-zero shift v we use the Bhattacharyya translation affinity

$$\Phi(G, v) = \sum_u \sqrt{G(u)G(u+v)}.$$

Writing $G = p^2$ with $p \geq 0$ and $\|p\|_2 = 1$ gives $\Phi(G, v) = p^\top S_v p$, where S_v is the permutation induced by translation by v , so discovery reduces to entropy minimization on the positive unit sphere under quadratic affinity constraints. The analytic and symmetry reductions leave $2^\ell - 1$ free probability coordinates, at most 15 for $\ell \leq 4$, with no remaining dependence on n . We had intended a neural parameterization of G , but at this point a generic network was unnecessary, so we searched the reduced space directly by multistart sequential quadratic programming: 28 instance solves over $\ell \leq 4$ and $\varepsilon \in \{0.30, 0.20, 0.10, 0.05\}$, 13,600 local optimizations in total. No run produced a configuration below the first MRRW benchmark, and the recovered optima matched the quasirandom tensor-product family to within 10^{-11} coordinatewise (Supplementary Methods 7.5).

The tensorization obstruction

The repeated null result is explained by an inequality that holds for every admissible configuration. Let $\varepsilon \in (0, 1)$, let V be the set of shifts at which the spectral walk condition of [12] is verified, and let

$$h(\varepsilon) = H_2\left(\frac{1}{2}(1 - \sqrt{1 - \varepsilon^2})\right) = H_2\left(\frac{1}{2}(1 - \sqrt{\delta(1 - \delta)})\right), \quad \delta = \frac{1 - \varepsilon}{2},$$

where H_2 is the binary entropy function; the second form is the first MRRW expression. Two properties of the construction enter: the admissible shifts span $\mathbb{F}_2^\ell$, and at each such shift a finite even spectral-walk order m satisfies a lower bound of the form $W_m(G, v) \geq c\varepsilon^m$ with $c \geq 1$, where W_m is the paired walk weight appearing in the configuration asymptotics of [12]. Comparing W_m termwise with the multinomial expansion of $\Phi(G, v)^m$ gives $W_m(G, v) \leq \Phi(G, v)^m$, so the walk condition forces the affinity floor $\Phi(G, v) \geq \varepsilon$ at a fixed m , with no exchange of the $n \rightarrow \infty$ and $m \rightarrow \infty$ limits. Conditioning each coordinate on the other turns that floor into an entropy floor: the conditional entropy of X_i given X_{-i} equals $\mathbb{E}[h(2\sqrt{p_Y(1-p_Y)})]$ while $\Phi(G, e_i)$ equals $\mathbb{E}[2\sqrt{p_Y(1-p_Y)}]$ for the same conditional marginals, so convexity of h and Jensen's inequality give $H(X_i | X_{-i}) \geq h(\Phi(G, e_i))$. The entropy chain rule then yields

$$H(G) \geq \sum_{i=1}^{\ell} h(\Phi(G, e_i)) \geq \ell h(\varepsilon), \quad \text{hence} \quad J_\ell(G) \geq h(\varepsilon).$$

The tensor-product configuration $G_\varepsilon = \text{Bernoulli}(p_\varepsilon)^{\otimes \ell}$ attains equality in the relaxed affinity-constrained problem. Under the stronger finite-walk hypotheses (A1)–(A2), however, the entropy-based rate estimate is strictly larger than $h(\varepsilon)$. Thus the computationally recovered tensor-product structure is explained by the tensorization inequality within the analyzed relaxation. This does not establish feasibility of G_ε for the original finite-walk construction, nor does it rule out improvements through configurations or dual constructions that lie outside these hypotheses. Full hypotheses, proofs and scope are given in Supplementary Methods 7.6 (Theorem 18 and Corollary 19); the algebraic reductions underlying the theorem were verified against independent brute-force implementations (Supplementary Methods 7.7).

The obstruction constrains only the configuration. It does not constrain alternative sign functions ϕ , spectral functions supported on several configuration orbits, other uses of the partial-Fourier dual families, or general dual feasible solutions, and the higher-order hierarchy itself remains complete. The computational search therefore localized the missing asymptotic freedom in those components rather than in G .

3 Sign-uncertainty discovery and certification

Fourier-eigenfunction formulation

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i \langle x, \xi \rangle} dx$$

and consider the +1 sign-uncertainty constant $A_+(d)$ over non-zero even integrable functions satisfying $\widehat{f} = f$, $f(0) = 0$, and $f(x) \geq 0$ for all sufficiently large $|x|$ [28, 29]. To construct upper bounds we restrict the search to finite-dimensional radial Laguerre polynomial–Gaussian eigenspaces. With

$$u = \pi|x|^2, \quad \alpha = \frac{d}{2} - 1,$$

the functions

$$\psi_n(u) = L_n^{(\alpha)}(2u)e^{-u}$$

satisfy $\widehat{\psi_n} = (-1)^n \psi_n$. Hence the truncated +1 eigenspace is

$$V_N = \text{span}\{\psi_0, \psi_2, \dots, \psi_{2N}\}.$$

Writing $f = P(u)e^{-u}$ converts the Fourier constraint into the linear requirement that P lie in the corresponding even Laguerre span. For integer d these generalized Laguerre polynomials have rational coefficients, which enables exact reconstruction.

From neural proposals to a convex search

The initial discovery stage searched a broader Fourier-eigenfunction family with trainable Gaussian scales and contact locations. Local and neural optimization rapidly improved the objective but became

unreliable when the coefficient systems approached condition numbers of 10^{17} – 10^{18} . Independent multiprecision reevaluation showed that apparently improving trajectories were not reproducible. We therefore treated learned widths and contacts only as proposals and used this failure to reconsider the search coordinates rather than to infer that the function class had been exhausted.

For the pure Laguerre family the rescaled function is simply the polynomial $P(u) = e^u f(u)$. Since $e^{-u} > 0$, fixing a candidate last-sign point u_0 reduces admissibility to

$$P(0) = 0, \quad P(u) \geq 0 \quad (u \geq u_0).$$

The feasible set in the Laguerre coefficients is convex and monotone in u_0 . Thus, the non-convex optimization over roots and contact locations can be replaced by a one-dimensional bisection in u_0 with a convex semi-infinite feasibility problem at each step.

Adaptive semi-infinite feasibility

At each bisection point we solve a margin-maximizing LP on an adaptive grid. A positive linear normalization,

$$\int_{u_0}^{\infty} P(u) e^{-u} du = 1,$$

removes the conic scale, and the leading Laguerre coefficient is constrained to have the sign required for positivity at infinity. The grid is only a relaxation. After each LP solve, roots of P' are located at high precision; any missed negative minimum is inserted as a cutting plane and the LP is resolved. Near the fixed-degree feasibility boundary, the active constraints identify candidate double contacts.

The exchange loop is used to locate the boundary and its contact structure, not to certify the quoted upper bound. The underlying space is not a Haar system, so uniqueness of the active contact set is not assumed. Instead, the convex semi-infinite formulation supplies a numerical bracket for sharpness within the chosen finite-dimensional space, while exact admissibility is established independently downstream. Failure of the numerical oracle can therefore reduce sharpness but cannot make an invalid certificate valid.

Exact rational reconstruction and verification

Let $\tau_1 < \dots < \tau_m$ denote the active tangencies returned by the convex search, rounded to nearby rational values. We reconstruct P exactly in the Laguerre basis by imposing

$$P(0) = 0, \quad P(\tau_i) = P'(\tau_i) = 0 \quad (1 \leq i \leq m),$$

together with a fixed positive top coefficient. In the square certificates, $n_b = 2m + 2$ Laguerre coordinates are retained. Solving over $\mathbb{Q}$ gives

$$P(u) = \prod_{i=1}^m (u - \tau_i)^2 R(u), \quad R \in \mathbb{Q}[u],$$

with every polynomial division checked to have zero remainder.

Positivity on the complete ray is then decided independently of the LP grid. We verify that the leading coefficient of R is positive, that $R(\bar{u}) > 0$, and that a Sturm chain has the same sign-variation count at $\bar{u}$ and at $+\infty$. Hence R has no zero on $[\bar{u}, \infty)$ and therefore

$$P(u) \geq 0 \quad (u \geq \bar{u}).$$

Since $\hat{f} = f$ and $f(0) = 0$ hold exactly by construction, $f = P(u)e^{-u}$ is admissible and

$$A_+(d) \leq \sqrt{\bar{u}/\pi}.$$

For the $d = 1$ degree-38 certificate, $m = 9$ and

$$\bar{u} = \frac{514\,997\,856\,761}{5 \cdot 10^{11}},$$

which yields

$$A_+(1) \leq 0.572588699 \dots$$

The $d = 2$ degree-22 certificate is constructed analogously and gives $A_+(2) \leq 0.756206236 \dots$. The final certificate contains only rational Laguerre data and integer/rational Sturm arithmetic; floating-point computation is confined to discovery of a useful contact structure.

Scope

The finite-dimensional convex reformulation does not contradict the Cohn–Dong–Gonçalves asymptotic ceiling for sublinear-degree polynomial–Gaussian families [13]. It shows instead that, at small dimension, the classical family had not been numerically exhausted by the previous root-parameterized searches. Detailed tripwires, discovery experiments, convergence diagnostics, contact degeneracies and higher-dimensional numerical limits are given in the Supplementary Methods.

4 Use of large language models

A large language model (Claude, Anthropic; Opus 4.5, 4.8 and Fable5) was used throughout the exploratory development of the computational pipeline as an interactive programming and research-assistance tool. Its contributions were concentrated in software engineering (making the scripts into a package) and numerical debugging: proposing and revising implementation strategies, generating and refactoring code, and diagnosing numerical failure modes in the certification pipeline, including floating-point underflow (e.g. float-64 versus float-32 in GPU representations) and dynamic-range collapse, basis-conditioning and rank-instability effects, quadrature overflow, and (formulation) inconsistencies between the discovery and certification stages (see Supplementary Tables 2 and 3). The model also served as a discussant for (formal) mathematical framing.

The mathematical content of this work is the author’s. The variational formulation, the reduction underlying the pipeline, the certification scheme, and all theorem statements were conceived, derived, and verified by the author; the model’s role with respect to the mathematics was limited to exposition, sanity-checking, and the surfacing of relevant literature, all of which the author confirmed. Where the model proposed an algorithmic optimisation, the proposal was accepted only after the author verified it. For reference; PyTorch and Flint are not installed on Anthropic servers so no computations were done by a LLM.

All model outputs were critically evaluated by the author, who bears sole responsibility for the correctness of every claim. All reported numerical bounds are the output of deterministic software. Crucially, the language model is not invoked by the final certification pipeline: the validity of certified bound rests entirely on the exact-arithmetic certificate and its independent verification, which can be checked without reference to any part of the development process. A structured contribution-provenance record, and the complete certification and verification software are provided in the Supplementary Methods.

References

1. Davies, A. *et al.* Advancing mathematics by guiding human intuition with AI. *Nature* **600**, 70–74 (2021).
2. Fawzi, A. *et al.* Discovering faster matrix multiplication algorithms with reinforcement learning. *Nature* **610**, 47–53 (2022).
3. Romera-Paredes, B. *et al.* Mathematical discoveries from program search with large language models. *Nature* **625**, 468–475 (2024).
4. Mundinger, K., Zimmer, M., Kiem, A., Spiegel, C. & Pokutta, S. *Neural Discovery in Mathematics: Do Machines Dream of Colored Planes?* in *Proceedings of the 42nd International Conference on Machine Learning* **267** (PMLR, 2025).
5. Engelsma, T. J. & Sutherland, A. V. *Narrow Admissible Tuples* <https://math.mit.edu/~primegaps/>. Entry for $k = 3655$ submitted by A. V. Sutherland, 27 June 2013. 2013.

6. Polymath, D. H. J. *Variants of the Selberg sieve, and bounded intervals containing many primes* 2014. arXiv: [1407.4897](https://arxiv.org/abs/1407.4897) [math.NT]. <https://arxiv.org/abs/1407.4897>.
7. Stadlmann, J. On primes in arithmetic progressions and bounded gaps between many primes. *Advances in Mathematics* **468**, 110190 (2025).
8. Stadlmann, J. *Bounded gaps between primes* Submitted 31 August 2026. 2026. arXiv: [2608.31126](https://arxiv.org/abs/2608.31126) [math.NT]. <https://arxiv.org/abs/2608.31126>.
9. Charton, F. *et al.* *A new bound for small gaps between primes* Preliminary draft, Axiom Math. Dated 3 September 2026. Lean formalization at <https://github.com/AxiomMath/PrimeGapsLib>. Accessed 14 September 2026. 2026.
10. OpenAI. *Improved short gaps between primes* https://cdn.openai.com/pdf/51126fac-1b68-4128-9666-c908bcc16033/short_gaps.pdf. Dated 30 August 2026. Proof attributed to GPT-6 Astra; Lean 4 formalization and numerical certificate at <https://github.com/openai/PrimeGaps186>. Accessed 14 September 2026. 2026.
11. Loyfer, E. & Linial, N. New LP-based upper bounds in the rate-vs.-distance problem for binary linear codes. *IEEE Transactions on Information Theory* **69**, 2886–2899 (2023).
12. Coregliano, L. N., Jeronimo, F. G., Jones, C., Linial, N. & Loyfer, E. Higher-order delarte dual lps: Lifting, constructions and completeness. *arXiv preprint arXiv:2501.04854* (2025).
13. Cohn, H., Dong, D. & Gonçalves, F. Sign uncertainty principles and low-degree polynomials. *arXiv preprint arXiv:2210.01684* (2022).
14. Raayoni, G. *et al.* Generating conjectures on fundamental constants with the Ramanujan Machine. *Nature* **590**, 67–73 (2021).
15. Udrescu, S.-M. & Tegmark, M. AI Feynman: A physics-inspired method for symbolic regression. *Science advances* **6**, eaay2631 (2020).
16. Trinh, T. H., Wu, Y., Le, Q. V., He, H. & Luong, T. Solving olympiad geometry without human demonstrations. *Nature* **625**, 476–482 (2024).
17. Tan, Y. K., Yang, J., Soos, M., Myreen, M. O. & Meel, K. S. *Formally certified approximate model counting in International Conference on Computer Aided Verification* (2024), 153–177.
18. Hales, T. *et al.* *A formal proof of the Kepler conjecture in Forum of mathematics, Pi* **5** (2017), e2.
19. Maynard, J. Small gaps between primes. *Annals of mathematics*, 383–413 (2015).
20. Parlett, B. N. *The Symmetric Eigenvalue Problem* (SIAM, 1998).
21. Golub, G. H. & Van Loan, C. F. *Matrix Computations* 4th ed. (Johns Hopkins University Press, 2013).
22. Feynman, R. P. Forces in Molecules. *Physical Review* **56**, 340–343 (1939).
23. Semay, C. The Hellmann–Feynman theorem, the comparison theorem, and the envelope theory. *Results in Physics* **5**, 322–323 (2015).
24. Kingma, D. P. & Ba, J. *Adam: A Method for Stochastic Optimization in International Conference on Learning Representations* (2015).
25. Liu, D. C. & Nocedal, J. On the Limited Memory BFGS Method for Large Scale Optimization. *Mathematical Programming* **45**, 503–528 (1989).
26. Golub, G. H. & Welsch, J. H. Calculation of Gauss Quadrature Rules. *Mathematics of Computation* **23**, 221–230 (1969).
27. Johansson, F. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers* **66**, 1281–1292 (2017).
28. Bourgain, J., Clozel, L. & Kahane, J.-P. *Principe d’Heisenberg et fonctions positives in Annales de l’institut Fourier* **60** (2010), 1215–1232.
29. Cohn, H. & Gonçalves, F. An optimal uncertainty principle in twelve dimensions via modular forms: H. Cohn, F. Gonçalves. *Inventiones mathematicae* **217**, 799–831 (2019).

5 Supplementary Methods: Maynard–Tao sieve

5.1 Variational definitions and normalization

Let

$$\mathcal{R}_k = \left\{ \mathbf{t} \in \mathbb{R}_{\geq 0}^k : \sum_{i=1}^k t_i \leq 1 \right\}.$$

For an admissible square-integrable trial function F supported on $\mathcal{R}_k$, define

$$I_k(F) = \int_{\mathcal{R}_k} F(\mathbf{t})^2 d\mathbf{t}$$

and

$$J_k^{(m)}(F) = \int_{\mathcal{R}_{k-1}} \left(\int_0^{1-\sum_{i \neq m} t_i} F(\mathbf{t}) dt_m \right)^2 d\mathbf{t}_{-m}.$$

The Maynard variational constant is

$$M_k = \sup_{F \neq 0} \frac{\sum_{m=1}^k J_k^{(m)}(F)}{I_k(F)}.$$

For symmetric F , all marginal functionals are equal and

$$M_k = k \sup_{F \neq 0} \frac{J_k^{(1)}(F)}{I_k(F)}.$$

Throughout the computational pipeline, the reported Rayleigh quotient is normalized according to this convention. We also use the classical Cauchy–Schwarz [1] ceiling

$$M_k \leq \frac{k}{k-1} \log k,$$

as a validation constraint on both single-channel and full numerical quotients. The enlarged-support calculations use the corresponding Polymath ε -variant, with the same discovery–certification separation. Because the exact normalization of the enlarged polytope and the external conversion from a certified variational threshold to a numerical H_m bound involve the sieve theorem and an admissible k -tuple rather than the NeuralCert certificate itself, those ingredients are kept distinct from the variational certificate. In particular, the certificate proves the stated lower bound for the explicit trial function; the final H_m conversion additionally uses the external Maynard–Tao/Polymath theorem and a separately supplied admissible-tuple diameter.

5.2 From exact replication to neural discovery

The Maynard experiments began from an exact finite-dimensional control rather than from a neural model. Following Lemmas 8.1 and 8.2 of [2], we used the symmetric polynomial family

$$F(\mathbf{t}) = \sum_{\ell=1}^d a_\ell (1 - P_1(\mathbf{t}))^{b_\ell} P_2(\mathbf{t})^{c_\ell}, \quad P_j(\mathbf{t}) = \sum_{i=1}^k t_i^j,$$

for which both I_k and $\sum_m J_k^{(m)}$ reduce to rational quadratic forms in the coefficient vector a . The finite-dimensional optimization is therefore a generalized Rayleigh–Ritz problem. This approach is consistent with Maynard’s observation that an extremizer should satisfy an eigenfunction equation for the underlying integral operator. The neural stage can therefore be interpreted as a nonlinear numerical search for a dominant variational eigenfunction, rather than as unconstrained function fitting. As a first control we reproduced Maynard’s explicit $k = 5$ trial

$$F(\mathbf{t}) = (1 - P_1)P_2 + \frac{7}{10}(1 - P_1)^2 + \frac{1}{14}P_2 - \frac{3}{14}(1 - P_1),$$

for which

$$Q_5(F) = \frac{1417255}{708216} > 2.$$

We subsequently reproduced the degree-constrained generalized eigenvalue calculation at $k = 105$. Exact rational assembly was retained throughout; the floating-point generalized eigensolve was used only to locate a Ritz vector, which was then rationalized and reevaluated in the exact quadratic forms.

The exact polynomial controls were next used to calibrate Monte Carlo estimators before any black-box optimization. Uniform simplex samples were generated from normalized independent exponential variables. For the marginal functional, two conditionally independent samples U, U' were drawn on the same fibre. This gives the unbiased product estimator

$$\mathbb{E}[F(U, \mathbf{W})F(U', \mathbf{W}) \mid \mathbf{W}] = \left(\frac{1}{C} \int_0^C F(u, \mathbf{W}) du \right)^2,$$

where $C = 1 - \sum W_i$. Squaring a single inner sample would add its conditional variance and bias the estimate upward. Agreement of the Monte Carlo estimates with the exact polynomial values, expressed through standardized residuals, was used as the first stochastic validation gate.

Only after these controls were in place was the fixed polynomial ansatz replaced by a trainable symmetric function. Early neural models used scaled power-sum coordinates

$$\phi_j(\mathbf{t}) = k^{j-1} \sum_i t_i^j$$

and optimized a stochastic Rayleigh quotient. These experiments established that the optimizer could recover useful symmetric structure without being given the final polynomial representation, but also exposed the variance, conditioning and representation-transfer problems that motivated the deterministic separable pipeline used in the main analysis.

Finally, further control showed that enlargement of the admissible domain does not by itself improve a fixed trial function. For example, Maynard’s Eq. (8.16), optimized for the simplex geometry, decreases from approximately 2.00 to 1.66 when evaluated at $\varepsilon = 1/4$. The gain from enlarged support therefore requires re-optimization to the modified geometry rather than simple reuse of a simplex-optimized extremizer. This provided an additional motivation for learning the trial function directly in each ε -geometry.

Recent computational work has also explored heuristic modifications of Maynard-type multidimensional sieves, including dynamically enlarged support regions and random-matrix-inspired perturbations [3], but their predicted prime-gap improvements remain heuristic. In contrast, the present framework separates exploratory optimization from certification of an explicit variational trial function and invokes only independently stated sieve-theoretic inputs for the final number-theoretic consequence.

5.3 Exploratory representation experiments

Before arriving at the final separable channel representation, five candidate compression mechanisms, most of which were ultimately rejected. These experiments were diagnostic rather than proof components: their role was to identify which representations preserved the variational structure well enough to justify further development. All bounds in the paper are certified independently of the outcomes described below.

5.3.1 Experiment 1: partitions

Exact Rayleigh–Ritz optimization in the full monomial-symmetric basis is effective at small k , but encounters a symmetric curse of dimensionality: although the partition basis is L^2 -dense, the number of basis elements up to degree D grows as the restricted partition count $p_k(D)$ and becomes rapidly prohibitive. This motivated attempts to retain only a small, variationally relevant subset of partitions.

Three selectors were tested: exact one-vector residual greedy expansion, a global floating-point Ritz oracle, and neural coefficient extraction through a power-sum-product representation. The residual score

$$\eta_\lambda = \frac{|\langle m_\lambda, (L - \lambda_S)F_S \rangle|}{\|m_\lambda\|_2}$$

improved the recorded exploratory bound by only about 2.5×10^{-3} after adding approximately 30 highest-scoring partitions, much less than adding a complete degree shell. The global floating-point and neural coefficient selectors were additionally destabilized by the ill-conditioned change of coordinates into the monomial-symmetric basis. The experiment therefore rejected sparse single-partition selection as an effective acceleration mechanism: the missing variational gain was collective and strongly correlated rather than localized in a few coordinates.

This failure suggested that the relevant structure was not sparse in the partition basis but instead closer to a low-complexity product or tensor representation, motivating the tensor-decomposition experiments below.

5.3.2 Experiment 2: symmetric tensor decompositions

The failure of sparse partition selection suggested that the optimizer might be compact in a tensor representation. We therefore fitted the symmetric CP family

$$F_{\text{CP}}(\mathbf{t}) = \sum_{q=1}^r c_q \prod_{i=1}^k \phi_q(t_i)$$

to the certified $k = 5$, degree-10 Ritz optimizer. The relative empirical L^2 reconstruction errors were

r	1	2	3	4	5	6
$\mathcal{E}_r$	0.1887	0.1536	0.1032	0.1028	0.1076	0.1089

and plateaued near 10% beyond rank three. The corresponding Monte Carlo Rayleigh estimates were statistically compatible with the exact target but noisy and non-monotone; values above the target were within the quoted Monte Carlo uncertainty and were never treated as improved lower bounds. Thus, increasing the symmetric CP rank beyond $r = 3$ produced essentially no further reduction in reconstruction error over the tested range. This rejected the simplest low-rank CP compression as a complete representation, but not the broader tensor-network hypothesis.

5.3.3 Experiment 3: structural signatures of the CP residual

For the rank-six symmetric CP fit, the relative empirical reconstruction error was

$$\frac{\|F^\star - F_{\text{CP},6}\|_2}{\|F^\star\|_2} = 0.10903.$$

We projected the residual onto the full degree-10 monomial-symmetric basis. The largest fitted coordinate signatures occurred in the one-part directions

$$(4), (3), (2), (5), (1)$$

and the two-part directions

$$(2, 1), (3, 1), (1, 1).$$

Since one-part monomial-symmetric functions are power sums, this pointed to low-order power-sum structure in the component not captured by the CP fit.

As a diagnostic, we formed the diagonal coordinate proxy

$$\zeta_\lambda = b_\lambda^2 \|m_\lambda\|_{2,\text{emp}}^2.$$

Approximately 83% of this proxy was associated with one-body partitions and 17% with two-body partitions, with negligible proxy weight at higher body counts. This must not be interpreted as an orthogonal L^2 -energy decomposition:

$$\sum_{\lambda} \zeta_{\lambda} = 6.71 \times 10^6,$$

whereas the empirical squared residual norm was only 1.43, revealing strong cancellation in the non-orthogonal partition basis.

The diagnostic nevertheless suggested that the residual was concentrated in structurally simple low-order directions rather than in a diffuse high-body tail. Since such components are in principle representable within a sufficiently flexible CP family, the observed rank saturation pointed more naturally to an optimization or conditioning limitation than to an immediate expressivity obstruction. One plausible explanation, explored in subsequent experiments, was that these directions were difficult to resolve by gradient-based fitting against Monte Carlo noise.

5.3.4 Experiment 4: explicit power-sum corrections

The residual signature motivated the augmented model

$$F_{\text{CP+P}} = F_{\text{CP}} + \beta_0 + \sum_{j=1}^J \beta_j \tilde{P}_j, \quad \tilde{P}_j = P_j / \mathbb{E}[P_j].$$

This experiment did not yield a reliable acceleration. The revised implementation failed first to reproduce its own CP-only baseline, the Monte Carlo estimator for J became high variance for some augmented functions, and the explicit power-sum branch overlapped strongly with directions already represented by the CP factors. Consequently, the small observed changes with increasing J could not be separated cleanly from optimization and implementation signal.

5.3.5 Experiment 5: body-restricted partition spaces

The final pre-separable experiment restricted the exact polynomial trial space simultaneously by total degree and partition length, or “body count”. For

$$\mathcal{V}(\{D_b\}) = \text{span}\{m_\lambda : \ell(\lambda) = b, |\lambda| \leq D_b\},$$

matrix entries were assembled by enumerating coalescences of the positive parts of two partitions rather than their full permutation orbits. The restriction is therefore specified by a degree cap D_b for each body count b ; at $k = 50$ no run included $b > 5$, so every space reported below is a genuine body-restricted subspace of the full monomial-symmetric space of the same total degree. At $k = 5$, where $\ell(\lambda) \leq k = 5$ holds automatically, the unrestricted degree-10 space contained 113 partitions and reproduced

$$M_5 \geq 2.0071443298,$$

approaching Bogaert’s Krylov value 2.0071451444. Restricting to partition length at most two reduced the space to 36 basis elements and gave a strictly weaker value, confirming the coalescence construction while showing that low-body truncation alone does not recover the full small- k optimizer.

At $k = 50$, exact matrix assembly remained feasible, but the generalized eigensolve became the dominant limitation. Rank-revealing whitening retained only a fraction of the nominal Gram modes; for a common degree cap of 15, 18, 20 and 22 on every body count the reductions were

$$408 \rightarrow 191, \quad 769 \rightarrow 278, \quad 1125 \rightarrow 344, \quad 1601 \rightarrow 421.$$

The retained block consistently approached the imposed conditioning threshold, $\kappa \sim 10^{12}$, indicating that the effective ceiling was set by float64 numerical resolution rather than by the nominal size of the partition space. This effect is visible in the non-monotone certified ladder obtained from these increasing nominal spaces:

$$3.36957, \quad 3.36618, \quad 3.36800, \quad 3.36435.$$

This does not contradict Rayleigh–Ritz monotonicity: the nominal trial spaces are nested, whereas the independently whitened effective subspaces need not be. Exact reevaluation still certifies the returned trial vector, but cannot restore directions discarded before the eigensolve.

The precision dependence is particularly clear in a matched 408-dimensional calculation at degree cap 15. Retaining the complete pencil at high precision gave the certified value

$$M_{50} \geq 3.6347,$$

whereas float64 spectral whitening retained only 191 modes and yielded 3.3696. Clearly, the lost variational gain was already present in the polynomial trial space but fell below the numerical resolution of the whitened Gram pencil. The loss is strongly space-dependent rather than systematic: on the 441-dimensional one-two-body space the high-precision solve gave 3.1862092 against 3.1862074 from 134 whitened modes, so there the discarded directions carried almost no variational weight. A larger space admitting three-body partitions up to degree 20 similarly reached 3.3904 with all 678 modes retained at high precision, while larger float64-whitened spaces did not systematically improve the bound. Supplementary Table 1 summarizes representative calculations.

Why float64 whitening was insufficient in the monomial-symmetric basis. One numerical difficulty in the early polynomial calculations was severe ill-conditioning of the denominator Gram matrix in the monomial-symmetric basis. The matrix is positive definite on a linearly independent trial space, but after conversion to double precision its smallest eigenvalues can fall below the numerical resolution of the eigensolver. Whitening then became problematic because the transformation

$$W = U_{\text{keep}} \text{diag}(\sigma_i^{-1/2})$$

amplified perturbations associated with small Gram eigenvalues, while directions below the chosen threshold were removed entirely. So, the optimization is no longer performed over the nominal polynomial space, but only over its numerically resolved spectral image. Exact rational re-evaluation of the returned Ritz vector still yields a valid lower bound for that particular vector, but it cannot recover variationally relevant directions that were discarded during whitening. With increasing k , larger float64-whitened spaces could produce weaker certified bounds than smaller spaces solved at high precision, demonstrating the numerical precision bottleneck. This motivated representations with Gram structure without spectral truncation.

Supplementary Table 1: Representative $k = 50$ partition-space calculations. The trial space is specified by a degree cap per body count b (partition length); no run included $b > 5$. Certified values refer to the returned explicit trial function. High-precision solves retain the full trial space, whereas float64 whitening removes Gram directions below the numerical resolution threshold.

Trial space (degree cap by body count)	Eigensolver	Dimension	Retained	Certified
$b \leq 5$, all degrees ≤ 15	high precision	408	408	3.6347
$b \leq 5$, all degrees ≤ 15	float64 whitening	408	191	3.3696
$b \leq 2$, degrees ≤ 40	float64 whitening	441	134	3.1862
$b \leq 3$, degrees (40, 40, 20)	high precision	678	678	3.3904
$b \leq 5$, all degrees ≤ 22	float64 whitening	1601	421	3.3644

Note. Bogaert reported $M_{50} \geq 3.9358660346$.

The experiment therefore identified two distinct obstacles in the monomial-symmetric representation: rapid growth of the partition basis and loss of numerically resolvable Gram directions. The limiting issue at $k = 50$ was not expressive power of the polynomial space itself, but the combination of basis growth and near-degeneracy. This motivated the move to representations whose complexity is controlled before the generalized eigensolve and whose structure is more directly compatible with exact certification.

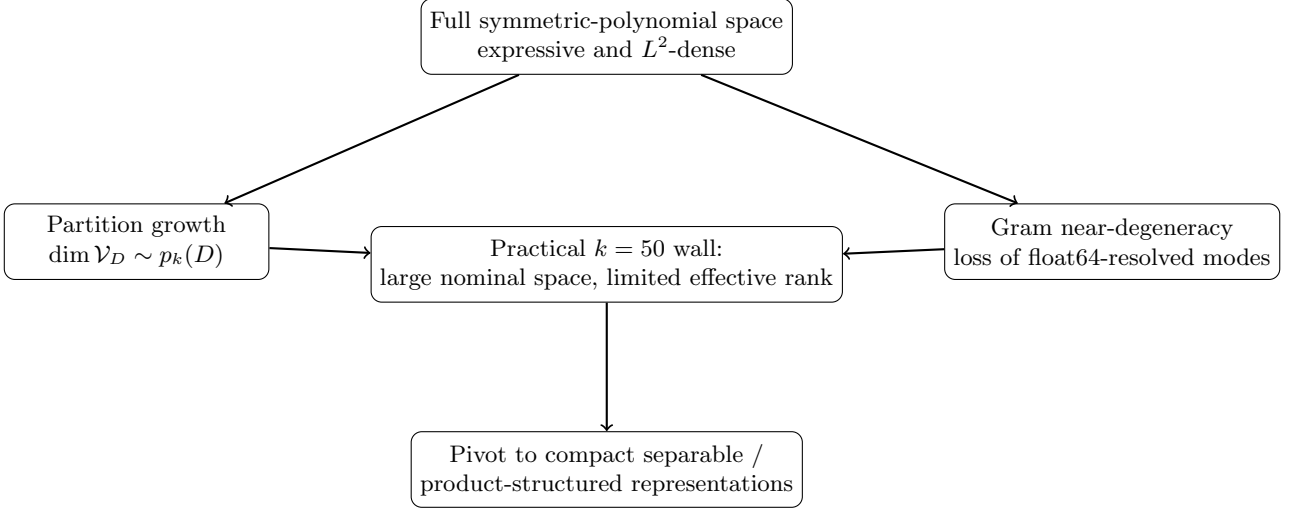


Figure 3: Scaling obstruction of the monomial-symmetric representation. Increasing polynomial degree expands an expressive symmetric trial space, but at $k = 50$ this produces both rapid partition growth and increasingly ill-conditioned Gram pencils. The resulting loss of numerically resolved directions motivates the later separable representation.

5.3.6 Consequence for the final representation

The abovementioned five experiments ruled out several superficially natural routes: sparse partition support, very-low-rank symmetric CP compression, unqualified interpretation of residual coefficients, additive power-sum patches, and brute-force growth of ill-conditioned polynomial spaces. The common failure mode was not lack of a strong candidate function but loss of structure during representation or optimization. This motivated the separable-channel formulation used in the main Methods, in which a small dictionary of one-dimensional functions is optimized directly, the generalized Ritz coefficients are profiled out, and independent certification operates only after the discovered function has been reduced to an explicit mathematical representation.

Scripts for the exploratory analyses described above are not included in NeuralCert. Together with additional exploratory outputs including full residual rankings, body-count and parity summaries, and repeated fit diagnostics, they are available upon request.

5.4 Neural channel model

The discovery stage uses the symmetric separable representation

$$F_{\theta,c}(t_1, \dots, t_k) = \sum_{j=1}^m c_j \prod_{i=1}^k g_{\theta,j}(t_i).$$

The one-dimensional channels are written

$$g_{\theta,j}(x) = r_{\theta,j}(x)e^{-\rho_j x}, \quad \rho_j > 0,$$

with trainable rates initialized across a broad geometric range. The residual factors are produced jointly by a shared multilayer perceptron with channel-specific outputs and input

$$(x, e^{-kx}).$$

For the positive-channel model,

$$r_{\theta,j}(x) = \exp(f_{\theta,j}(x, e^{-kx})), \quad g_{\theta,j}(x) = \exp(f_{\theta,j}(x, e^{-kx}) - \rho_j x).$$

Positivity permits cancellation-free log-domain convolution. The architecture differs from a conventional DeepSets model: permutation invariance follows from the product structure of the separable expansion, while the neural network supplies a dictionary of one-dimensional factors.

Listing 1: Positive neural channel dictionary. A shared network produces the residual factors, while the exponential envelopes are represented analytically.

```

PROCEDURE CHANNELS(x, k, theta, eta, rho_max)
  rho    <- MIN(SOFTPLUS(eta), rho_max)
  features <- STACK(x, EXP(-k*x))
  log_raw <- SHARED_MLP_theta(features)
  g       <- EXP(log_raw - OUTER(x, rho))
  RETURN g, log_raw, rho
END

FUNCTION TRIAL(t, c, theta)
  RETURN SUM_j c[j] * PRODUCT_i CHANNEL_j(t[i], theta)
END

```

5.5 Factored convolution powers and Gauss–Jacobi recursion

For a channel pair (j, ℓ) , write

$$h_{j\ell}(x) = r_{j\ell}(x)e^{-a_{j\ell}x}, \quad r_{j\ell}(x) = r_{\theta,j}(x)r_{\theta,\ell}(x), \quad a_{j\ell} = \rho_j + \rho_\ell.$$

Every convolution power is represented as

$$\nu_p^{(j\ell)}(s) := h_{j\ell}^{*p}(s) = s^{p-1}e^{-a_{j\ell}s}e^{\sigma_p^{(j\ell)}}\psi_p^{(j\ell)}(s).$$

The initialization is

$$\sigma_1^{(j\ell)} = 0, \quad \psi_1^{(j\ell)}(s) = r_{j\ell}(s).$$

We suppress the channel-pair superscript below.

For positive integers p and q , substitution into

$$\nu_{p+q}(s) = \int_0^s \nu_p(x)\nu_q(s-x) dx$$

and the change of variables $x = su$ give

$$\nu_{p+q}(s) = s^{p+q-1}e^{-as}e^{\sigma_p+\sigma_q} \int_0^1 \psi_p(su)\psi_q(s(1-u))u^{p-1}(1-u)^{q-1} du.$$

Let

$$B(p, q) = \int_0^1 u^{p-1}(1-u)^{q-1} du = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

We retain the beta factor in the residual recursion and therefore use

$$\sigma_{p+q} = \sigma_p + \sigma_q,$$

together with

$$\psi_{p+q}(s) = B(p, q) \int_0^1 \psi_p(su)\psi_q(s(1-u)) \frac{u^{p-1}(1-u)^{q-1}}{B(p, q)} du.$$

Thus the integral is an expectation with respect to the normalized Beta(p, q) density. The factor $B(p, q)$ occurs once, in ψ_{p+q} , and is not also included in σ_{p+q} .

The expectation is evaluated by Gauss–Jacobi quadrature:

$$\psi_{p+q}(s) \approx B(p, q) \sum_{r=1}^{n_q} \hat{w}_r^{(p,q)} \psi_p(su_r^{(p,q)}) \psi_q(s(1-u_r^{(p,q)})),$$

where

$$0 < u_r^{(p,q)} < 1, \quad \hat{w}_r^{(p,q)} > 0, \quad \sum_{r=1}^{n_q} \hat{w}_r^{(p,q)} = 1.$$

Listing 2: Binary-chain evaluation of the factored convolution power in logarithmic coordinates.

```

PROCEDURE LOG_CONVOLUTION_POWER(log_raw, j, 1, k)
  xi[1](s) <- log_raw_j(s) + log_raw_1(s)

  FOR EACH (p,q) IN BINARY_ADDITION_CHAIN(k-1)
    (u,w) <- NORMALIZED_GAUSS_JACOBI(p,q)
    FOR EACH representation node s
      a <- LOG_POSITIVE_INTERPOLATE(xi[p], s*u)
      IF q = 1
        b <- EVALUATE_BASE_PAIR_EXACTLY(s*(1-u), j, 1)
      ELSE
        b <- LOG_POSITIVE_INTERPOLATE(xi[q], s*(1-u))
      END
      xi[p+q](s) <- LOG_BETA(p,q)
        + LOGSUMEXP_r(LOG(w[r])+a[r]+b[r])
    END
  END
  RETURN xi[k-1]
END

```

The nodes and normalized weights incorporate the beta-weight concentration associated with the convolution orders p and q .

An optional numerical normalization can transfer residual amplitude into the scalar scale. If $\tilde{\psi}_{p+q}$ denotes the residual produced by the recursion above, choose a positive scalar C_{p+q} , independent of s , and set

$$\psi_{p+q}(s) = \frac{\tilde{\psi}_{p+q}(s)}{C_{p+q}}, \quad \sigma_{p+q} = \sigma_p + \sigma_q + \log C_{p+q}.$$

This leaves the represented convolution power unchanged. Taking $C_{p+q} = 1$ recovers the unnormalized recursion. For positive residuals, the factor $B(p, q)$ is handled through $\log B(p, q)$ in the log-domain recursion; any additional normalization subtracts $\log C_{p+q}$ from the log residual and adds it to σ_{p+q} .

The power $h^{*(k-1)}$ is assembled by a binary addition chain, reducing convolution depth from $k-2$ sequential stages to $O(\log k)$.

5.6 Log-domain recursion

For positive channels define $\xi_p(s) = \log \psi_p(s)$. The recursion becomes

$$\xi_{p+q}(s) = \log B(p, q) + \text{logsumexp}_r \left[\log \hat{w}_r^{(p,q)} + \xi_p(su_r^{(p,q)}) + \xi_q(s(1 - u_r^{(p,q)})) \right].$$

Values that would underflow in linear arithmetic remain representable as large negative logarithms, and the positive quadrature sum is evaluated without cancellation. This is important because loss of a small component during an early convolution can propagate into entire entries of the subsequent Gram matrices.

5.7 Representation and integration grids

Residual functions are stored on an endpoint-inclusive Chebyshev–Lobatto grid

$$0 = s_0 < s_1 < \cdots < s_{N-1} = L, \quad L = 1 + \varepsilon.$$

Every convolution target $s_i u_r$ or $s_i(1 - u_r)$ lies in $[0, s_i]$, so no extrapolation is required. Four numerical point sets are kept distinct: the Chebyshev–Lobatto representation grid; Gauss–Jacobi rules for the beta-weighted convolution; Gauss–Legendre rules for the primitives

$$G_j(y) = \int_0^y g_j(x) dx, \quad H_{j\ell}(y) = \int_0^y g_j(x) g_\ell(x) dx;$$

and a geometrically graded outer mesh for the final entries of A and B . The latter must resolve the pair-dependent factor $x^{k-2} e^{-a_{j\ell} x}$.

Listing 3: Construction of the four numerical point sets used by the discovery evaluator.

```

PROCEDURE BUILD_GRIDS(k, epsilon, N, n_jacobi, n_inner, n_outer)
  L <- 1 + epsilon; U <- 1 - epsilon
  s[i] <- L*(1-COS(pi*i/(N-1)))/2, i=0,...,N-1

  FOR EACH (p,q) IN BINARY_ADDITION_CHAIN(k-1)
    jacobi[p,q] <- NORMALIZED_GAUSS_JACOBI(n_jacobi,p,q)
  END
  inner <- GAUSS_LEGENDRE(n_inner, interval=[0,1])

  x_star <- MIN(X,(k-2)/a_ref)
  edges <- GEOMETRIC_MESH(x_star/32,X)
  outer <- COMPOSITE_GAUSS_LEGENDRE(edges,n_outer)
  // The same positive-weight outer rule is used for every channel pair.
  RETURN s, jacobi, inner, outer
END

```

5.8 Preservation of Gram structure

The exact denominator matrix is a Gram matrix and therefore satisfies $A \succeq 0$. To preserve this structure numerically, a target $z \in [s_i, s_{i+1}]$ is interpolated using the positive stencil

$$\psi(z) \approx \omega_i(z)\psi(s_i) + \omega_{i+1}(z)\psi(s_{i+1}), \quad \omega_i, \omega_{i+1} \geq 0, \quad \omega_i + \omega_{i+1} = 1.$$

In log-space, this interpolation is evaluated by a two-term log-sum-exp. Interpolating ψ itself, rather than $\log \psi$, is what makes the stencil multilinear and hence compatible with the Gram form. When the inner quadrature nodes are common to all channel pairs, the shared positive-weight outer rule yields the discrete Gram representation

$$A_{j\ell}^{(N)} = \sum_{\alpha} w_{\alpha} \Phi_{\alpha j} \Phi_{\alpha \ell}, \quad w_{\alpha} > 0,$$

and hence $A^{(N)} \succeq 0$ up to floating-point round-off. The shared mesh therefore preserves a common discrete Gram representation rather than merely improving entrywise quadrature accuracy.

In the saturated regime, where the substitution used for the channel primitives places the inner nodes at pair-dependent positions, the common discrete Gram representation is preserved only up to inner-quadrature error. Numerical adequacy is then assessed using the positive-semidefiniteness, two-grid and refinement diagnostics described below; these diagnostics are safeguards rather than rigorous bounds on the quadrature error. Since the reported bounds come from the exact certifier and not from the discrete pencil, a failure of this structure degrades the discovery objective rather than the validity of any certificate.

5.9 Diagonal equilibration and rank-revealing Ritz solution

With

$$D = \text{diag}(A_{11}, \dots, A_{mm}),$$

the generalized Rayleigh quotient is invariant under

$$A \mapsto D^{-1/2} A D^{-1/2}, \quad B \mapsto D^{-1/2} B D^{-1/2}.$$

After equilibration, write

$$\tilde{A} = U \text{diag}(\alpha_1, \dots, \alpha_m) U^{\top}.$$

Directions satisfying

$$\alpha_i \leq \tau_{\text{rank}} \max_j \alpha_j$$

are removed before whitening. On the retained space,

$$W = U_{\text{keep}} \text{diag}(\alpha_i^{-1/2}),$$

and the leading eigenpair of $W^{\top} \tilde{B} W$ determines the profiled quotient and mixing vector. No ridge term is introduced; rank truncation is treated as a numerical-resolution decision and monitored by rank-stability tests.

Listing 4: Log-domain assembly of one pair of Gram entries. With common inner quadrature nodes, the shared positive-weight outer rule gives an exact discrete Gram representation; in the saturated regime, pair-dependent inner quadrature preserves this structure only up to inner-quadrature error.

```

FUNCTION LOG_INTERPOLATE(z, xi)
  (i,i+1,omega0,omega1) <- POSITIVE_TWO_POINT_STENCIL(z)
  RETURN LOGADDEXP(LOG(omega0)+xi[i], LOG(omega1)+xi[i+1])
END

PROCEDURE GRAM_PAIR(j, l)
  a <- rho[j] + rho[l]
  xi <- LOG_CONVOLUTION_POWER(log_raw,j,l,k)
  log_Ghat, log_Hhat <- GAUSS_LEGENDRE_PRIMITIVES(j,l)
  // Saturated inner rules may depend on the channel pair (j,l).

  log_A[j,l] <- LOGSUMEXP_x((k-2)*LOG(x)-a*x
    + LOG_INTERPOLATE(x,xi)
    + LOG(L-x)+LOG_INTERPOLATE(L-x,log_Hhat)+LOG(w_x))
  log_B[j,l] <- LOGSUMEXP_x((k-2)*LOG(x)-a*x
    + LOG_INTERPOLATE(x,xi)+2*LOG(L-x)
    + LOG_INTERPOLATE(L-x,log_Ghat_j)
    + LOG_INTERPOLATE(L-x,log_Ghat_l)+LOG(w_x))
  RETURN log_A[j,l], log_B[j,l]
END

```

Listing 5: Diagonally equilibrated, rank-revealing generalized Ritz solve.

```

PROCEDURE TOP_RITZ(log_A, log_B, tau_rank)
  d[j] <- log_A[j,j]
  Ahat[j,l] <- EXP(log_A[j,l]-(d[j]+d[l])/2)
  Bhat[j,l] <- EXP(log_B[j,l]-(d[j]+d[l])/2)

  (alpha,U) <- SYMMETRIC_EIGENDECOMPOSITION(Ahat)
  keep <- {i : alpha[i] > tau_rank*MAX(alpha)}
  REQUIRE keep is nonempty
  W <- U[:,keep] * DIAG(alpha[keep]^(-1/2))

  (lambda,z) <- LEADING_EIGENPAIR(TRANPOSE(W)*Bhat*W)
  v <- W*z
  v <- v/SQRT(TRANPOSE(v)*Ahat*v)
  RETURN k*lambda, v, CARDINALITY(keep), d
END

```

5.10 Hellmann–Feynman gradients and optimization

Numerical solution of the generalized Ritz problem may require removal of directions associated with numerically negligible eigenvalues of $A(\theta)$. Let

$$\mathcal{S}(\theta)$$

denote the retained coefficient subspace after this rank-revealing truncation, and define the corresponding numerical Ritz value by

$$\hat{R}(\theta) = \max_{\substack{v \in \mathcal{S}(\theta) \\ v \neq 0}} k \frac{v^\top B(\theta)v}{v^\top A(\theta)v}.$$

Because $\mathcal{S}(\theta)$ may change with θ , the map $\hat{R}(\theta)$ need not be differentiable when a singular direction crosses the truncation threshold. Moreover, the standard Hellmann–Feynman or envelope identity does not automatically apply across such a parameter-dependent change of retained subspace.

We therefore freeze the retained subspace during each differentiable optimization block. At outer iteration s , let

$$\mathcal{S}_s := \mathcal{S}(\theta_s)$$

and define the fixed-subspace profile

$$R_s(\theta) = \max_{\substack{v \in \mathcal{S}_s \\ v \neq 0}} q_v(\theta), \quad q_v(\theta) = k \frac{v^\top B(\theta)v}{v^\top A(\theta)v}.$$

Let $v_s \in \mathcal{S}_s$ be the normalized leading generalized Ritz vector at the anchor point:

$$B(\theta_s)v_s = \lambda_s A(\theta_s)v_s \quad \text{on } \mathcal{S}_s, \quad v_s^\top A(\theta_s)v_s = 1.$$

Provided that $A(\theta)$ remains positive definite on $\mathcal{S}_s$ in a neighbourhood of θ_s and that the leading restricted generalized eigenvalue is simple, the fixed-subspace envelope identity gives

$$\nabla_\theta R_s(\theta_s) = \nabla_\theta q_{v_s}(\theta_s).$$

Thus the Ritz vector may be frozen when differentiating the quotient, but only while the retained subspace is held fixed.

For every fixed $v \in \mathcal{S}_s$,

$$q_v(\theta) \leq R_s(\theta),$$

and at the anchor point

$$q_{v_s}(\theta_s) = R_s(\theta_s).$$

Consequently, an inner step satisfying

$$q_{v_s}(\theta_{s+1}) \geq q_{v_s}(\theta_s)$$

obeys the fixed-subspace minorize–maximize relation

$$R_s(\theta_{s+1}) \geq q_{v_s}(\theta_{s+1}) \geq q_{v_s}(\theta_s) = R_s(\theta_s).$$

This monotonicity statement concerns R_s , with $\mathcal{S}_s$ fixed. It does not by itself imply monotonicity of the adaptively truncated value $\hat{R}(\theta)$ after the numerical rank is recomputed.

Neural channels are first optimized with Adam using learning-rate warm-up, cosine decay, and gradient clipping. Candidate solutions are subsequently polished by block minorize–maximize L-BFGS. Within each such block, the retained subspace, truncation mask, and frozen Ritz vector are held fixed. The rank-revealing decomposition is recomputed only between blocks.

After this recomputation, a candidate outer step is accepted only if the adaptively evaluated Ritz value satisfies

$$\hat{R}(\theta_{s+1}) \geq \hat{R}(\theta_s) - \tau_{\text{acc}},$$

where τ_{acc} is a prescribed numerical acceptance tolerance. If a change in retained rank causes this test to fail, the candidate is rejected and the step size or optimization block is restarted. This acceptance

Listing 6: Profiled optimization with a frozen Ritz vector.

```

FUNCTION FROZEN_QUOTIENT(theta, v)
  (A_theta,B_theta) <- ASSEMBLE_GRAM(theta)
  RETURN k*(TRANPOSE(v)*B_theta*v)/(TRANPOSE(v)*A_theta*v)
END

PROCEDURE OPTIMIZE_CHANNELS(theta)
  FOR EACH increasing pair (channel rank m, grid size N)
    GROW_DICTIONARY_BY_PERTURBED_DUPLICATES(theta,m)
    FOR EACH Adam iteration, with warm-up and cosine decay
      IF this is a reporting point
        VALIDATE_CURRENT_STATE(theta)  // before the update
      END
      (_,v) <- TOP_RITZ(ASSEMBLE_GRAM(theta))
      theta <- CLIPPED_ADAM_ASCENT(
        FROZEN_QUOTIENT(theta,STOP_GRADIENT(v)))
    END
  END

  REPEAT for each block minorize-maximize step
    (_,v) <- TOP_RITZ(ASSEMBLE_GRAM(theta))
    theta <- LBFGS_MAXIMIZE(FROZEN_QUOTIENT(theta,STOP_GRADIENT(v)))
    VALIDATE_CURRENT_STATE(theta)
    UPDATE incumbent only if its two-grid score improves
  END
  RETURN theta
END

```

test is an explicit numerical safeguard; it is not presented as a consequence of the Hellmann–Feynman identity.

Finite-difference gradient tests are performed with the retained subspace and truncation mask frozen. A finite-difference stencil that changes the retained numerical rank crosses a nonsmooth truncation boundary and is therefore not used as a test of the Hellmann–Feynman gradient. Such cases are instead flagged as rank-transition events. Away from these events, the test checks the complete differentiation path through the neural channels, factored convolution recursion, integration, and matrix assembly.

Optimization proceeds through increasing channel ranks

$$m_1 < \cdots < m_S$$

and representation resolutions

$$N_1 < \cdots < N_S.$$

Existing channels are copied when the channel rank is increased, and new channels are initialized as perturbed duplicates. Because pairwise assembly scales as $m(m+1)/2$, more iterations are allocated to the smaller early models.

5.11 Validation-aware model selection

At each reporting point the candidate must satisfy:

1. agreement between the training objective and an independently recomputed Rayleigh quotient;
2. numerical positive semidefiniteness of the denominator matrix;
3. retention of a prescribed minimum fraction of the channel span after rank truncation;
4. stability of the leading quotient under a more conservative rank threshold;
5. compliance of every diagonal ratio kB_{jj}/A_{jj} with the applicable rigorous upper bound;
6. compliance of the full quotient with the corresponding rigorous upper bound; and

Listing 7: Validation and two-grid model selection applied at each reporting point.

```

FUNCTION RIGOROUS_CEILING(k, epsilon)
  IF epsilon = 0
    RETURN k*LOG(k)/(k-1)
  ELSE
    RETURN k*LOG(2*k-1)/(k-1)
  END
END

PROCEDURE VALIDATE_CURRENT_STATE(theta)
  R_bound <- RIGOROUS_CEILING(k,epsilon)

  (R_train,A,B,v,rank) <- INDEPENDENT_RECOMPUTATION(theta)
  REQUIRE AGREES(R_train,FROZEN_QUOTIENT(theta,v))
  REQUIRE MIN_EIGENVALUE(A) >= -tau_psd*MAX_EIGENVALUE(A)
  REQUIRE rank >= required_fraction*NUMBER_OF_CHANNELS(theta)
  REQUIRE STABLE_UNDER_STRONGER_RANK_TRUNCATION(A,B)
  REQUIRE EACH k*B[j,j]/A[j,j] <= R_bound
  REQUIRE R_train <= R_bound

  R_hi <- RECOMPUTE_ON_GRID(theta,1.5*N)
  REQUIRE RELATIVE_DIFFERENCE(R_train,R_hi) <= tau_snapshot
  score <- MIN(R_train,R_hi)
  STORE theta only if score improves the validated incumbent
  RETURN score
END

```

7. reproduction on a finer representation grid within the prescribed snapshot tolerance.

When two grids are used during training, candidates are ranked by

$$R_{\text{score}} = \min\{R_N, R_{N_{\text{hi}}}\}.$$

A substantial negative eigenvalue of A is treated as failure of the numerical representation rather than repaired by discarding the negative branch. Rank-stability tests separately guard against whitening directions below the accuracy of the assembled matrices.

5.12 Closed-form controls and representation refinement

Every run begins with closed-form controls. For $g(x) = 1$ and $\varepsilon = 0$,

$$R_{\text{const}} = \frac{2k}{k+1},$$

and for $g(x) = x$,

$$R_x = \frac{3k}{3k+1}.$$

The first exercises convolution, outer integration and scaling; the second also probes interpolation of a nonconstant residual convolution shape.

After training, candidates are evaluated on grids N , $1.25N$ and $1.5N$ and fitted to

$$R_N = R_\infty - cN^{-p}.$$

The fitted limit is diagnostic only. the export is marked unconverged when the sequence is noncontracting, violates a rigorous ceiling or fails the refinement tolerance. No extrapolated discovery value is treated as a certified lower bound.

The final separable discovery solver emerged through a sequence of numerical and structural corrections. Supplementary Table 2 summarizes the principal failure modes identified during development and the corresponding fixes. These historical iterations are not separate proof components; they document how the numerical invariants enforced in the final solver were established.

Supplementary Table 2 lists the different development steps for the discovery script.

Supplementary Table 2: Reconstructed development history of the `maynard_separable_rayleigh_v9_factored` lineage (neural discovery).
Version numbers map to successive saved iterations.

#	Correction	Cause / symptom
1	Baseline v9 (factored): $\nu_p = s^{p-1} e^{-as} e^{\sigma} \psi_p$ per pair; stable <code>inv_softplus</code> ; per-pair max-rescale; adaptive log-space GL outer windows; diagonal preconditioning ($\text{diag } A = 1$); <code>parts()</code> protocol.	Foundation. <code>inv_softplus</code> fixes $\text{softplus}^{-1} = \log(\text{expm1}) \rightarrow +\infty$ for rate ≥ 710 (all $k \geq 355$); per-pair rescale replaces a global max that flushed small pairs to round-off.
2	σ double-count fix: divide exact base re-evaluations by $e^{-\sigma_1}$.	Exact base evals carried e^{σ_1} while σ_1 was also added in <code>new_sig</code> ; the spurious factor is not of Gram form $\Rightarrow$ indefinite A .
3	<code>train()</code> validation-ordering fix: validate before <code>opt.step()</code> .	<code>R_{train}</code> read pre-step, <code>report()</code> post-step; <code>agree</code> measured the Adam step, rejecting nearly all snapshots (fallback to iter 700).
4	Log-space channel protocol: <code>log_parts</code> ; <code>--channel-sign positive/free</code> .	Prerequisite for the log-space recursion; needs sign-definite channels.
5	Log-space ξ recursion: $\xi_{p+q} = \log B(p, q) + \text{logsumexp}(\log \hat{w} + \xi_p + \xi_q)$.	The $(\min \psi_1 / \max \psi_1)^{k-1}$ collapse drove ψ past the float64 floor; round-off put random signs on diagonal pairs ($\psi \geq 0$), making A indefinite.
6	Shared graded outer mesh (one geometric Gauss rule for all pairs).	Per-pair windows gave each entry its own rule $\Rightarrow A$ the Gram matrix of nothing; $\lambda_{\min}/\lambda_{\max} \approx -1$ and did not improve under refinement.
7	Log-singularity fix: $G(y) = y \hat{G}(y)$; interpolate $\log \hat{C}$, reattach $\log y$ analytically.	$G(y)$ vanishes linearly $\Rightarrow \log G \rightarrow -\infty$ (clamped); interpolating that oscillated $\Rightarrow B \approx 10^{45}$.
8	Gram-preserving positive interpolation: two-point stencil, weights ≥ 0 , via <code>logaddexp</code> ; <code>--interp positive/spectral</code> .	Interpolating the log gives $\prod \psi_i^{M_i}$ (geometric), not $\sum M_i \psi_i$ (linear) $\Rightarrow$ not multilinear $\Rightarrow$ not Gram $\Rightarrow$ indefinite A .
9	Hard gates (PSD, $R \leq k$, gradient) + removed false “PSD by construction” + parameter-independent mesh.	$R = 118, 979, 5.2 \times 10^8$ were optimised toward; positive-entrywise $\neq$ PSD; moving mesh gave a spurious HF-gradient gap.
10	Resolution diagnostics (ξ span/jump), $g = x$ preflight ($3k/(3k+1)$), interp-dependent n_{rep} , sharp bound $\frac{k}{k-1} \ln k$, Richardson gate, N provenance.	$g \equiv 1$ is exact for any N in positive mode (blind to under-resolution); $R \leq k$ too loose to catch a $6 \rightarrow 49$ climb.
11	Refinement-stable snapshots + strikes + $1.5 \times$ validation twin.	Optimiser maximises $R_{\text{true}} + \text{error}(\theta)$; refinement-instability is the exploitation signature and must gate snapshots.
12	Two-tier snapshots + $\min(R_N, R_{1.5N})$ ranking + fitted-order Richardson + grid continuation + result reorder.	One tolerance did two jobs and rejected every honest state; trained-channel order $p \approx 1.6 \neq 2$, so fixed-order Richardson mis-extrapolated.
13	$\varepsilon > 0$ bound fix: $R_{\text{bound}} = k$ for $\varepsilon > 0$; tight bound only for $\varepsilon = 0$.	$M_{k,\varepsilon,1/2}$ legitimately exceeds the vanilla bound ($4.02 > 3.99$ at $k = 50$), so the tight bound falsely aborted. Later aligned this with proposition 6.5 from Polymath8b to
		$R_{\text{bound}} = \frac{k}{k-1} \log(2k-1) \quad (\varepsilon > 0).$
		There is a more stringent bound in Polymath8b Remark 6.6 but this suffices as hard-coded discovery threshold / safeguard.
14	Per-pair diagnostic + <code>--dump-violation</code> + <code>--patience</code> ; k -aware panels; excursion guard; <code>polish()</code> incumbent seeding.	Fixed 8-panel mesh under-resolved x^{k-2} at $k \geq 250$ (e^{152} vs $2n_{\text{out}}=128$); <code>polish()</code> started at $-\infty$ so step 1 installed $R = 70$ as best.
15	<code>--trunc-floor</code> prevention + rank-stability detection + strike-counter fix.	$R = 56$ lived in A -directions at $\lambda/\lambda_{\max} \sim 10^{-6}$, below assembly accuracy 10^{-3} ; whitening amplified error (again!) $\sim 10^3$. <code>strikes</code> reset before the check $\Rightarrow$ never accrued.

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Listing 8: Closed-form preflights and post-training representation refinement.

```

PROCEDURE PREFLIGHT_AND_REFINE(theta,k,epsilon,N)
  REQUIRE NUMERICAL_QUOTIENT(g=1) AGREES WITH CLOSED_FORM_CONSTANT(k,epsilon)
  IF epsilon = 0
    REQUIRE NUMERICAL_QUOTIENT(g=x) AGREES WITH 3*k/(3*k+1)
  END

  FOR Nq IN {N, CEIL(1.25*N), CEIL(1.5*N)}
    R[Nq] <- VALIDATED_QUOTIENT(theta,Nq)
  END

  IF successive differences contract with a common sign
    FIT R[Nq] = R_infinity - c*Nq^(-p)
    REQUIRE 0.8 <= p <= 4
    REQUIRE ABS(R[1.5*N] - R_infinity) / ABS(R_infinity) <= tau_refine
    REQUIRE R_infinity <= RIGOROUS_CEILING(k,epsilon)
  ELSE
    MARK candidate as numerically unresolved
  END
  RETURN R, R_infinity, p
END

```

Supplementary Table 2 – continued

#	Correction	Cause / symptom
16	Stability recalibration (25%, floored) + absolute $R \leq k$ backstop + excursion baseline anchor + grow-step guard + Richardson already-converged path.	5% flagged honest wobble; best = $-\infty$ let a run-away pass at iter 0; a 0.375 baseline made 3.66 look 9.7 $\times$; a post-grow state crashed the run; a converged $k=500$ run got a false FAIL.

5.13 Aggregate modular evaluation

After projection, let q_j, q_ℓ be rational channel polynomials and

$$h_{j\ell}(u) = q_j(u)q_\ell(u).$$

Rather than reconstructing every exact entry of the Gram matrices, the aggregate certifier accumulates the weighted pair contributions modulo machine-word primes:

$$S_A(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_\ell A_{j\ell}^{\text{int}} \pmod{p}, \quad S_B(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_\ell B_{j\ell}^{\text{int}} \pmod{p}.$$

All polynomial powers, coefficient extractions, pair weights and channel sums are evaluated in $\mathbb{F}_p[x]$. The machine-word primes have no denominator occurring in the modular formulas with vanishing modulo p , ensuring that every required inverse in $\mathbb{F}_p$ is defined.

Polynomial arithmetic is performed with exact compiled integer and modular kernels. Earlier exact-certification implementations used Kronecker substitution: integer polynomial coefficients were packed into large integers, multiplied using GMP-backed arbitrary-precision arithmetic, and unpacked exactly; polynomial powers were then formed by binary exponentiation. In the later FLINT-backed implementations, polynomial multiplication, powering and modular reduction are delegated to FLINT [4], accessed through `python-flint`. FLINT selects its internal multiplication strategy according to operand size and representation. These implementation choices affect runtime and memory use but not the mathematical certificate, which depends only on the exact resulting elements of $\mathbb{F}_p[x]$ and the subsequent CRT reconstruction.

Only the two scalar aggregate integers S_A, S_B are reconstructed. Their exact quadratic forms are

$$c^\top A c = \frac{S_A}{D_A}, \quad c^\top B c = \frac{S_B}{D_B},$$

with global denominators fixed before the modular sweep.

Let

$$N_{\text{CRT}} = k s_{\text{max}}, \quad r_{\text{max}} = (k-1) s_{\text{max}}, \quad E = k-1 + r_{\text{max}} + m_{\text{max}},$$

and

$$L_{\text{cm}} = \text{lcm}\{k-1, \dots, k-1 + r_{\text{max}} + m_{\text{max}}\}.$$

The resulting global denominators are

$$D_A = (k + N_{\text{CRT}})! d_h^k d_c^2,$$

and

$$D_B = (k - 2 + r_{\text{max}})! L_{\text{cm}} \rho_d^E T_{\text{den}} d_h^{k-1} d_c^2.$$

All factors are therefore known before reconstruction.

The numerator-side coefficient sum can be expressed as a polynomial correlation. If $T_{j\ell}(x)$ denotes the pair-dependent integrated-channel polynomial and

$$V_p(x) = \sum_{v=0}^{r_{\text{max}}+m_{\text{max}}} (k-1+v)^{-1} x^v \in \mathbb{F}_p[x],$$

then the required shifted inner products are obtained from multiplication with a reversed form of $T_{j\ell}$, implementing the transposition principle.

Polynomial powers are formed deterministically from exact arithmetic. Classical sparse-polynomial powering algorithms provide related powering recurrences [5], but in the present application the factorially weighted pair polynomials are typically dense. The principal computational gain therefore comes from aggregate modular evaluation, exact coefficient extraction and CRT reconstruction rather than from sparse polynomial storage itself.

5.14 Known denominators and deterministic CRT reconstruction

The modular stage reconstructs signed integer numerators rather than generic rational numbers. Rigorous bounds

$$|S_A| \leq \mathcal{B}_A, \quad |S_B| \leq \mathcal{B}_B$$

are obtained using

$$\|h^r\|_1 \leq \|h\|_1^r$$

together with explicit bounds for factorial ratios, powers of the rational support ratio, integrated channel coefficients and rational mixing weights. If

$$M_P = \prod_{p \in P} p$$

and

$$M_P > 2 \max(\mathcal{B}_A, \mathcal{B}_B),$$

the centered CRT representative in $(-M_P/2, M_P/2]$ is unique. The implementation uses an additional safety margin, selecting primes until the product exceeds four times the larger bound. Held-out primes provide an implementation-level check but are not needed for mathematical uniqueness.

The certified quotient is

$$R_{\text{cert}} = k(1 + \varepsilon) \frac{S_B/D_B}{S_A/D_A}.$$

Once the rational channels and mixing vector are fixed, this value is independent of the neural discovery stage.

5.15 Development and validation history of the CRT certifier

The exact certifier was developed through a sequence of increasingly large stress tests in which numerical agreement between the floating-point proposal stage and the exact reconstruction stage was treated as a falsification criterion. Supplementary Table 3 records the principal failure modes and the corresponding corrections. These historical iterations are not independent proof components; they document how the final implementation acquired the numerical invariants and trust boundaries used throughout the reported certifications.

Supplementary Table 3: Development history of the exact multimodular (CRT) certifier for $M_{k,\varepsilon,1/2}$. Each version records the change and the failure mode that motivated it. With one exception (the FFT accuracy floor, v4), every defect was a silent disagreement between the floating-point pass and the exact pass at a specific weight vector c , invisible at small k and exposed only by scale, conditioning, or polynomial degree.

Ver	Change	Failure mode / motivation
v1	Initial multimodular backend. $\hat{h}^{k-1}$ is never formed over $\mathbb{Z}$; per 62-bit prime, one <code>nmod</code> powering plus two weight-table multiplies execute both linear functionals as single coefficient extractions: $[x^{n_{\max}}](bh^k \cdot W_{\text{rev}})$ for A , and for B the pair-dependent t -weights combined as $\sim m_{\max}$ scalar-poly ops against per-prime tables $G^{(m)}$ folding in $\Lambda = \text{lcm}(k-1, \dots, k-1+r_{\max}+m_{\max})$, the ρ -power clearing, and batched harmonic inverses. Reconstruction is aggregate-only—two integers S_A, S_B with known-denominator scalings—via deterministic centered CRT against exact a priori bounds, plus held-out verification primes. Float pass decoupled from the exact engine ($A = \int_0^1 h^{*k}$, $B = \int_0^\rho h^{*(k-1)} Q_j Q_l (1-s)$ by renormalized binary powering). Greedy c -weighted pruning on the float pencil (<code>--prune-tol</code>); only surviving pairs reach the CRT engine.	Establishes the architecture: avoid the one unavoidable giant integer, reconstruct only two aggregate scalars, keep the float discovery pass cheap and separate from exact arithmetic.
v2	<code>_ritz</code> no longer calls the Cholesky-based generalized solver: it equilibrates by the diagonal, eigendecomposes the metric A , truncates to eigenvalues above 10^{-12} of the top one, and solves the standard symmetric problem in whitened coordinates, embedding zeros for dead channels. Self-test adds exact agreement with <code>scipy.eigh(B,A)</code> on a PD case and a finite λ on a singular metric.	<code>scipy.eigh(B,A)</code> requires A positive definite; at $k = 500$ the float Gram is numerically semidefinite and Cholesky fails (<code>LinAlgError: leading minor not positive definite</code>). Restricting to $\text{range}(A)$ is the correct regularization: null-space components contribute ≈ 0 to $c^T A c$.
v3	B-side of the float DP rescaled by $u = s/\rho$: run the $(k-1)$ -fold convolution power on $h_\rho(t) = h(\rho t)$, so $B = \rho^{k-1} \int_0^1 h_\rho^{*(k-1)}(u) T(\rho u) du$, with ρ^{k-1} carried in the log-scale ($\varepsilon = 0$ degenerates to the old path). Driver interlock: float-predicted R checked against discovery R_{NN} before any exact work.	At $k = 500$ the certified value collapsed to $\sim 10^{-12}$ (valid but worthless). $h^{*(k-1)}$ concentrates like s^{k-2} near $s = 1$, so the whole B-region $s \leq \rho = \frac{12}{13}$ sat $\rho^k \sim e^{-40}$ below the renormalized peak—beneath float64 accuracy. B was noise; the pruner dropped 48 channels “for free” and the exact engine certified a garbage frozen c .
v4	Truncated convolution <code>_tconv</code> forced to <code>direct</code> <code>np.convolve</code> , removing the FFT path. Large- k float-DP regression test against exact $g \equiv 1$ closed forms at $k = 200, 500$.	FFT convolution carries an <i>absolute</i> error $\sim 10^{-16}$ relative to the vector peak, while h^{*j} spans s^{j-1} internally; entries sit up to 2^{-j} below peak and are buried by noise by $j \sim 55$, then amplified through each squaring (closed-form check failed by e^{+1000}). Direct summation keeps per-entry <i>relative</i> accuracy; the self-convolution integrand peaks at the interior $t = s/2$, representable to $j \sim 511$ (float64 ceiling near $k \sim 1000$).

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Table 3, continued

Ver	Change	Failure mode / motivation
v5	Product-exact prime budgeting (gen_primes_for_bound): accumulate the actual prime product until it exceeds $4 \cdot$ bound rather than assuming 61 bits/prime. Miller–Rabin citation corrected (least strong pseudoprime to bases $2 \dots 37$ is 3.19×10^{23} ; Sorenson–Webster 2017, OEIS A014233). Self-test cross-checks generated primes against FLINT’s proven fmpz.is_prime ; new strong-cancellation test ($c = (1, -1)$, near-identical channels).	Collaborator review round: tighten certification claims (exact prime count, correct primality bound, FLINT belt-and-braces), cover the small-signed-aggregate CRT branch, and <i>measure</i> which proposed speedups were real (mul_low benched slower than the full FFT multiply; native modular dot kernel deferred).
v6	Added --float-only survey mode: projection + float DP + pruning per degree, stopping before any exact/CRT work.	Cheap degree scouting at large k : projection L_∞ and float-predicted R answer “is this k reachable at this basis degree” in minutes per degree, without spending the prime budget.
v7	Stable Legendre-basis float evaluation. project_channels_v3 returns both monomial integers (exact engine) and dyadic Legendre numerators (float engine); the float engine evaluates channels and the B-weight antiderivative via the shifted-Legendre recurrence, with a non-finite guard that raises with a diagnosis.	At $d = 160$ the float pass produced NaN: channels were evaluated by np.polyval on monomial coefficients of size $\sim 8^d \sim 10^{150}$, so Horner formed an $O(1)$ value as an alternating sum of 10^{150} -scale terms (~ 150 orders of cancellation), poisoning the convolutions to $\pm\infty$ and hence $\text{off}_A = \max(-\infty) = -\infty$, $(-\infty) - (-\infty) = \text{NaN}$. The Legendre recurrence ($ \tilde{P}_n \leq 1$ on $[0, 1]$) is well-conditioned at any degree.
v8	Auto-escalating rank truncation in the whitened Ritz, anchored to R_{NN} : _ritz gains a rank_tol threaded through greedy_prune ; the driver escalates $10^{-10} \rightarrow 10^{-8} \rightarrow 10^{-6} \rightarrow 10^{-4}$ until the predicted R is physical, or uses a fixed --rank-tol .	At $d = 40$, $k = 500$ the float R came out 7.4×10^{21} . Gram entries were cancellation-limited to $\sim 10^{-8}$ but the rank cut was $10^{-12} \lambda_{\text{max}}$; directions above the cut yet below the accuracy floor are noise, and whitening by $1/\sqrt{\text{noise}}$ makes $\lambda = B_{\text{noise}}/A_{\text{noise}}$ explode. The true pencil is bounded by $M_{k,\varepsilon}$ (single digits).
v9	Worker refactor: the B-functional inner sum $S_R = \sum_m tw_m/(k-1+R+m)$ computed for all R by <i>one</i> correlation multiply $IT = \text{INV} \cdot TW_{\text{rev}}$ against a single global per-prime inverse polynomial, then combined in a short loop against the base ladder. The $(m_{\text{max}}+1)$ -polynomial G_{rev} tables are eliminated; per-prime tables become three ladders regardless of degree. Added --resume : append-only (p, S_A, S_B) checkpoint.	At $d = 160$, $m_{\text{max}} = 2(d+1) = 322$, so the v1 G_{rev} tables were $323 \times 160k \approx 413$ MB per worker and $\sim 5 \times 10^7$ Python ops per prime for table construction alone—dominating the sweep across $\sim 28k$ primes. The correlation form is degree-oblivious; resume makes the multi-hour sweep preemption-tolerant (spot instances).

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Table 3, continued

Ver	Change	Failure mode / motivation
v10	Exact mantissa-preserving c rationalization plus a frozen- c gate. $c_q = \lceil \text{Fraction}(\text{float}(v)) \rceil$ (full 53-bit mantissa per component) replaces uniform-absolute dyadic(v , bits) rounding; the gate re-evaluates the float quotient at the rationalized c and aborts if it moved by $> 1\%$. Resume caches gain a SHA-256 payload fingerprint and refuse stale residues.	After an 11-hour $k = 500$ sweep the certified value was 8.9×10^{-5} . Uniform absolute 2^{-48} rounding annihilated small-but-essential components of the balanced Ritz c , whose entries span dozens of orders ($c_j \sim A_{jj}^{-1/2}$, $A_{jj} \sim (\sup)^{2k}$, so channels differing by 1.1 in sup-norm have diagonals differing by $1.1^{1000} \sim 10^{41}$). The float pass evaluated the true c (≈ 4.19); the exact engine certified the mutilated one.
v11	Fixed the Gauss–Jacobi overflow in the shared quadrature builder: replace <code>scipy.roots_jacobi</code> with an overflow-free Golub–Welsch construction (Jacobi tridiagonal eigenproblem; weights = squared first-eigenvector components normalized to sum 1, so the diverging μ_0 cancels). (<i>Shared-infrastructure fix; gates whether large-k discovery npz files can be produced at all.</i>)	<code>roots_jacobi</code> computes $\mu_0 = 2^{a+b+1} B(a+1, b+1)$ in linear space; with $p - 1 \sim 2499$ at $k = 2500$ the Beta/Gamma factors overflow float64 ($\Gamma(171)$ already overflows), returning NaN weights and crashing the $g \equiv 1$ preflight. Golub–Welsch needs only recurrence coefficients (no Γ); validated to $\sim 10^{-14}$ vs <code>roots_jacobi</code> for $p \leq 60$ and finite/correct to $p = 6000$.
v12	Diagonal preconditioning of the float Gram: <code>matrices()</code> subtracts $(\log A_{jj} + \log A_{ii})/2$ from each log-entry (so $\hat{A}$ ’s diagonal is exactly 1, off-diagonals $O(1)$ by Cauchy–Schwarz), applies the same congruence to B , and returns $\log A_{\text{diag}}$. Regression: a 195-nat diagonal spread must keep all channels alive with float-vs-CRT agreement 10^{-5} .	A single global $\exp(\log A - \text{off})$ offset underflowed entries $< \text{off} - 745$ to exactly 0; at $k = 1000$ the diagonal spans ~ 6900 nats, so the Ritz solve collapsed to rank 2 (observed: 30 of 32 channels dropped “for free” at 0.0 loss). Per-channel log-diagonal preconditioning keeps every channel alive regardless of k .
v13	Frame conversion of the imported discovery c on load. When the npz carries $\log A_{\text{diag}}$, the loader maps discovery’s exported (preconditioned-frame) c to the true channel frame via $c_{\text{true}} = c \cdot \exp(-\frac{1}{2} \log A_{\text{diag}}) \cdot \text{channel_norms}$ (both factors were then thought necessary: the k -th-power diagonal scale is independent of the L^2 scale). Later corrected: discovery assembles its Gram pencil on the L^2 -normalized channels $g_j / \ g_j\ _2$, and the certifier rebuilds exactly these channels by projecting <code>g_fine/channel_norms</code> . The coefficient on the normalized channels is therefore $c_{\text{true}} = \exp(-\frac{1}{2} \log A_{\text{diag}}) \hat{c}$, with no channel-norm factor. A conversion to the raw channels would require $\ g_j\ _2^{-k}$, not $\ g_j\ _2$. The conversion is now shared by all exact backends (<code>certification/frames.py</code>). Validity was never affected, since any explicit rational c yields a valid lower bound, and the re-Ritz path (v14) never used the channel-norm factor.	At $k = 53$ discovery reported 3.994 but the certifier gave 3.594. The pencil’s frame-invariant top eigenvalue was 3.9948; discovery exports c in the frame of its <i>preconditioned</i> Gram ($\hat{c} = D^{1/2} c_{\text{true}}$), while the certifier evaluates c on the <i>true</i> channels. The deflation grows with the $\log A_{\text{diag}}$ spread (308 nats at $k = 53$, ~ 6900 at $k = 1000$).

continued on next page

Table 3, continued

Ver	Change	Failure mode / motivation
v14	Frame conversion of the re-Ritz c (second instance of the same bug). The driver captures $\log A_{\text{diag}}$ from the float engine and maps <code>_ritz</code> 's output back to the true frame ($c_{\text{true}} = \exp(-\frac{1}{2} \log A_{\text{diag}}) \hat{c}$) on the active channels before rationalizing; the frozen- c gate is made frame-consistent.	After v13, <code>--use-nn-c</code> still gave 3.594. <code>_ritz</code> whitens the preconditioned Gram internally, so its eigenvector lives in the preconditioned frame—and the driver certified it as if true-frame. Certifying $\hat{c}$ as true-frame gives 3.5938 (the observed value); the mapped c gives 3.9939, and the end-to-end run then certified 3.9944 (gap $+4 \times 10^{-4}$ to discovery). Affected <i>every</i> re-Ritz-path certification against a preconditioned-export npz.

5.16 Representation-preserving transfer

The discovery solver uses diagonal preconditioning. If

$$A_{\text{disc}} = D^{-1/2} A_{\text{true}} D^{-1/2}$$

and $\hat{c}$ is the mixing vector in discovery coordinates, then

$$c_{\text{true}} = D^{-1/2} \hat{c}.$$

Any additional channel normalization is likewise restored before certification.

During projection and floating-point evaluation the channels are represented in a shifted-Legendre basis,

$$q_j(u) = \sum_{r=0}^d a_{jr} \tilde{P}_r(u),$$

and evaluated through the three-term recurrence. For exact certification the same rational expansion is transformed symbolically into integer monomial coefficients with a common denominator. If

$$h(u) = \sum_{r=0}^s h_r u^r,$$

define its factorially weighted coefficient polynomial by

$$\hat{h}(x) = \sum_{r=0}^s r! h_r x^r.$$

For enlarged support,

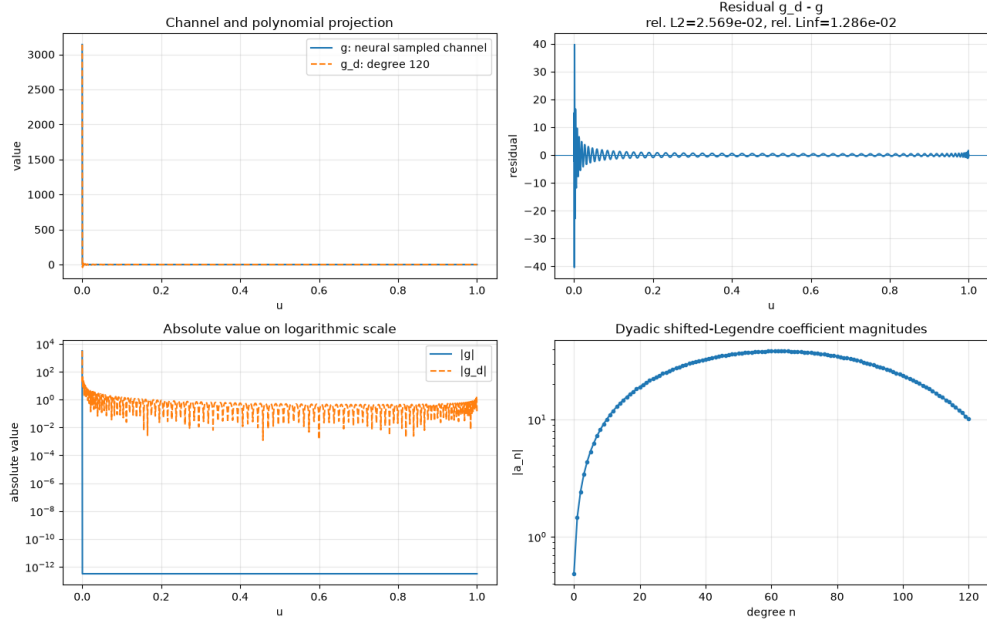
$$u = \frac{t}{1 + \varepsilon}, \quad r_\varepsilon = \frac{1 - \varepsilon}{1 + \varepsilon},$$

and the associated Jacobian and powers of $1 + \varepsilon$ and r_ε are retained explicitly. The mixing vector is embedded into the rationals by preserving each binary floating-point component as an exact dyadic rational, avoiding loss of small but cancellation-relevant entries. Before modular evaluation, the frozen rational vector is re-evaluated in the floating-point quadratic forms and required to reproduce the Ritz candidate within tolerance. This is a transfer check, not part of the proof.

5.17 Loss-controlled channel reduction

Exact certification cost is quadratic in the number of surviving channels. Starting from the full floating-point Gram pencil, channels are removed by backward elimination. For each candidate removal, the generalized Rayleigh–Ritz problem is re-solved. If λ_0 is the original value and λ_S the value on the active subset,

$$\frac{\lambda_0 - \lambda_S}{\lambda_0} \leq \tau_{\text{prune}}$$



Supplementary Figure 1: **Diagnostic large- k failure of the early discovery pipeline.** For $k = 3600$, the discovery stage reported the spurious value $R = 8.1375743146$. Nearly all Ritz weight collapsed onto a single channel, shown here together with its degree-120 polynomial projection, residual structure, logarithmic-scale amplitude and dyadic shifted-Legendre coefficient magnitudes. Although the discovery objective was stable under grid refinement and remained below the Cauchy–Schwarz ceiling, the certification failed independent reconstruction and exposed the representation inconsistency; rejecting this candidate.

is required. At each step the channel whose removal leaves the largest quotient is discarded. This differs from coefficient thresholding because importance is measured directly in the reoptimized variational objective. The reduced rational trial function is then certified exactly; pruning changes only cost and attainable bound, not mathematical validity.

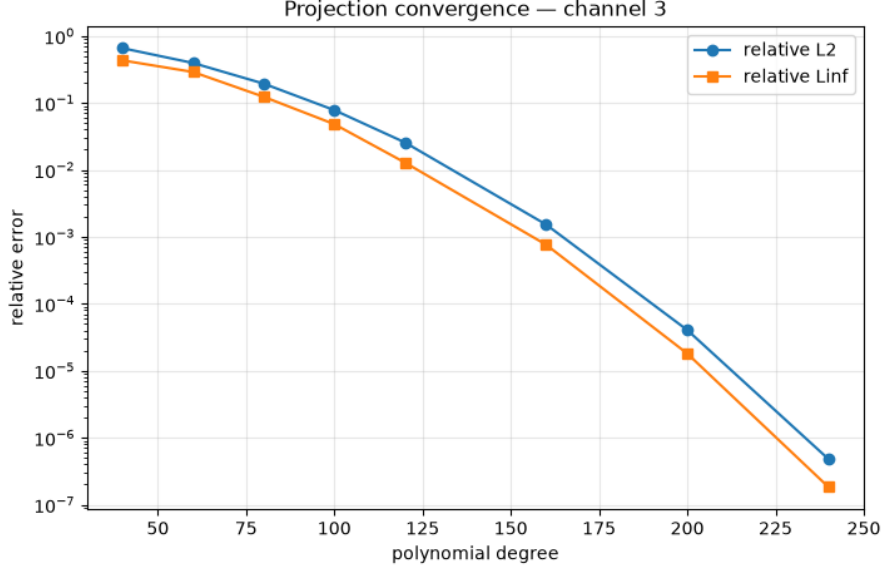
5.18 Diagnostic large- k failure and rational distillation

The large- k experiments exposed a failure mode that was not detected by the original discovery diagnostics. In an early run at $k = 3600$, the multi-channel optimizer collapsed onto a single sharply localized channel and reported a Rayleigh value above 8 (See Supplementary Figures 1 and 2). The value was stable under refinement of the representation grid and remained below the aforementioned ceiling ($R_k \leq \frac{k}{k-1} \log k$) so the candidate initially appeared numerically credible. Also, across the validated separable runs, increasing k was accompanied by a progressive reduction in the number of channels needed to represent the best surviving candidates. This trend was empirical.

Upon certification, it appeared difficult to adequately represent the one-channel discovery solution. For a single exported channel q , the scale-invariant quantity

$$R_{1D} = k \frac{(\int q)^2}{\int q^2}$$

disagreed by orders of magnitude with the stored diagonal ratio kB_{jj}/A_{jj} . Since multiplication of q by an arbitrary scalar cannot change either ratio, this discrepancy cannot be explained by a normalization or frame convention. For the sharpest channel, the support of the exported function also gave the Cauchy–Schwarz bound $R_{1D} \leq k\sigma$, where σ is the effective support measure; the stored discovery value violated the corresponding scale by more than two orders of magnitude.



Supplementary Figure 2: **Polynomial projection convergence for the single Ritz weight holding channel of $k = 3600$.** Relative L^2 and L^∞ projection errors decrease with polynomial degree, illustrating that the transfer error can be reduced systematically once a numerically resolved channel has been obtained.

The discrepancy was caused by a scale-law inconsistency in the discovery evaluation of (increasingly) narrow channels. The convolution recursion operated in a normalized variable, whereas the exported channel lived in the physical coordinate. In the intermediate-rate regime the error followed approximately the missing-Jacobian law

$$R_{\text{stored}} \approx \frac{\rho}{k} R_{\text{direct}},$$

with ρ the channel rate. At the highest rates a second resolution effect appeared, and the sharpest exported channel was itself truncated. The optimizer therefore concentrated its Ritz weight on the channel for which the numerical artifact was largest. The apparent rank-one structure was consequently not, by itself, evidence of a genuine extremizer.

The comparison is exact when the channel is genuinely compactly supported: if $\text{supp}(g) \subseteq [0, \sigma]$ with $k\sigma \leq 1$, then $[0, \sigma]^k \subseteq \mathcal{R}_k$, so $I_k(F) = (\int g^2)^k$ and $J_k^{(1)}(F) = (\int g)^2 (\int g^2)^{k-1}$, giving $R = k(\int g)^2 / \int g^2$ identically. For a channel with only an *effective* support the identity acquires an uncontrolled tail term, so the scalar test is applied as an empirical safeguard against gross inflation rather than as an exact identity.

This observation is important because the artifact was close to the true large- k scale. The sharp-channel artifact approached a value proportional to the first inner quadrature weight, and for the quadrature order used in the run it crossed 8 in the same k -range in which the true variational constant approaches 8. Grid stability and upper bounds were therefore insufficient to identify artifact from valid.

The final discovery solver adds an independent scalar-channel gate. For each sufficiently localized channel it compares kB_{jj}/A_{jj} with a direct one-dimensional quadrature of

$$k \frac{(\int g_j)^2}{\int g_j^2},$$

using a quadrature path independent of the convolution assembly. Wide channels are additionally checked against support-aware Cauchy–Schwarz bounds. Narrow channels get the hard $\mu \times$ inflation rejection. Candidates failing these tests are rejected before export.

Listing 9: Checks used to diagnose unresolved or spuriously inflated neural channels before transitioning to the rational large- k route. The all-width moment comparison is a conservative empirical discovery safeguard and is not part of the exact certificate.

```

FUNCTION LOG_INNER_MEAN(log_residual, rate, Y, z_cap)
  IF rate*Y < z_cap
    u <- Y*x_GL
    RETURN LOGSUMEXP_r(LOG(w_GL[r])+log_residual(u[r])-rate*u[r])
  ELSE
    z <- z_cap*x_GL
    // z=rate*u resolves the exponential boundary layer explicitly.
    RETURN LOG(z_cap/(rate*Y))
      +LOGSUMEXP_r(LOG(w_GL[r])+log_residual(z[r]/rate)-z[r])
  END
END

PROCEDURE LARGE_K_CHANNEL_CHECKS(g,A,B)
  rho <- SOFTPLUS(eta)
  (I1[j],I2[j],sigma[j]) <- FINE_GRADED_ONE_DIMENSIONAL_QUADRATURE(g_j)
  R_direct[j] <- k*I1[j]^2/I2[j]
  R_conv[j] <- k*B[j,j]/A[j,j]

  REQUIRE R_conv[j] <= (1+tau_moment)*R_direct[j] FOR EVERY j
    // conservative empirical discovery filter
  IF k*sigma[j] <= 1 AND R_conv[j] > mu*R_direct[j]
    REJECT unresolved single-channel inflation
  END
  IF k*sigma[j] > 1 AND R_conv[j] > (1+tau_support)*k*sigma[j]
    FLAG the channel as support-inconsistent
  END

  curvature <- MEAN_SQUARED_CURVATURE(LOG_RAW, coordinate=rho*x)
  RETURN curvature // Adam search penalty only
END

```

Conditional on candidates that survive these discovery checks, rational distillation provides a second and logically independent filter. Richer multi-cluster rational fits can reduce pointwise approximation error while failing probability-mass, conditioning or Fourier-stability checks. Because small one-dimensional transform errors are raised to the $(k-1)$ st power in the large- k evaluation, pointwise agreement alone does not preserve the Maynard functional.

In the stable distilled representation, the surviving family reduces to the single rational pole

$$g(t) = \frac{1}{c + (k-1)t}.$$

Its Rayleigh quotient is then optimized and evaluated independently of the neural model. Thus the rank-one collapse observed in the failed neural run and the rank-one structure of the certified rational family have opposite logical status: the former was an optimization response to a numerical artifact, whereas the latter survives reconstruction and independent certification. Neither the failed discovery value nor the neural rank collapse is used as evidence for the final bound.

5.19 Neural-to-rational structural distillation

At large k , the neural candidate is compressed into the clustered rational family

$$g_{\text{rat}}(t) = \sum_{a=1}^r \sum_{j=1}^{p_{\max}} u_{a,j} (c_a + nt)^{-j}, \quad n = k-1. \quad (39)$$

For fixed cluster locations $c = (c_1, \dots, c_r)$, the coefficients enter linearly and are determined by variable projection,

$$u^*(c) = \arg \min_u \sum_i \omega_i \left[g_{\text{NN}}(t_i) - \sum_{a,j} u_{a,j} (c_a + nt_i)^{-j} \right]^2. \quad (40)$$

The outer optimizer therefore varies only the nonlinear pole locations. Negligible coefficients are removed and the highest retained power at each pole determines an inferred multiplicity. Higher powers are confluent directions, since

$$\frac{\partial^q}{\partial c^q}(c + nt)^{-1} = (-1)^q q! (c + nt)^{-(q+1)}.$$

Each distilled candidate is subsequently reconstructed in an independent deterministic evaluator. Fourier-domain cross terms are reduced to

$$F_p(\xi) = \int_0^1 e^{i\xi t} (c + nt)^{-p} dt, \quad (41)$$

with, for $p \geq 2$,

$$F_p(\xi) = -\frac{e^{i\xi}(c + n)^{1-p} - c^{1-p} - i\xi F_{p-1}(\xi)}{n(p-1)}. \quad (42)$$

Because upward recursion can amplify round-off at large $|\xi|$, the Fourier computation is restricted to a stability window and requires the characteristic function to decay below a prescribed threshold before truncation. Products involving distinct clusters are reduced by partial fractions; cross-cluster cancellation is monitored and ill-conditioned configurations are rejected.

Further checks include conservation of probability mass, agreement of the first moment with its analytic value, control of the negative Fourier-inversion noise floor, rank diagnostics for the Gram matrix, and the analytic ceiling

$$R_k \leq \frac{k}{k-1} \log k.$$

A candidate that violates any check is rejected rather than assigned a numerical objective. A surviving structure is locally refined in the deterministic evaluator, so the final rational value is a newly optimized Rayleigh quotient rather than the neural objective or the pointwise fit.

5.20 Probabilistic reduction for the rational trial

For $k \geq 3$, let $n = k - 1$, choose an exact rational $c > 0$, and define

$$F(t_1, \dots, t_k) = \prod_{i=1}^k g(t_i) \mathbf{1}_{\mathcal{R}_k}(t), \quad g(t) = \frac{1}{c + nt}, \quad (43)$$

where $\mathcal{R}_k = \{t_i \geq 0 : \sum_i t_i \leq 1\}$. Put $w = g^2$ and

$$m_0 = \int_0^1 w(t) dt = \frac{1}{c(c+n)}.$$

Let $S = S_n$ denote the sum of n independent variables with density w/m_0 on $[0, 1]$. Then

$$G(\rho) = \int_0^\rho g(t) dt = \frac{\log(1 + n\rho/c)}{n}, \quad H(\rho) = \int_0^\rho w(t) dt = \frac{1}{n} \left(\frac{1}{c} - \frac{1}{c + n\rho} \right). \quad (44)$$

Peeling one coordinate from the Maynard integrals gives

$$I_k(F) = m_0^n D, \quad J_k^{(1)}(F) = m_0^n N, \quad (45)$$

where

$$N = \mathbb{E}[G(1 - S)^2 \mathbf{1}_{S < 1}], \quad D = \mathbb{E}[H(1 - S) \mathbf{1}_{S < 1}], \quad (46)$$

and consequently $M_k \geq kN/D$.

Let

$$\hat{w}(\theta) = \int_0^1 e^{i\theta t} w(t) dt, \quad \varphi(\theta) = \left(\frac{\hat{w}(\theta)}{m_0} \right)^n,$$

and, for $X \in \{G^2, H\}$,

$$\Psi_X(\theta) = \int_0^1 X(\rho) e^{i\theta \rho} d\rho.$$

Listing 10: Weighted variable-projection distillation of a neural channel, followed by independent Rayleigh optimization and evaluation in the resulting clustered rational family.

```

PROCEDURE DISTILL_NEURAL_CHANNEL(t, g_NN, omega, initial c, multiplicities mu, k)
  n <- k-1
  PARAMETERIZE c as positive and strictly ordered

  FUNCTION PROFILED_FIT_ERROR(c)
    X[:,a,j] <- (c[a]+n*t)^(-j), j=1,...,mu[a]
    Xw <- DIAG(SQRT(omega))*X
    yw <- SQRT(omega)*g_NN
    scale each column of Xw to unit norm
    u <- LEAST_SQUARES(Xw,yw)
    undo the column scaling in u
    RETURN SUM_i omega[i]*(g_NN[i]-(X*u)[i])^2
  END

  c <- LBFGS_MINIMIZE(PROFILED_FIT_ERROR,c)
  u <- PROFILE_LINEAR_COEFFICIENTS(c)
  PRUNE terms with negligible scaled contribution
  INFER cluster multiplicities from the retained powers
  RETURN c, u, inferred multiplicities, relative fit error
END

PROCEDURE REFINE_RATIONAL_FAMILY(k, cluster_locations c, multiplicities mu)
  n <- k-1; L <- 1+epsilon
  basis <- {(c[a]+n*t)^(-j) : j=1,...,mu[a]}
  PARAMETERIZE c as positive and strictly ordered

  FOR EACH channel pair
    IF both channels belong to one cluster
      F[1](theta) <- SINE_COSINE_INTEGRAL_FORMULA(theta,c,n,L)
      FOR p=2,...,p_max
        F[p] <- -(EXP(i*theta*L)*(c+n*L)^(1-p)-c^(1-p))
                  -i*theta*F[p-1])/(n*(p-1))
      END
    ELSE
      REDUCE the product by partial fractions
      REJECT excessive cross-cluster cancellation
    END

    theta_max <- n*(amplification_budget*(p_max-1)!)^(1/(p_max-1))
    REQUIRE the characteristic function is negligible at theta_max
    phi(theta) <- (F_pair(theta)/F_pair(0))^n FOR |theta|<=theta_max
    density <- REAL(FFT(phi))/period
    REQUIRE conservation of mass, first moment, and noise-floor control
    INTEGRATE A[j,1] and B[j,1] against density
  END

  PROFILE the linear coefficients by TOP_RITZ(A,B)
  OPTIMIZE c by Brent (one cluster) or finite-difference LBFGS (multiple)
  RETURN the best candidate re-evaluated on a finer grid
END

```

The required expectation has the Fourier representation

$$I_X = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi(\theta) e^{-i\theta} \Psi_X(\theta) d\theta. \quad (47)$$

The integrability needed for inversion follows from integration-by-parts decay of both factors.

5.21 Aliasing bound

Fix $P > 2$ and $\Delta\theta = 2\pi/P$. For the full trapezoidal sum T_X , Poisson summation and compact support imply

$$0 \leq T_X - I_X \leq X_{\max} \mathbb{P}(S > P - 1). \quad (48)$$

The one-sided sign follows from nonnegativity of the aliased density. The residual probability is bounded by certified moments. With $\widetilde{M}_0 = 1$ and

$$\widetilde{M}_r = \sum_{j=1}^r \binom{r-1}{j-1} n \mathbb{E}[t^j] \widetilde{M}_{r-j}, \quad (49)$$

the supplied compound-Poisson domination gives

$$\mathbb{P}(S > y) \leq \min_{1 \leq r \leq r_{\max}} \frac{\widetilde{M}_r}{y^r}. \quad (50)$$

$E[S^r]$ expands over set partitions with falling-factorial coefficients $n^{\underline{b}} \leq n^b$, and the compound-Poisson moments have exactly $n^{\underline{b}}$ with all terms nonnegative.

5.22 Fourier truncation bound

Integration by parts yields, for $\theta \neq 0$,

$$\left| \frac{\widehat{w}(\theta)}{m_0} \right| \leq \frac{A}{|\theta|}, \quad A = \frac{2(c+n)}{c}, \quad (51)$$

and hence $|\varphi(\theta)| \leq (A/|\theta|)^n$. The omitted positive frequencies are divided into a finite band and an analytic tail. In the finite band, indices are partitioned into blocks $[a_j, b_j]$ and Arb wide-ball evaluation supplies certified suprema σ_j for $|\widehat{w}/m_0|$ on each block. Thus

$$|T_X^{\text{band}}| \leq \frac{\Delta\theta}{\pi} X_{\text{int}} \sum_j (b_j - a_j + 1) \sigma_j^n. \quad (52)$$

For

$$M_0 = \max(\lfloor \Theta_{\text{ball}}/\Delta\theta \rfloor, n_{\text{near}}), \quad \Theta_{\text{ball}} = 2A,$$

the remaining tail satisfies

$$|T_X^{\text{tail}}| \leq \frac{\Delta\theta}{\pi} X_{\text{int}} \left(\frac{A}{\Delta\theta} \right)^n \frac{M_0^{1-n}}{n-1}. \quad (53)$$

Taking $M_0 \geq n_{\text{near}}$ ensures that the analytic tail begins where the computed sum ends.

5.23 Arb evaluation and directed rounding

Every scalar entering the computation is produced by Arb ball arithmetic. The transform $\widehat{w}/m_0$ is evaluated from a closed form involving sine and cosine integrals away from the origin and by direct rigorous integration in the region where the special-function representation is ill-conditioned. The transforms Ψ_X and the moments entering the aliasing bound are likewise evaluated by rigorous integration. The pole and logarithmic branch point lie at $t = -c/n < 0$, outside the real integration interval; subdivision is used whenever an evaluation enclosure approaches a singularity or branch cut.

All error terms remain in ball arithmetic until the final quotient is assembled. Floating-point exports are explicitly nudged until Arb comparisons prove outward rounding. If N_{ball} and D_{ball} denote the truncated-sum enclosures and E_N, E_D the certified totals of the aliasing and truncation errors, then

$$R_{\text{low}} = \text{lb} \left(\frac{k \text{ lb}(N_{\text{ball}} - E_N)}{\text{ub}(D_{\text{ball}} + E_D)} \right). \quad (54)$$

The stored claim is additionally rounded downward so that independent recomputation with different valid Arb subdivisions or enclosure radii cannot strengthen the stated result.

5.24 Certified rank-one ladder and uniform finite-range bound

For $n = k - 1$ and $c > 0$, let

$$g_c(t) = \frac{1}{c + nt},$$

and denote by $R_k^{(1)}$ the value attained by this rank-one rational family. Every such value is a lower bound for M_k .

The fixed headroom value used below was selected from the independently optimized rank-one sweep rather than imposed a priori. Writing

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)},$$

the median optimized value of η moves steadily toward approximately -0.475 as k increases:

range of k	median η
$k < 10^3$	-0.4256
$10^3 \leq k < 10^5$	-0.4593
$10^5 \leq k < 10^7$	-0.4715
$10^7 \leq k < 10^9$	-0.4747

This convergence motivated the fixed choice

$$\eta^* = -0.4746,$$

which agrees most closely with the independently optimized scale in the large- k regime containing the principal threshold results.

Fixing η incurs very little loss for the bulk of the optimized sweep. Interpolating the fixed-headroom ladder to the k -values of the direct rank-one sweep, the optimized certified value exceeds the corresponding fixed- η value by a median of only

$$4.2 \times 10^{-5},$$

with maximum positive difference

$$4.2 \times 10^{-3}.$$

The comparison is not uniformly positive: the minimum difference is

$$-4.2 \times 10^{-2},$$

so at least one nominally optimized sweep row is outperformed by the fixed-headroom construction at the same k . This is consistent with the isolated high- k outlier near $\eta = -0.32$, and is one reason the direct sweep is interpreted as a record of the numerical optimization landscape rather than as a certified sequence of exact one-parameter optima.

To obtain a uniform finite-range statement rather than isolated certified dimensions, we constructed an approximately log-uniform grid beginning at $k = 100$, with target spacing

$$\Delta \log k = 0.02.$$

At each grid point, c was determined by solving

$$\eta(c, k) = \eta^* = -0.4746$$

using the closed-form moments of the rank-one probability measure. The resulting explicit trial function was then certified independently using the ball-arithmetic procedure of Section 5.23. All retained rungs passed certification.

For consecutive certified grid points $k_i < k_{i+1}$, monotonicity of M_k implies that every integer $k \in [k_i, k_{i+1}]$ satisfies

$$M_k \geq M_{k_i} \geq R_{\text{cert}}^{(1)}(k_i) \geq \log k - \left[\log k_{i+1} - R_{\text{cert}}^{(1)}(k_i) \right].$$

Thus the relevant covering deficit on the interval $[k_i, k_{i+1}]$ is

$$\delta_i = \log k_{i+1} - R_{\text{cert}}^{(1)}(k_i).$$

The largest interval deficit over the certified ladder is

$$\max_i \delta_i = 0.306754936,$$

attained on the penultimate full-size interval

$$k_i = 1,834,543,653, \quad k_{i+1} = 1,880,000,000.$$

The maximum does not occur on the final interval because the last grid step was deliberately shortened to terminate at the round endpoint

$$k = 1.88 \times 10^9.$$

Its logarithmic width is only

$$\Delta \log k = 0.004476,$$

substantially smaller than the nominal spacing 0.02. The final truncated interval therefore contributes a smaller covering deficit than the preceding full-size rung.

The distinction between pointwise and interval-covering deficits is also important. On the certified grid points themselves, the largest observed pointwise deficit is

$$\max_i \left[\log k_i - R_{\text{cert}}^{(1)}(k_i) \right] = 0.286812601.$$

Hence the certified rungs satisfy the sharper pointwise statement

$$R_{\text{cert}}^{(1)}(k_i) \geq \log k_i - 0.287.$$

The larger constant 0.307 is required only to cover every intermediate integer k between successive certified rungs.

Consistency of the threshold certificates with the ladder. The four threshold certificates reported in the Results lie inside the range covered by this ladder, and were produced independently of it: the pole parameter c was optimised at each threshold rather than selected from the fixed-headroom relation, and each was certified as an isolated trial function. They therefore provide an internal check on the finite-range behaviour rather than additional coverage.

Supplementary Table 4 records the certified deficit $\log k - R_{\text{cert}}^{(1)}$ at each threshold, together with the value obtained from the empirical finite-size relation

$$\log k - R_k^{(1)} \approx C_* - \frac{1.047}{\log k}, \quad C_* \simeq 0.3343, \quad (55)$$

which was introduced in Section 5.25 from the optimised rank-one sweep. Agreement is within 3×10^{-3} at every point, across more than five orders of magnitude in k , and the final row shows that the upper end of the certified ladder follows the same relation.

Two remarks on the status of this comparison. First, it is a consistency check and not a proof component: the constant C_* is a numerical evaluation of the limiting formula, and the coefficient 1.047 is an empirical finite-size fit for which Section 5.26 establishes no quantitative rate. Neither enters any certified inequality. Second, the agreement is nevertheless informative precisely because nothing

Supplementary Table 4: Certified deficits at the four thresholds and at the upper end of the fixed-headroom ladder, compared with the finite-size relation Eq. (55). Thresholds were optimised and certified independently of the ladder; the final row is the pointwise deficit at the last ladder rung. The relation was not fitted to these values.

k	$\log k$	$R_{\text{cert}}^{(1)}$	$\log k - R_{\text{cert}}^{(1)}$	$\mathcal{C}_* - 1.047/\log k$	difference
3 655	8.203851	8.0000648690	0.203787	0.206677	-0.002891
208 910	12.249659	12.0000019900	0.249657	0.248828	+0.000829
11 655 069	16.271252	16.0000003924	0.271251	0.269953	+0.001298
644 589 002	20.284123	20.0000009143	0.284123	0.282683	+0.001439
1 880 000 000	21.354538	(ladder rung)	0.286813	0.285271	+0.001542

was tuned to produce it: the thresholds were optimised for a different purpose (crossing the integer values 8, 12, 16 and 20), the ladder was generated from a fixed headroom target, and the relation (55) was extracted from a third sweep. That three independently constructed families of certified rank-one trials trace the same $\mathcal{C}_* - b/\log k$ curve supports the interpretation of the finite- k behaviour as the pre-asymptotic regime of the stable-law limit rather than as a numerical coincidence. The single deviation of appreciable size, at $k = 3655$, is in the conservative direction and is consistent with the relation being fitted at larger k .

The interval constant is close to saturated at the upper end of the computed range. The final four full-size interval deficits are

$$0.306613, \quad 0.306660, \quad 0.306708, \quad 0.306755,$$

increasing by approximately

$$4.7 \times 10^{-5}$$

per rung. Continuing the same fixed-headroom ladder for only several further full-size steps would therefore push the covering deficit above 0.307. Accordingly, rounding the observed maximum upward gives the finite-range statement

$$M_k \geq \log k - 0.307, \quad 100 \leq k \leq 1,880,000,000.$$

No extension of the constant 0.307 beyond this explicitly certified range is claimed.

5.25 Rank-one scale selection and the headroom invariant

Under the probability measure proportional to $g_c(t)^2 dt$, let $S = \sum_{i < k} t_i$ and write

$$c^{-1} = \log n + a, \quad n = k - 1.$$

Uniformly for $a = O(\sqrt{\log n})$, the closed-form moments of this measure imply

$$1 - \mathbb{E}[S] = c(a + 1 - \log c^{-1}) + O\left(\frac{c^2 \log n}{n}\right), \quad \text{sd}(S) = \sqrt{c} \left(1 + O\left(\frac{\log n}{n}\right)\right). \quad (56)$$

For fixed a , $\log c^{-1} = \log \log n + O(1/\log n)$. Consequently,

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)} = \sqrt{c}(a + 1 - \log c^{-1}) + O\left(\frac{\log n}{n}\right). \quad (57)$$

Fixing $\eta = -\gamma_h$, multiplying Eq. (57) by $c^{-1/2}$ and substituting $a = c^{-1} - \log n$ gives the implicit scale equation

$$c^{-1} + \gamma_h c^{-1/2} - \log c^{-1} = \log n - 1 + O\left(\frac{(\log n)^{3/2}}{n}\right). \quad (58)$$

The left-hand side is strictly increasing in c^{-1} for $c^{-1} \geq 1$, so Eq. (58) determines the fixed-headroom scale to the stated accuracy. Writing $x = c^{-1}$, it gives $x = \log n + O(\sqrt{\log n})$, hence $\log x = \log \log n + O((\log n)^{-1/2})$ and $\sqrt{x} = \sqrt{\log n} - \gamma_h/2 + O(\log \log n / \sqrt{\log n})$. Substituting these expansions yields

$$c^{-1} = \log n - \gamma_h \sqrt{\log n} + \log \log n - 1 + \frac{\gamma_h^2}{2} + O\left(\frac{\log \log n}{\sqrt{\log n}}\right). \quad (59)$$

The constant $\gamma_h^2/2$ arises from the shift of $c^{-1/2}$ relative to $\sqrt{\log n}$. The remainder decays only like $\log \log n / \sqrt{\log n}$ and is not small over the certified range ($\log \log n / \sqrt{\log n} \approx 0.66$ at $\log n \approx 21.35$). The ladder was therefore computed from the exact closed-form moments, which agree with Eq. (58) to the stated accuracy, and not from the truncated expansion (59). Thus the empirically selected coefficient $\gamma_h \simeq 0.4746$ is not an independent asymptotic fit: it is the same quantity as $-\eta^*$ and follows directly from the fixed-headroom parameterization.

Equations (58) and (59) describe the fixed-headroom parameterization of the certified ladder; they are not a sharper asymptotic optimizer law. Indeed, fixed $\eta = -\gamma_h$ corresponds to the shift

$$a(L) = c^{-1} - L = \log L - 1 - \gamma_h \sqrt{L} + \frac{\gamma_h^2}{2} + O\left(\frac{\log L}{\sqrt{L}}\right), \quad L = \log n,$$

so $a(L) \rightarrow -\infty$ and fixed η eventually leaves every fixed-shift family. The leading-order part of $a(L)$ is not monotone: its derivative $1/L - \gamma_h/(2\sqrt{L})$ vanishes at

$$\sqrt{L} = \frac{2}{\gamma_h},$$

where it attains its maximum, and it decreases thereafter. Over the certified range, however, the induced shift obtained from Eq. (58) remains close to the small negative values where the numerical limiting defect $\mathcal{C}(a)$ is minimized. This explains the effectiveness of the headroom surrogate over the finite range studied here: it selects rational trials lying close to the same near-optimal fixed-shift regime, even though the two parameterizations diverge asymptotically.

At the upper end of the certified ladder, the measured pointwise deficit is approximately

$$\log k - R(F_{k,c_k}) \simeq 0.2868,$$

whereas numerical evaluation of the limiting fixed-shift formula gives

$$\mathcal{C}_* \simeq 0.3343.$$

The difference is approximately

$$0.3343 - 0.2868 = 0.0475.$$

Over the finite range considered in the numerical sweep, this is close to the magnitude of the empirically fitted correction

$$\frac{1.047}{\log k} \simeq 0.049 \quad (\log k \simeq 21.35).$$

Equivalently, the observed deficits are numerically consistent with the finite-range relation

$$\log k - R(F_{k,c_k}) \approx \mathcal{C}_* - \frac{1.047}{\log k}.$$

This relation is an empirical consistency check only. In particular, the coefficient 1.047 has not been derived analytically and is not identified with the coefficient of a proved second-order expansion.

The distinction is important because the stable-law theorem concerns the fixed-shift family

$$c^{-1} = \log n + a, \quad a \in \mathbb{R} \text{ fixed},$$

whereas the fixed-headroom ladder corresponds to a shift $a = a(\log n)$ that eventually tends to $-\infty$. Therefore the finite-size expansion for fixed a cannot be applied directly to the optimized headroom ladder. Even within the fixed-shift family, the proof establishes only

$$\log k - R(F_{k,c_k}) = \mathcal{C}(a) + o(1),$$

because no sufficiently sharp convergence rate has been proved for

$$\frac{L_n}{P_n} - \frac{L_a}{P_a}.$$

Consequently, neither the numerical agreement above nor the fixed-headroom parameterization establishes a rigorous $1/\log k$ correction. The coefficient 1.047 should therefore be reported solely as a numerical finite-size observation.

5.26 Spectrally positive 1-stable limit

Section 6 gives the complete asymptotic analysis of the single-channel rational family. For

$$n = k - 1, \quad c_k^{-1} = \log n + a,$$

with fixed $a \in \mathbb{R}$, the rescaled boundary variable converges in distribution to

$$X_a = (a + 1) - \Lambda,$$

where Λ is the spectrally positive 1-stable random variable defined by

$$\mathbb{E}e^{is\Lambda} = \exp \left\{ \int_0^\infty (e^{isx} - 1 - isx\mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2} \right\}.$$

The proof supplements weak convergence with uniform density, small-ball, tail, and logarithmic moment estimates, which are needed because the Rayleigh quotient contains logarithmic observables at the simplex boundary.

Theorem 9 consequently gives

$$R(F_{k,c_k}) = \log k - \mathcal{C}(a) + o(1),$$

where

$$\mathcal{C}(a) = a - 2\mathbb{E}[\log X_a \mid X_a > 0].$$

Thus, with

$$\mathcal{C}_* := \inf_{a \in \mathbb{R}} \mathcal{C}(a),$$

Corollary 10 yields

$$M_k \geq \log k - \mathcal{C}_* - o(1).$$

Here $\mathcal{C}_*$ is the optimal asymptotic defect only within the fixed-shift family

$$c_k^{-1} = \log(k - 1) + a, \quad a \in \mathbb{R}.$$

The proof does not establish optimality over arbitrary sequences c_k , nor does it require the infimum to be attained. Numerical evaluation suggests

$$\mathcal{C}_* \approx 0.3343,$$

but this decimal value should be regarded as a numerical estimate unless the global infimum is separately certified. A rigorous enclosure of $\mathcal{C}(a_0)$ at one fixed shift a_0 already suffices for an explicit asymptotic lower bound.

Finally, although the finite- n identity in Section 6 isolates a possible $1/\log k$ correction, the required convergence rate for

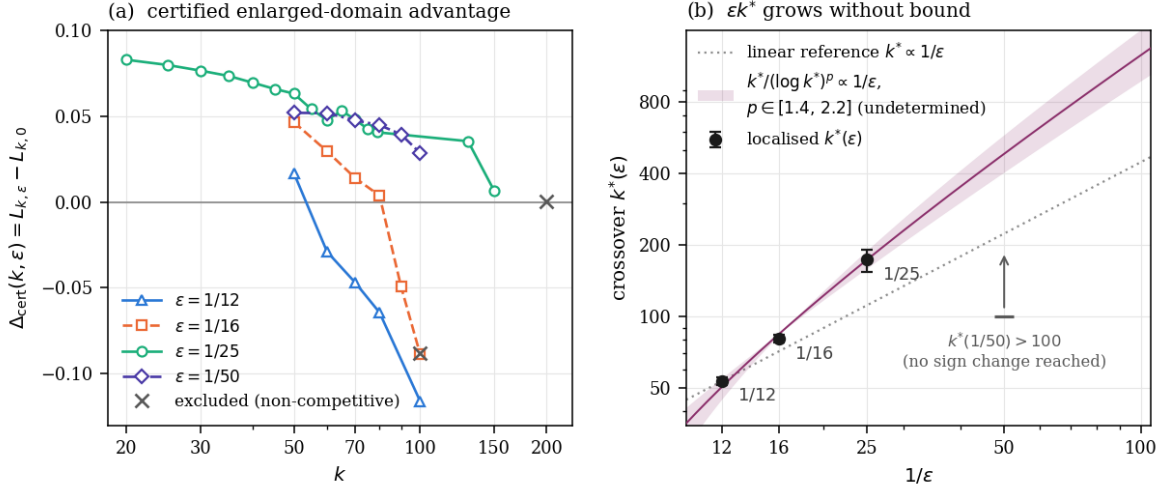
$$\frac{L_n}{P_n} - \frac{L_a}{P_a}$$

has not been proved. Accordingly, no second-order asymptotic coefficient is claimed here.

Relation to the optimized rank-one sweep. The optimized rank-one sweep explores a different asymptotic parameter regime, including an empirically observed $\sqrt{\log n}$ correction. Those numerical observations are not used in the fixed-shift theorem above.

5.27 Enlarged-support crossover analysis

Across the ladder $50 \leq k \leq 220$, the advantage conferred by ε fluctuates at small k , then decreases and changes sign at a finite crossover $k_*(\varepsilon)$, which we locate at $k_*(1/12) \approx 53.7$, $k_*(1/16) \approx 80.7$ and $k_*(1/25) \in [155, 192]$. We conjecture that such a crossover exists for every $\varepsilon > 0$ and that $k_*(\varepsilon)$ grows superlinearly in $1/\varepsilon$, so that $\varepsilon k_*(\varepsilon) \rightarrow \infty$ (Supplementary Table 9).



Supplementary Figure 3: **Enlarged-support crossover.** (a) The certified difference $\Delta_{\text{cert}}(k, \varepsilon) = L_{k, \varepsilon} - L_{k, 0}$, a difference of certified lower bounds rather than of the exact constants, against k for four values of ε . It is positive at small k , decreases with k , and changes sign at a finite scale for $\varepsilon = 1/12$ and $1/16$; no sign change is reached for $\varepsilon = 1/50$ within the computed range. Grey crosses mark certified but non-competitive runs excluded from crossover inference (Section 5.27). (b) The localised crossovers $k^*(\varepsilon)$ against $1/\varepsilon$ on logarithmic axes. The $\varepsilon = 1/12$ and $1/16$ points are linear interpolations between the certified brackets; the $\varepsilon = 1/25$ marker is the midpoint of the extrapolative range $[155, 192]$, with the bar spanning that range. The points lie above the linear reference $k^* \propto 1/\varepsilon$ (dotted) and pull away from it, giving $\varepsilon k^* = 4.48, 5.04, 6.94$ and motivating $\varepsilon k^*(\varepsilon) \rightarrow \infty$. The band shows the family $k^*/(\log k^*)^p \propto 1/\varepsilon$ over the range $p \in [1.4, 2.2]$ admitted by the data, with the constant refitted at each p ; the exponent is not determined by these points.

Write $L_{k, \varepsilon}$ for the certified lower bound produced for the enlarged-domain construction and $L_{k, 0}$ for the corresponding vanilla lower bound, and define

$$\Delta_{\text{cert}}(k, \varepsilon) = L_{k, \varepsilon} - L_{k, 0}.$$

For the reliable computed configurations, this difference is positive at smaller k , decreases with k , and changes sign at a finite scale. The certified pairs used to localize the observed crossovers include

ε	k_{left}	k_{right}
1/12	50	60
1/16	80	90

with certified differences $+0.016961$ and -0.028948 for $\varepsilon = 1/12$, and $+0.003771$ and -0.049093 for $\varepsilon = 1/16$. Linear interpolation gives

$$k^*(1/12) \approx 53.7, \quad k^*(1/16) \approx 80.7.$$

For $\varepsilon = 1/25$, the last reliable positive certified points are $k = 130$ and 150 , with differences $+0.035443$ and $+0.006581$; the supplied analysis reports the broader extrapolative range

$$k^*(1/25) \in [155, 192]$$

rather than treating a degraded $k = 200$ pair as a direct bracket. No sign change was reached for $\varepsilon = 1/50$ within the reliable computed range.

All certified runs are in Supplementary Table 9. Several runs are excluded from crossover inference because they satisfy numerical certification of the particular trial function but are demonstrably noncompetitive discovery outcomes. These include a reproducible local failure at $(k, \varepsilon) = (100, 1/25)$, degraded paired runs at $k = 200$, and incomplete or nonmonotone optimization at $k = 140$ and 220 . The exclusion criterion is therefore not failure of the certificate: it is failure of the discovery stage to

provide a competitive construction. Monotonicity in k , the discovery-to-certification gap and retained effective rank provide inexpensive diagnostics for these cases.

The observed crossover locations give increasing values of $\varepsilon k^*(\varepsilon)$ as ε decreases (see Supplementary Figure 3), motivating the conjecture that for each fixed $\varepsilon > 0$ a finite crossover exists and

$$\varepsilon k^*(\varepsilon) \rightarrow \infty \quad (\varepsilon \rightarrow 0).$$

The available points are compatible with a scaling variable

$$x = \frac{\varepsilon k}{(\log k)^p},$$

but do not determine p : the supplied fits admit approximately $p \in [1.4, 2.2]$, and the natural $p = 1$ and $p = 2$ alternatives cannot be distinguished by the current data. We therefore do not assign a specific exponent.

Scope of the crossover statement. A crucial distinction is that Δ_{cert} is a difference of certified lower bounds. Therefore $\Delta_{\text{cert}}(k, \varepsilon) < 0$ does not prove $M_{k, \varepsilon} < M_k$; it proves only that the certified vanilla construction found by the pipeline outperforms the certified enlarged-domain construction found at that k . Extending the observed crossover to the exact variational constants is consequently conjectural. Likewise, interpolated or extrapolated values of $k^*(\varepsilon)$ are descriptive summaries of the computed ladder, not independently certified roots.

5.28 Supplementary data: optimized rank-one sweep

Supplementary Data `maynard_R_sweep_eta.csv` contains the direct rank-one optimization sweep for

$$g_c(t) = \frac{1}{c + (k-1)t},$$

with one optimized pole parameter c for each reported k . The file records the discovery value R , the independently certified lower bound, the discovery–certification gap, the Cauchy–Schwarz ceiling, the optimized scale c , the standardized simplex headroom

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)},$$

and numerical diagnostics used during optimization. All retained solutions have rank one in both numerator and denominator Gram representations.

The sweep contains 729 attempted configurations over $100 \leq k \leq 2 \times 10^9$. Of these, 727 pass certification and two are explicitly marked **FAILED**; failed rows are retained in the data for provenance but are excluded from mathematical claims. The certified discovery gap is small throughout the successful runs, with median 9.9×10^{-4} and maximum 2.3×10^{-3} . The optimized headroom is not fixed: its variation across the sweep is used diagnostically to identify the approximately constant headroom regime that motivates the geometric ladder below. This file is therefore a record of the numerical optimization landscape, not itself the basis for the uniform finite-range theorem.

5.29 Supplementary data: fixed-headroom geometric ladder

Supplementary Data `maynard_geometric_eta_sweep.csv` contains the certified fixed-headroom ladder used to convert isolated rank-one trial functions into a uniform bound over a continuous range of integer k . At each rung, c is chosen from

$$\eta(c, k) = -0.4746,$$

and the resulting explicit rank-one trial is certified independently. The file contains 828 certified rungs from $k = 100$ to $k = 1.88 \times 10^9$; every row has certification status **PASS**. The target headroom is reproduced to numerical precision throughout the sweep.

The ladder is approximately uniform in $\log k$, with asymptotic spacing $\Delta \log k \simeq 0.02$; the small deviations at the lower end and final endpoint arise from integer rounding. For consecutive certified rungs $k_i < k_{i+1}$, monotonicity of M_k gives

$$M_k \geq R_{\text{cert}}(k_i) \quad (k_i \leq k \leq k_{i+1}),$$

and therefore

$$M_k \geq \log k - [\log k_{i+1} - R_{\text{cert}}(k_i)].$$

The largest interval deficit in the supplied data is

$$0.3067549357,$$

attained between

$$k_i = 1\,834\,543\,653, \quad k_{i+1} = 1\,871\,603\,894.$$

Rounding this deficit upward yields the reported uniform statement

$$M_k \geq \log k - 0.307$$

throughout the certified range. The file additionally records the pointwise deficit $\log k - R_{\text{cert}}$, the discovery–certification gap, the Cauchy–Schwarz ceiling and the scale diagnostics used to relate the finite ladder to the fixed-shift asymptotic analysis.

5.30 Conversion to admissible-tuple bounds

A certified inequality $M_k > 4m$ supplies the variational input to the Maynard–Tao bounded-gap theorem. To obtain a numerical bound on H_m , one additionally requires an admissible k -tuple and its diameter. We therefore distinguish the variational certificate from the admissible-tuple construction and write D_k for the diameter of the explicit admissible k -tuple used in the conversion.

For the first threshold, $M_{3655} > 8$, a stronger admissible-tuple diameter is already recorded in the Engelsma–Sutherland database (<https://math.mit.edu/~primegaps/>). The entry for $k = 3655$, submitted by Sutherland on 27 June 2013, gives

$$D_{3655} \leq 33118.$$

Our independently constructed hybrid sieve gives the slightly weaker control

$$D_{3655} \leq 33270,$$

and is retained as a reproducibility check.

For $k = 3655$ and $k = 208910$, admissible tuples were constructed with a hybrid shifted-Schinzel/greedy sieve. The algorithm first applies a Schinzel-type presieve, then greedily removes a least-populated residue class for successive primes, extracts the narrowest block of k survivors, and finally applies local diameter-reducing moves. Crucially, the discovery trajectory is not trusted: the final tuple is rechecked from scratch for every prime $p \leq k$.

Each constructed tuple is exported as an `admissible-tuple-certificate/1` record containing the ordered gap sequence and, for every prime $p \leq k$, an explicitly missing residue class. A standalone verifier shares no construction code path. It regenerates the primes up to k , reconstructs the tuple from the gap list, checks its cardinality and diameter, verifies that the witness list contains exactly the primes $p \leq k$, and confirms by direct enumeration that every stated residue class is absent. Since a k -element set cannot occupy all residue classes modulo any prime $p > k$, these checks establish admissibility.

The resulting independently verified hybrid-sieve bounds are

$$D_{3655} \leq 33270, \quad D_{208910} \leq 2718108.$$

For $k = 3655$, the stronger database value $D_{3655} \leq 33118$ is used in the final bounded-gap comparison, while our independently constructed tuple serves as a reproducibility control.

For the larger thresholds, explicit window sieving is unnecessary. Let

$$\mathcal{H}_k = \{p_{\pi(k)+1}, \dots, p_{\pi(k)+k}\},$$

the first k primes strictly exceeding k . Every element of $\mathcal{H}_k$ is prime and greater than k , so residue class $0 \bmod p$ is empty for every prime $p \leq k$; for $p > k$, a k -element set cannot meet all p residue classes. Hence $\mathcal{H}_k$ is admissible by construction. Segmented prime enumeration gives

$$D_{11655069} \leq 214099720, \quad D_{644589002} \leq 14541349288.$$

The code additionally evaluates explicit prime-counting and prime-location bounds as an independent large- k consistency check.

Combining these tuple diameters with the certified variational thresholds $M_k > 4m$ gives

$$H_m \leq D_k,$$

and hence the numerical bounded-gap statements reported in the Results. The tuple-construction code, certificate files, and independent verifier are supplied as Supplementary Software/Data.

5.31 Standalone verification and certificate schema

The large- k certificate is a self-contained JSON record containing the exact rational trial parameter c , Fourier discretization parameters, intermediate summaries and the claimed lower bound. Stored intermediate numerical enclosures are not trusted. The verifier reconstructs the analytic Fourier-tail constant, regenerates the far-field block partition, recomputes wide-ball suprema, moment bounds, Fourier nodes, trapezoidal sums, aliasing and truncation errors, and independently assembles R_{low} in ball arithmetic.

The certificate proves only the Rayleigh quotient of the fixed admissible trial function encoded by the exact rational c . It does not prove optimality of c , re-prove the external sieve criterion or general upper bounds on M_k , or certify admissible-tuple diameters. Monte-Carlo comparisons and external small- k values are falsification checks rather than proof components.

6 Asymptotics of the single-channel rational family

We now explain analytically why the single-channel rational trial functions used in the large- k computations remain within a bounded distance of the logarithmic scale $\log k$.

For $k \geq 2$, write

$$n := k - 1,$$

and consider the symmetric trial function

$$F_{k,c}(t_1, \dots, t_k) = \mathbf{1}_{\{t_i \geq 0, \sum_{i=1}^k t_i \leq 1\}} \prod_{i=1}^k \frac{1}{c + nt_i}, \quad c > 0. \quad (60)$$

Its overall scalar normalization is immaterial.

The exact probabilistic reduction established in Section 5.20 gives

$$R(F_{k,c}) = k \frac{N_{k,c}}{D_{k,c}}, \quad (61)$$

where

$$w(t) = \frac{1}{(c + nt)^2}, \quad m_0 = \int_0^1 w(t) dt = \frac{1}{c(c + n)},$$

and, if $T_1, \dots, T_n$ are independent with density

$$\frac{w(t)}{m_0} = \frac{c(c + n)}{(c + nt)^2}, \quad 0 \leq t \leq 1,$$

and

$$S_n := T_1 + \dots + T_n,$$

then

$$N_{k,c} = \mathbb{E}[G(1 - S_n)^2; S_n < 1], \quad (62)$$

$$D_{k,c} = \mathbb{E}[H(1 - S_n); S_n < 1], \quad (63)$$

with

$$G(\rho) = \frac{1}{n} \log \left(1 + \frac{n\rho}{c} \right), \quad (64)$$

$$H(\rho) = \frac{1}{n} \left(\frac{1}{c} - \frac{1}{c + n\rho} \right). \quad (65)$$

The relevant scale for c is

$$\frac{1}{c_k} = \log(k-1) + a, \quad a \in \mathbb{R} \text{ fixed.} \quad (66)$$

The numerical optimization of the rational family independently exhibits this $c_k \asymp 1/\log k$ scaling. The result below identifies the limiting $O(1)$ defect from $\log k$ for every fixed shift a .

6.1 The limiting stable law

We first isolate the random variable governing the boundary of the simplex. Define

$$X_n := \frac{1 - S_n}{c}. \quad (67)$$

Thus

$$S_n < 1 \iff X_n > 0.$$

It is convenient to rescale a single summand by writing

$$V := \frac{T}{c}.$$

A direct change of variables gives the density

$$q_n(v) = \frac{c+n}{(1+nv)^2}, \quad 0 \leq v \leq \frac{1}{c}. \quad (68)$$

Hence

$$X_n = \frac{1}{c} - \sum_{j=1}^n V_j, \quad (69)$$

where $V_1, \dots, V_n$ are independent with density (68).

We use the following normalization of the totally right-skewed 1-stable law.

Definition 1 (The limiting 1-stable variable). *Let Λ be the infinitely divisible random variable with characteristic function*

$$\mathbb{E} e^{is\Lambda} = \exp \left\{ \int_0^\infty (e^{isx} - 1 - isx \mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2} \right\}. \quad (70)$$

This is a spectrally positive stable law of index 1, in the Lévy–Khintchine normalization used below. Its law has a continuous, everywhere positive density on $\mathbb{R}$.

Because Λ has only positive jumps, its Laplace transform is finite in the damping direction. The following elementary computation is used repeatedly: it controls the left tail of Λ , supplies the integrability needed in Lemma 11, and provides a one-dimensional quadrature route to the constants P_a , L_a , Q_a introduced below.

Lemma 2 (Laplace transform and left tail of Λ). *For every $\theta > 0$,*

$$\mathbb{E} e^{-\theta\Lambda} = \exp\{\theta \log \theta - \theta + \gamma_E \theta\}, \quad (71)$$

where γ_E is the Euler–Mascheroni constant. To avoid collision we use γ_E instead of γ_h from Section 5.25. Consequently, for all $y \geq 0$,

$$\mathbb{P}(\Lambda \leq -y) \leq \exp\{-e^{y-\gamma_E}\}, \quad (72)$$

and in particular

$$\kappa := \mathbb{E}[(-\Lambda)^+] < \infty. \quad (73)$$

Proof. Analytic continuation of (70) to $s = i\theta$ gives

$$J(\theta) := \log \mathbb{E} e^{-\theta\Lambda} = \int_0^\infty (e^{-\theta x} - 1 + \theta x \mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2},$$

the integral being absolutely convergent for $\theta > 0$. Differentiating under the integral sign,

$$J'(\theta) = \int_0^1 \frac{1 - e^{-\theta x}}{x} dx - \int_1^\infty \frac{e^{-\theta x}}{x} dx = (\log \theta + \gamma_E + E_1(\theta)) - E_1(\theta) = \log \theta + \gamma_E,$$

where E_1 is the exponential integral and we used the standard identity $\int_0^z (1 - e^{-t}) t^{-1} dt = \gamma_E + \log z + E_1(z)$. Since $J(0^+) = 0$, integration gives (71).

For (72), Markov's inequality applied to $e^{-\theta\Lambda}$ yields, for every $\theta > 0$,

$$\mathbb{P}(\Lambda \leq -y) \leq e^{-\theta y} \mathbb{E} e^{-\theta\Lambda} = \exp \{ \theta (\log \theta - 1 + \gamma_E - y) \}.$$

The exponent is minimized at $\theta = e^{y - \gamma_E}$, where it equals $-\theta$, giving (72). Finally $\mathbb{E}[(-\Lambda)^+] = \int_0^\infty \mathbb{P}(\Lambda \leq -y) dy \leq \int_0^\infty \exp\{-e^{y - \gamma_E}\} dy < \infty$. $\square$

Lemma 3 (Stable limit). *Suppose*

$$\frac{1}{c} = \log n + a$$

with fixed $a \in \mathbb{R}$. Then

$$X_n \xrightarrow{d} X_a := (a + 1) - \Lambda. \quad (74)$$

Proof. The characteristic function of X_n is

$$\phi_n(s) = e^{is/c} (\mathbb{E} e^{-isV})^n.$$

Introduce the finite Lévy measure

$$\mu_n(dx) := nq_n(x) dx = \frac{n(c+n)}{(1+nx)^2} \mathbf{1}_{\{0 \leq x \leq 1/c\}} dx,$$

of total mass $\mu_n([0, 1/c]) = n$. For every fixed $x > 0$,

$$\frac{n(c+n)}{(1+nx)^2} \longrightarrow \frac{1}{x^2}.$$

For fixed s and sufficiently large n , choose the logarithm of $1 + z_n(s)/n$ near 1 and use the corresponding characteristic exponent for $\phi_n(s)$. With this convention, we write

$$\log \phi_n(s) = \frac{is}{c} + n \log \left(1 + \frac{z_n(s)}{n} \right), \quad z_n(s) := \int (e^{-isx} - 1) \mu_n(dx). \quad (75)$$

Fix s . Using $|e^{-isx} - 1| \leq \min(2, |s|x)$ and splitting the integral at $x = 1/|s|$,

$$|z_n(s)| \leq |s| \int_0^{1/|s|} x \mu_n(dx) + 2 \mu_n((1/|s|, 1/c]) = O_s(\log n), \quad (76)$$

by the elementary evaluation (78) below. Thus $z_n(s)/n \rightarrow 0$, and the principal branch of the logarithm satisfies

$$n \log \left(1 + \frac{z_n}{n} \right) = z_n + O \left(\frac{|z_n|^2}{n} \right) = z_n + O_s \left(\frac{\log^2 n}{n} \right) = z_n + o(1).$$

(The hypothesis needed here is $z_n = o(\sqrt{n})$, not boundedness of z_n ; (76) supplies it.) Hence

$$\begin{aligned} \log \phi_n(s) &= \frac{is}{c} + \int_0^{1/c} (e^{-isx} - 1) \mu_n(dx) + o(1) \\ &= is \left[\frac{1}{c} - \int_0^1 x \mu_n(dx) \right] \\ &\quad + \int_0^{1/c} (e^{-isx} - 1 + isx \mathbf{1}_{\{x \leq 1\}}) \mu_n(dx) + o(1). \end{aligned} \quad (77)$$

The compensated integral converges, by dominated convergence on compact subintervals and elementary control at 0 and ∞ , to

$$\int_0^\infty (e^{-isx} - 1 + isx \mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2}.$$

It remains to identify the drift. Substituting $u = nx$,

$$\int_0^1 x \mu_n(dx) = \left(1 + \frac{c}{n}\right) \int_0^n \frac{u du}{(1+u)^2} = \left(1 + \frac{c}{n}\right) \left[\log(1+n) - 1 + \frac{1}{1+n} \right]. \quad (78)$$

Since $c = 1/(\log n + a)$,

$$\frac{1}{c} - \int_0^1 x \mu_n(dx) \longrightarrow a + 1.$$

Consequently

$$\log \phi_n(s) \longrightarrow is(a+1) + \int_0^\infty (e^{-isx} - 1 + isx \mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2}. \quad (79)$$

By Definition 1, the limiting characteristic function is that of $(a+1) - \Lambda$. Lévy's continuity theorem gives (74). $\square$

6.2 Uniform density and logarithmic moment control

Weak convergence alone is insufficient for the logarithmic observables appearing in the Rayleigh quotient, because $\log x$ is singular at $x = 0$. We therefore record two uniform estimates.

Lemma 4 (Uniform Fourier integrability and bounded densities). *For fixed a , with $c^{-1} = \log n + a$, there exists a constant $C_a < \infty$ such that, for all sufficiently large n ,*

$$\int_{\mathbb{R}} |\phi_n(s)| ds \leq C_a. \quad (80)$$

Consequently X_n possesses a density f_n satisfying

$$\sup_{n \geq n_0} \|f_n\|_\infty < \infty. \quad (81)$$

In particular, uniformly for $0 < \delta \leq 1$,

$$\mathbb{P}(0 < X_n \leq \delta) \ll_a \delta. \quad (82)$$

Proof. Since translation does not affect the modulus of the characteristic function,

$$|\phi_n(s)| = |\psi_n(s)|^n, \quad \psi_n(s) := \mathbb{E} e^{-isV},$$

and $|\psi_n|$ is even in s , so we may assume $s > 0$.

An explicit two-interval bound. For $\beta > 0$ let

$$I_1(\beta) := \left[0, \frac{\pi}{4}\beta\right], \quad I_2(\beta) := \left[\frac{3\pi}{4}\beta, \frac{5\pi}{4}\beta\right].$$

These are disjoint, and for $v \in I_1(\beta)$, $v' \in I_2(\beta)$ we have $v' - v \in [\frac{\pi}{2}\beta, \frac{5\pi}{4}\beta]$. Taking $\beta = 1/s$ therefore gives $s(v' - v) \in [\pi/2, 5\pi/4]$, on which $\cos \leq 0$ and hence $1 - \cos(s(v - v')) \geq 1$. Writing V' for an independent copy of V and discarding all other configurations,

$$1 - |\psi_n(s)|^2 = \mathbb{E}[1 - \cos(s(V - V'))] \geq 2 \mathbb{P}(V \in I_1(1/s)) \mathbb{P}(V \in I_2(1/s)). \quad (83)$$

Both intervals lie in $[0, 1/c]$ once $5\pi/(4s) \leq 1/c$, which holds for all $s \geq 1$ and n large, since $1/c = \log n + a \rightarrow \infty$.

From (68), for $0 \leq \alpha < \beta \leq 1/c$,

$$\mathbb{P}(V \in [\alpha, \beta]) = \frac{c+n}{n} \left[\frac{1}{1+n\alpha} - \frac{1}{1+n\beta} \right]. \quad (84)$$

Put $b := \pi n/(4s)$. Then (84) gives

$$\mathbb{P}(V \in I_1(1/s)) = \frac{c+n}{n} \cdot \frac{b}{1+b}, \quad \mathbb{P}(V \in I_2(1/s)) = \frac{c+n}{n} \cdot \frac{2b}{(1+3b)(1+5b)}.$$

Low frequencies $1 \leq s \leq n$. Here $b \geq \pi/4$, so $b/(1+b) \geq (\pi/4)/(1+\pi/4) > \frac{1}{3}$ and $2b/\{(1+3b)(1+5b)\} \geq 2b/\{(3+4/\pi)b \cdot (5+4/\pi)b\} \gg 1/b \gg s/n$. With (83),

$$1 - |\psi_n(s)|^2 \gg \frac{s}{n}, \quad 1 \leq s \leq n, \quad (85)$$

and therefore, for some $\kappa > 0$ independent of n ,

$$|\phi_n(s)| = (|\psi_n(s)|^2)^{n/2} \leq \left(1 - \frac{\kappa s}{n}\right)^{n/2} \leq e^{-\kappa s/2} \quad (1 \leq s \leq n). \quad (86)$$

Intermediate frequencies $n \leq s \leq 8(c+n)$. Now $b \leq \pi/4$, so $b/(1+b) \gg b$ and $2b/\{(1+3b)(1+5b)\} \gg b$, whence $1 - |\psi_n(s)|^2 \gg b^2 \gg n^2/s^2$. Since $s = O(n)$ throughout this range,

$$|\phi_n(s)| \leq e^{-\kappa' n}. \quad (87)$$

High frequencies. Integration by parts in

$$\psi_n(s) = \int_0^{1/c} e^{-isv} \frac{c+n}{(1+nv)^2} dv$$

gives

$$|\psi_n(s)| \leq \frac{2(c+n)}{s}. \quad (88)$$

Thus, for $s \geq 8(c+n)$,

$$|\phi_n(s)| \leq \left(\frac{1}{4}\right)^n \left(\frac{8(c+n)}{s}\right)^n \leq \left(\frac{1}{4}\right)^n, \quad (89)$$

and more precisely $|\phi_n(s)| \leq (2(c+n)/s)^n$, whose integral over $s \geq 8(c+n)$ is $O((c+n)4^{-n})$, hence uniformly bounded.

Combining (86), (87), and (89), together with the trivial bound $|\phi_n(s)| \leq 1$ for $|s| \leq 1$, proves (80). Fourier inversion then gives

$$\|f_n\|_\infty \leq \frac{1}{2\pi} \|\phi_n\|_{L^1},$$

which proves (81); (82) follows immediately. $\square$

Lemma 5 (Right-tail bound and logarithmic uniform integrability). *For fixed a , the random variables*

$$\log X_n \mathbf{1}_{\{X_n > 0\}} \quad \text{and} \quad (\log X_n)^2 \mathbf{1}_{\{X_n > 0\}}$$

are uniformly integrable. Consequently, writing

$$P_n := \mathbb{P}(X_n > 0), \quad (90)$$

$$L_n := \mathbb{E}[\log X_n; X_n > 0], \quad (91)$$

$$Q_n := \mathbb{E}[(\log X_n)^2; X_n > 0], \quad (92)$$

we have

$$P_n \longrightarrow P_a := \mathbb{P}(X_a > 0) > 0, \quad (93)$$

$$L_n \longrightarrow L_a := \mathbb{E}[\log X_a; X_a > 0], \quad (94)$$

$$Q_n \longrightarrow Q_a := \mathbb{E}[(\log X_a)^2; X_a > 0]. \quad (95)$$

Proof. The negative logarithmic tail is controlled by Lemma 4. For $y \geq 0$,

$$\mathbb{P}(0 < X_n < e^{-y}) \ll e^{-y}.$$

Hence the negative parts of $\log X_n$ have uniformly bounded moments of every fixed order.

For the positive tail, note from (69) that

$$X_n > x \iff \sum_{j=1}^n V_j < \frac{1}{c} - x.$$

For $\lambda > 0$, Chernoff's inequality gives

$$\mathbb{P}(X_n > x) \leq \exp\left(\lambda\left(\frac{1}{c} - x\right)\right) (\mathbb{E}e^{-\lambda V})^n. \quad (96)$$

We claim that there is a constant C_a , independent of n and λ , such that

$$n(1 - \mathbb{E}e^{-\lambda V}) \geq \lambda \left(\log \frac{n}{\lambda} - C_a\right), \quad 1 \leq \lambda \leq n. \quad (97)$$

Substituting $u = nv$ in (68) and writing $\beta := \lambda/n \in (0, 1]$,

$$n(1 - \mathbb{E}e^{-\lambda V}) = (c+n) \int_0^{n/c} (1 - e^{-\beta u}) \frac{du}{(1+u)^2}. \quad (98)$$

Rather than splitting the range, evaluate the integral exactly. Integration by parts gives, for the untruncated integral,

$$\int_0^\infty (1 - e^{-\beta u}) \frac{du}{(1+u)^2} = \beta \int_0^\infty \frac{e^{-\beta u}}{1+u} du = \beta e^\beta E_1(\beta).$$

To make the estimate uniform over the entire range $0 < \beta \leq 1$, note that $e^\beta E_1(\beta) - \log(1/\beta)$ is continuous on $(0, 1]$ and tends to $-\gamma_E$ as $\beta \downarrow 0$. Consequently,

$$\sup_{0 < \beta \leq 1} |e^\beta E_1(\beta) - \log(1/\beta)| < \infty.$$

The discarded tail satisfies $\int_{n/c}^\infty (1+u)^{-2} du \leq c/n$, contributing at most $(c+n)c/n = O(1) \leq O(\lambda)$ since $\lambda \geq 1$. Multiplying by $(c+n) = n(1+c/n)$ and using $\beta = \lambda/n$, with $(c/n) \log(n/\lambda) \leq (c/n) \log n = O(n^{-1})$, yields

$$n(1 - \mathbb{E}e^{-\lambda V}) = \lambda \left(\log \frac{n}{\lambda} - \gamma_E + O(1)\right),$$

which is (97). (The coefficient 1 in front of $\log(n/\lambda)$ is essential: it is what cancels $1/c = \log n + a$ in (96). A cruder split using $1 - e^{-y} \geq y/2$ produces the coefficient $1/2$ and the argument fails.)

Since $\log z \leq z - 1$ for $z > 0$,

$$n \log \mathbb{E}e^{-\lambda V} \leq -n(1 - \mathbb{E}e^{-\lambda V}).$$

Using $c^{-1} = \log n + a$ in (96) therefore gives

$$\mathbb{P}(X_n > x) \leq \exp\{\lambda(\log \lambda + A_a - x)\}, \quad 1 \leq \lambda \leq n, \quad (99)$$

with $A_a := a + C_a$, enlarging C_a if necessary so that $C_a \geq 0$.

Choosing

$$\lambda = e^{x - A_a - 1}$$

whenever this lies in $[1, n]$ makes the exponent equal to $-\lambda$, giving the double-exponential bound

$$\mathbb{P}(X_n > x) \leq \exp\{-e^{x - A_a - 1}\}, \quad A_a + 1 \leq x \leq \log n + A_a + 1. \quad (100)$$

For $x < A_a + 1$ the trivial bound $\mathbb{P} \leq 1$ suffices, and for $x > \log n + A_a + 1$ we have $x > 1/c$ (as $C_a > -1$), so $\mathbb{P}(X_n > x) = 0$ because $X_n \leq 1/c$ almost surely.

Thus both the positive and negative logarithmic tails are uniformly integrable. Combining this with Lemma 3, and noting that the limiting stable law has a continuous positive density, so that $\mathbb{P}(X_a = 0) = 0$ and $P_a > 0$, gives (93)–(95). $\square$

6.3 Asymptotics of the Rayleigh quotient

We now return to the exact Rayleigh quotient.

Since $1 - S_n = cX_n$, equations (64) and (65) give, on $\{X_n > 0\}$,

$$G(1 - S_n) = \frac{1}{n} \log(1 + nX_n), \quad (101)$$

$$\frac{H(1 - S_n)}{m_0} = \frac{(c + n)X_n}{1 + nX_n}. \quad (102)$$

Therefore

$$R(F_{k,c}) = c \left(1 + \frac{1}{n}\right) \left(1 + \frac{c}{n}\right) \frac{\mathbb{E}[\log^2(1 + nX_n); X_n > 0]}{\mathbb{E}[\frac{(c+n)X_n}{1+nX_n}; X_n > 0]}. \quad (103)$$

Let

$$\ell := \log n, \quad c = \frac{1}{\ell + a}.$$

Lemma 6 (Denominator asymptotics). *For fixed a ,*

$$\mathbb{E}\left[\frac{(c+n)X_n}{1+nX_n}; X_n > 0\right] = P_n + O(n^{-1/2}). \quad (104)$$

Remark 7. The explicit rate, rather than a bare $o(1)$, is what the proof of Theorem 9 requires: the denominator is divided into a numerator of size ℓ^2 , so an additive error η here propagates to an error of size $\ell\eta$ in R . An $o(1)$ statement would only give $o(\ell)$, which is insufficient.

Proof. For $0 < X_n \leq 1/c$,

$$0 \leq 1 - \frac{(c+n)X_n}{1+nX_n} = \frac{1 - cX_n}{1+nX_n} \leq \frac{1}{1+nX_n}.$$

Choose

$$\delta_n = n^{-1/2}.$$

On $0 < X_n \leq \delta_n$, Lemma 4 gives probability $O(\delta_n)$. On $X_n > \delta_n$,

$$\frac{1}{1+nX_n} \leq \frac{1}{n\delta_n} = n^{-1/2}.$$

Hence the difference between the left-hand side of (104) and P_n is $O(n^{-1/2})$. $\square$

Lemma 8 (Numerator asymptotics). *For fixed a ,*

$$\mathbb{E}[\log^2(1 + nX_n); X_n > 0] = \ell^2 P_n + 2\ell L_n + Q_n + O(\ell^2 n^{-1/2}). \quad (105)$$

Proof. Let $\delta_n = n^{-1/2}$.

Region $X_n > \delta_n$. Here

$$\log(1 + nX_n) = \ell + \log X_n + r_n, \quad 0 \leq r_n = \log\left(1 + \frac{1}{nX_n}\right) \leq \frac{1}{nX_n} \leq n^{-1/2},$$

so that

$$\log^2(1 + nX_n) = (\ell + \log X_n)^2 + 2(\ell + \log X_n)r_n + r_n^2.$$

By Lemma 5 the quantities $\mathbb{E}[|\log X_n|; X_n > 0]$ are bounded uniformly in n , so

$$\mathbb{E}[|2(\ell + \log X_n)r_n| + r_n^2; X_n > \delta_n] = O(\ell n^{-1/2}).$$

Region $0 < X_n \leq \delta_n$. The uniform density bound of Lemma 4 gives

$$\begin{aligned} \mathbb{E}[(\ell + \log X_n)^2; 0 < X_n \leq \delta_n] &\leq C \int_0^{\delta_n} (\ell + \log x)^2 dx \\ &= C\delta_n [(\ell + \log \delta_n)^2 - 2(\ell + \log \delta_n) + 2] = O(\ell^2 n^{-1/2}), \end{aligned} \quad (106)$$

using $\ell + \log \delta_n = \ell/2$. Likewise, since $nX_n \leq n\delta_n = n^{1/2}$ on this region,

$$\mathbb{E}[\log^2(1 + nX_n); 0 < X_n \leq \delta_n] \leq C \int_0^{\delta_n} \log^2(1 + nx) dx = O(\ell^2 n^{-1/2}).$$

Adding the two regions and recognising $\mathbb{E}[(\ell + \log X_n)^2; X_n > 0] = \ell^2 P_n + 2\ell L_n + Q_n$ proves (105). $\square$

6.4 Main asymptotic theorem

Theorem 9 (Asymptotic defect of the rational trial family). *Fix $a \in \mathbb{R}$. For all sufficiently large k , put*

$$n = k - 1, \quad c_k = \frac{1}{\log n + a}.$$

For all sufficiently large k , $c_k > 0$. Let F_{k,c_k} be the single-channel rational trial function (60). Then

$$R(F_{k,c_k}) = \log k - \mathcal{C}(a) + o(1), \quad k \rightarrow \infty, \quad (107)$$

where

$$\boxed{\mathcal{C}(a) = a - 2 \mathbb{E}[\log X_a \mid X_a > 0]} \quad (108)$$

and

$$X_a = (a + 1) - \Lambda$$

with Λ defined by (70).

Equivalently,

$$\lim_{k \rightarrow \infty} (\log k - R(F_{k,c_k})) = \mathcal{C}(a). \quad (109)$$

Proof. Write Num and Den for the numerator and denominator in (103). By Lemmas 6 and 8,

$$\text{Num} = \ell^2 P_n + 2\ell L_n + Q_n + O(\ell^2 n^{-1/2}) = O(\ell^2), \quad \text{Den} = P_n + O(n^{-1/2}),$$

and $P_n \rightarrow P_a > 0$ by Lemma 5, so Den is bounded away from 0 for large n . Hence

$$\frac{\text{Num}}{\text{Den}} = \frac{\ell^2 P_n + 2\ell L_n + Q_n}{P_n} + O(\ell^2 n^{-1/2}).$$

Multiplying by $c(1 + \frac{1}{n})(1 + \frac{c}{n}) = c(1 + O(n^{-1}))$ and using $c = (\ell + a)^{-1}$,

$$R(F_{k,c}) = \frac{\ell^2 + 2\ell L_n/P_n + Q_n/P_n}{\ell + a} + O(\ell n^{-1/2}). \quad (110)$$

A direct rearrangement yields the finite- n expansion

$$\begin{aligned} \ell - R(F_{k,c}) &= a - 2 \frac{L_n}{P_n} \\ &\quad - \frac{Q_n/P_n - 2aL_n/P_n + a^2}{\ell + a} + O(\ell n^{-1/2}). \end{aligned} \quad (111)$$

By Lemma 5,

$$\frac{L_n}{P_n} \longrightarrow \frac{L_a}{P_a} = \mathbb{E}[\log X_a \mid X_a > 0],$$

while the fraction on the second line of (111) tends to zero. Hence

$$\ell - R(F_{k,c}) \longrightarrow a - 2\mathbb{E}[\log X_a \mid X_a > 0].$$

Finally,

$$\log k - \log(k - 1) = O(n^{-1}),$$

so ℓ may be replaced by $\log k$. This proves (107) and (109). $\square$

6.5 Consequence for M_k

Define

$$\mathcal{C}_* := \inf_{a \in \mathbb{R}} \mathcal{C}(a). \quad (112)$$

Corollary 10 (Asymptotic lower bound for M_k). *The Maynard variational constant satisfies*

$$\liminf_{k \rightarrow \infty} (M_k - \log k) \geq -\mathcal{C}_*. \quad (113)$$

Equivalently,

$$M_k \geq \log k - \mathcal{C}_* - o(1). \quad (114)$$

Proof. For every $\varepsilon > 0$, choose a fixed $a_\varepsilon \in \mathbb{R}$ such that

$$\mathcal{C}(a_\varepsilon) \leq \mathcal{C}_* + \varepsilon.$$

Since M_k is the supremum over admissible trial functions, Theorem 9 gives

$$M_k \geq R(F_{k,c_k}) = \log k - \mathcal{C}(a_\varepsilon) + o(1)$$

for the choice

$$c_k^{-1} = \log(k-1) + a_\varepsilon.$$

Thus

$$\liminf_{k \rightarrow \infty} (M_k - \log k) \geq -\mathcal{C}_* - \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves (113). $\square$

The infimum in (112) may be restricted to a half-line bounded above, since $\mathcal{C}$ grows without bound in the positive direction.

Lemma 11 (Growth of $\mathcal{C}$ for large positive shift). *With $\kappa = \mathbb{E}[(-\Lambda)^+] < \infty$ as in (73), there is a_0 such that for all $a \geq a_0$,*

$$\mathcal{C}(a) \geq a - 2 \log(2(a+1+\kappa)) \xrightarrow{a \rightarrow \infty} \infty. \quad (115)$$

In particular $\inf_{a \in \mathbb{R}} \mathcal{C}(a) = \inf_{a \leq a_1} \mathcal{C}(a)$ for some finite a_1 .

Proof. Since $X_a^+ = ((a+1) - \Lambda)^+ \leq (a+1) + (-\Lambda)^+$ for $a \geq -1$,

$$\mathbb{E}[X_a; X_a > 0] \leq (a+1) + \kappa.$$

Also $P_a = \mathbb{P}(\Lambda < a+1) \rightarrow 1$, so $P_a \geq \frac{1}{2}$ for $a \geq a_0$. By Jensen's inequality applied to the concave function $\log$ under the conditional law,

$$\mathbb{E}[\log X_a \mid X_a > 0] \leq \log \mathbb{E}[X_a \mid X_a > 0] = \log \frac{\mathbb{E}[X_a; X_a > 0]}{P_a} \leq \log(2(a+1+\kappa)),$$

which gives (115). $\square$

Remark 12 (Behaviour as $a \rightarrow -\infty$, and attainment of the infimum). Numerical results suggest that $\mathcal{C}(a) \rightarrow \infty$ as $a \rightarrow -\infty$ as well, and that the infimum in (112) is attained. These assertions are not proved here. The proposed mechanism is that conditioning on $\{X_a > 0\} = \{\Lambda < a+1\}$ for very negative a forces X_a to concentrate at 0: writing $y = -(a+1)$, the ratio $\mathbb{P}(\Lambda < -y-t)/\mathbb{P}(\Lambda < -y)$ is expected to behave like $\exp\{-c e^y t\}$ for small $t > 0$, so that conditionally X_a is approximately exponential with rate of order $e^{-(a+1)}$, giving $\mathbb{E}[\log X_a \mid X_a > 0] = (a+1) + O(1)$ and hence $\mathcal{C}(a) = -a + O(1)$. A rigorous proof requires a two-sided left-tail estimate for Λ ; Lemma 2 supplies only the upper bound (72), and the Jensen argument of Lemma 11 is too lossy in this regime because P_a is doubly-exponentially small. We therefore state (112) as an infimum; none of the results below depend on attainment.

Remark 13 (Scope of the optimization). The quantity $\mathcal{C}_*$ is the optimal asymptotic constant *among the fixed-shift scalings*

$$c_k^{-1} = \log(k-1) + a, \quad a \in \mathbb{R}.$$

The theorem does not by itself prove that the same constant is optimal over every possible sequence $c_k > 0$ in the full single-channel rational family. Such a statement would additionally require showing that every asymptotically competitive sequence satisfies

$$c_k^{-1} - \log k = O(1),$$

and controlling non-convergent or divergent shift sequences. We do not need this stronger statement for the lower bound (114).

6.6 Numerical interpretation

Theorem 9 is an analytic statement and does not depend on the numerical discovery or certification pipeline. Numerical quadrature of the limiting stable law can, however, be used to evaluate $\mathcal{C}(a)$ and to locate a favorable fixed shift. Two independent routes are available: direct quadrature against the density of Λ , and inversion of the explicit Laplace transform (71). Our numerical evaluation indicates a minimum near a small negative value of a and an asymptotic defect of approximately

$$\mathcal{C}_* \approx 0.3343. \quad (116)$$

The value in (116) is a numerical estimate of the infimum, not a certified enclosure established by this asymptotic analysis. Certifying a finite- k Rayleigh quotient and certifying the limiting constant are separate tasks.

For an explicit asymptotic lower bound it suffices to choose a single fixed shift a_0 and establish, by rigorous interval quadrature with controlled truncation errors, an upper bound $\mathcal{C}(a_0) \leq U$. Theorem 9 then gives

$$\liminf_{k \rightarrow \infty} (M_k - \log k) \geq -U,$$

without any proof of global minimality. By contrast, a rigorous enclosure of $\mathcal{C}_*$ itself also requires a lower bound valid for every $a \in \mathbb{R}$. Any numerical search restricted to a finite interval must therefore be supplemented by rigorous control of both excluded tails, including $a \rightarrow -\infty$. The upper bound on the left tail of Λ proved above does not by itself supply that control.

The finite- k optimization exhibits the same qualitative scaling:

$$c_k \sim \frac{1}{\log k}, \quad R(F_{k,c_k}) = \log k - O(1).$$

The fact that the discrepancy $\log k - R(F_{k,c_k})$ varies only slowly over more than five orders of magnitude in k is therefore explained by the stable-law limit rather than being an empirical coincidence.

6.7 Finite-size corrections

Expansion (111) also isolates the finite-size correction. All error terms accumulated in its derivation are $O(\ell n^{-1/2})$, hence $o(\ell^{-m})$ for every fixed m ; the replacement of $\log n$ by $\log k$ contributes $O(n^{-1})$. Consequently

$$\log k - R(F_{k,c}) = a - 2 \frac{L_n}{P_n} - \frac{Q_n/P_n - 2aL_n/P_n + a^2}{\ell + a} + O(\ell n^{-1/2}), \quad (117)$$

Define

$$d_n(a) := \frac{L_n}{P_n} - \frac{L_a}{P_a}, \quad B(a) := \frac{Q_a}{P_a} - 2a \frac{L_a}{P_a} + a^2.$$

Since $P_n \rightarrow P_a > 0$, $L_n \rightarrow L_a$, and $Q_n \rightarrow Q_a$, the expansion above implies

$$\log k - R(F_{k,c}) = \mathcal{C}(a) - 2d_n(a) - \frac{B(a)}{\ell} + o(\ell^{-1}). \quad (118)$$

Here $d_n(a) \rightarrow 0$, but no quantitative rate for this convergence has been proved. Thus $d_n(a)$ may dominate the displayed $1/\ell$ term, and (118) is not yet a two-term asymptotic expansion with a determined coefficient.

A sufficient additional estimate is

$$d_n(a) = o(\ell^{-1}),$$

in which case

$$\log k - R(F_{k,c}) = \mathcal{C}(a) - \frac{B(a)}{\log k} + o((\log k)^{-1}).$$

More generally, if $\ell d_n(a) \rightarrow b(a)$, the coefficient of $1/\log k$ is $-[B(a) + 2b(a)]$. A bound of only $d_n(a) = O(\ell^{-1})$ does not determine that coefficient. Accordingly, a quantitative stable-limit argument must establish an appropriate rate for these conditional logarithmic moments, or their first-order asymptotics; a distributional convergence rate alone requires an additional argument to handle the logarithmic singularity. Any fitted coefficient of $1/\log k$ remains a numerical finite-size observation until such an estimate is supplied.

6.8 Interpretation

The asymptotic mechanism can be summarized as follows. Squaring the single rational channel induces the one-dimensional density

$$q_n(v) = \frac{c + n}{(1 + nv)^2},$$

whose centered n -fold sum converges to a spectrally positive 1-stable law. Spectral positivity refers to the jumps; the limiting law has support on all of $\mathbb{R}$. Choosing

$$c^{-1} = \log n + a$$

centers the boundary variable

$$X_n = \frac{1}{c} - \sum_{j=1}^n V_j$$

at a non-degenerate limit. Conditional on the event $X_n > 0$, which is exactly the simplex event $S_n < 1$, the numerator of the Rayleigh quotient contains

$$\log^2(1 + nX_n) = (\log n + \log X_n + o(1))^2,$$

whereas the denominator converges to the same survival probability. The leading $\log^2 n$ term is therefore divided by $c^{-1} \sim \log n$, producing a Rayleigh quotient of size $\log n$. The conditional logarithmic moment of the limiting stable variable supplies the non-trivial $O(1)$ defect.

Thus the rational family is not merely a finite-dimensional numerical ansatz that happens to perform well at large k . Its observed scaling is the finite- k manifestation of a stable-law limit, and the family gives the explicit asymptotic lower bound

$$M_k \geq \log k - \mathcal{C}_* - o(1).$$

7 Supplementary Methods: higher-order Delsarte hierarchy

Notation follows the Methods. Throughout, $H(\cdot)$ denotes Shannon entropy in bits and $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ the binary entropy function; G is a normalized configuration on $\mathbb{F}_2^\ell$, i.e. a probability distribution, and $\Phi(G, v) = \sum_u \sqrt{G(u)G(u+v)}$ is the translation affinity at a shift $v \neq 0$.

7.1 Scale-up of the hierarchy to $r \geq 3$

The finite-length hierarchy was first implemented at $r = 2$, where $\text{GL}(2, 2) \cong S_3$ acts as the full symmetric group on the three non-zero vectors of $\mathbb{F}_2^2$ and orbit representatives can be generated by sorting. This shortcut cannot be generalized to $r = 3$: $|\text{GL}(3, 2)| = 168$ whereas $|S_7| = 5040$. Sorting therefore over-merges genuinely distinct Fano-plane orbits; at $n = 6$, 45 true orbits collapse to 30 sorted classes. We instead enumerate $\text{GL}(r, 2)$ explicitly and construct genuine type orbits under the group action.

For $S \subset [r]$, partial Fourier transforms factor after splitting $\mathbb{F}_2^r = \mathbb{F}_2^S \oplus \mathbb{F}_2^{S^c}$:

$$K_\alpha^S(\beta) = \prod_{w \in \mathbb{F}_2^{S^c}} K_{\alpha(\cdot, w)}^{(|S|)}(\beta(\cdot, w)),$$

with zero value unless the S^c -marginals of α and β agree. Each factor is a lower-dimensional full Krawtchouk transform, so the implementation recurses through the same exact row generator. Counting pairs of matrices of types (α, β) in the two possible orders gives the reciprocity

$$\binom{n}{\beta} K_\alpha(\beta) = \binom{n}{\alpha} K_\beta(\alpha).$$

Combined with $\text{GL}(r, 2)$ equivariance, this allows a row to be generated from one representative expansion per valid orbit. At $r = 3$, $n = 14$, $d = 6$, this reduces the number of expansions from 1235 to 112.

The resulting $r = 3$ implementation reproduced the $r = 2$ program when run at $r = 2$ and produced certified strict level-3 separations, including $V_2(9, 5) = 44/9 > V_3(9, 5) = 4$ and $V_2(12, 5) = 24.260255 \dots > V_3(12, 5) = 16$.

7.2 Exact dual certification and implementation checks

The floating-point solver is used as a proposal mechanism for an active dual set. The corresponding basic system is reconstructed at arbitrary precision, rounded to dyadic rationals, and verified against every dual inequality. The repair identity quoted in the Methods supplies an exactly feasible direction and an explicit rational repair cost, so the certified value is an exact rational upper bound on the level- r optimum obtained by weak duality.

Three independent anchors were used. First, full and partial Krawtchouk rows were compared with brute-force character sums over $\mathbb{F}_2^{r \times n}$ for small n . Second, $r = 1$ was checked against an independent classical Delsarte implementation. Third, the published $r = 2$ values of Loyfer and Linial were reproduced, including 24.260255... at (13, 6), 131.720587... at (16, 6), 32 at (16, 8) and (17, 8), and 256 at (17, 6) and (20, 8).

Structural invariant tests detected defects that value reproduction did not. An early active-set implementation factorized a matrix shifted by one column relative to the solved system, yielding 32.0000887... instead of 32 at (17, 8). Two precision-state errors in dyadic rounding and polishing partially masked this defect. These issues were corrected before the experiments reported here; all final bounds use the repaired exact-verification path.

7.3 Lift experiments and finite-length structural map

Coregliano et al. construct an explicit lift of a level-one dual to level ℓ with objective V_1^ℓ [6]. Because the level- ℓ hierarchy bounds $|C|^\ell$, the lift must be compared in this normalization. An initial level-one-normalized comparison reproduced V_1 but violated higher-level feasibility by +9949, which served as a normalization control before the systematic sweep.

The lift has no mass on partial families below the full transform. We therefore compared the full-Fourier-only hierarchy with the complete partial-Fourier program. Across twenty tested instances the full-only optimum never exceeded V_1^r . At $r = 2$ it was frequently exactly equal to the lift, for example

$$36 = 6^2 \quad (9, 5), \quad 576 = 24^2 \quad (11, 5), \quad 144 = 12^2 \quad (10, 5),$$

while the partial-Fourier program improved strictly. At $r = 3$, by contrast, the full-only hierarchy itself generally improved on the lift. The correction to a lifted level-one solution is therefore not a common object across levels.

A separate scan at approximately fixed relative distance showed why these finite instances were not used for asymptotic learning. At $\delta = 0.40$, the accessible finite- n rates were 0.33–0.50, compared with the first-MRRW value 0.0815, and the level-one-to-level-two gain decreased with n at fixed d . Extrapolation from the accessible n range would therefore have mixed strong finite-size effects with the asymptotic target.

7.4 Normalization of the configuration objective

For a level- ℓ dual,

$$|C|^\ell \leq f(0),$$

so an asymptotic bound obtained from a normalized configuration G is

$$R(C) \leq \frac{H(G)}{\ell} + O\left(\frac{\log n}{n}\right),$$

and $H(G)$ must not be compared directly with an ordinary code-rate bound. Re-deriving this normalization exposed two discrepancies in the cited preprint. First, the leading small- ε coefficient of the first MRRW expression is $\frac{1}{2}\varepsilon^2 \log_2(1/\varepsilon)$ rather than $\frac{1}{4}\varepsilon^2 \log_2(1/\varepsilon)$: with $p_\varepsilon = \frac{\varepsilon^2}{4} + O(\varepsilon^4)$ and $\log_2(1/p_\varepsilon) = 2 \log_2(1/\varepsilon) + 2 + O(\varepsilon^2)$, one gets $h(\varepsilon) = \frac{1}{2}\varepsilon^2 \log_2(1/\varepsilon) + O(\varepsilon^2)$. Second, the closed form stated in their Lemma 6.10(i) is a lower bound on the walk count rather than an equality, since its proof passes through $\left(\frac{m}{F}\right)^2 \geq \binom{m}{F}$. Neither affects the qualitative conclusion of that work: the stated rate bound is off by a factor 2ℓ relative to $H(G)/\ell$ — the ℓ being the rate normalization above and the 2 the same halving — but the factor is applied on both sides of the comparison made there.

7.5 Reduced configuration search

With $G = p^2$, the configuration problem becomes

$$\min_{\|p\|_2=1, p \geq 0} \frac{1}{\ell} H(p^2)$$

subject to quadratic translation-affinity constraints $p^\top S_v p \geq \varepsilon$. The largest search space considered ($\ell = 4$) has 15 free probability coordinates. We therefore used direct multistart sequential quadratic programming with analytic gradients rather than a neural network. The campaign comprised 28 instance solves over $\ell \leq 4$ and $\varepsilon \in \{0.30, 0.20, 0.10, 0.05\}$, using 400–600 randomized starts per instance (13,600 local optimizations in total). One start per instance used the CJJ vertex-uniform configuration and the remaining starts were randomized positive-sphere points with varying concentration.

No run produced a configuration below the first MRRW benchmark. Recovered objectives differed from MRRW by between -3.2×10^{-14} and -2.1×10^{-16} , consistent with floating-point noise, and the recovered configurations agreed coordinatewise with the tensor-product quasirandom family to within 10^{-11} .

7.6 The tensorization obstruction

7.6.1 Setting and hypotheses

Let G be a normalized configuration on $\mathbb{F}_2^\ell$, that is, $G(u) \geq 0$ and $\sum_u G(u) = 1$. Write $H(G)$ for its Shannon entropy in bits, H_2 for binary entropy, and

$$\Phi(G, v) = \sum_u \sqrt{G(u)G(u+v)}$$

for its translation affinity. By Cauchy–Schwarz, $0 \leq \Phi(G, v) \leq 1$.

For an even integer $m \geq 2$ and $v \neq 0$, the *paired walk weight* is

$$W_m(G, v) = \sum_F \frac{m!}{\prod_u F(u)!} \prod_u G(u)^{F(u)}, \quad (119)$$

where the sum runs over $F : \mathbb{F}_2^\ell \rightarrow \mathbb{Z}_{\geq 0}$ with $\sum_u F(u) = m$ and $F(u) = F(u+v)$ for all u , and $0^0 = 1$. For fixed G, v, m , this is the leading coefficient in the configuration asymptotics of [6]:

$$A_v^m \Lambda_{n,G}(X) = n^m W_m(G, v) + o(n^m), \quad X \in \text{config}_{n,\ell}^{-1}(nG). \quad (120)$$

Here A_v is the adjacency operator of the column-translation graph and $\Lambda_{n,G}$ is the indicator of the configuration fibre. The combinatorial definition (119) applies to every probability distribution G ; no limit in n is needed for Lemmas 14–17 or Theorem 18.

To interpret (120) for a fixed G , let

$$N_G = \{n \in \mathbb{N} : nG \in \text{Config}_{n,\ell}\}$$

and assume that N_G is infinite. This holds whenever G has rational entries. Along $n \rightarrow \infty$ in N_G , every positive entry satisfies $nG(u) = \Omega(n)$, as required by the configuration asymptotics. This lattice condition is needed only for the connection to finite configuration fibres, not for the entropy inequality below.

Fix $\varepsilon \in (0, 1)$. We isolate the following two hypotheses:

- (A1) *Spanning*. The set $V \subseteq \mathbb{F}_2^\ell \setminus \{0\}$ of tested shifts spans $\mathbb{F}_2^\ell$.
- (A2) *Finite-walk lower bound*. For every $v \in V$ there exist an even integer $m_v \geq 2$ and a constant $c_v \geq 1$ such that

$$W_{m_v}(G, v) \geq c_v \varepsilon^{m_v}.$$

The orders m_v and constants c_v are fixed independently of n .

The spanning condition is equivalent to requiring that, for every $i \neq 0$, some $v \in V$ satisfies $\langle i, v \rangle = 1$: failure to span is equivalent to containment in $i^\perp$ for some non-zero i . Condition (25) of [6] is a sufficient finite- n condition for the cited spectral construction. When it holds for a fixed G and a fixed even walk order along infinitely many admissible blocklengths, (120) implies (A1)–(A2) with $c_v = 2^{2^\ell - 1}$. Indeed, there are only finitely many possible choices of witnessing shifts, so they can be held fixed on an infinite subsequence; division by n^m and passage to the limit then gives (A2). We use only $c_v \geq 1$. Conversely, (A2) alone does not assert finite- n feasibility of the spectral construction.

7.6.2 The obstruction

Lemma 14 (Walk-affinity comparison). *For every normalized configuration G , every $v \neq 0$ and every even integer $m \geq 2$,*

$$W_m(G, v) \leq \Phi(G, v)^m.$$

The inequality is strict whenever $\Phi(G, v) > 0$.

Proof. Translation by $v \neq 0$ is a fixed-point-free involution of $\mathbb{F}_2^\ell$, so its orbits are the $2^{\ell-1}$ unordered pairs $P = \{u, u+v\}$. A multiplicity function F admissible in (119) is determined by $a_P := F(u) = F(u+v)$, with $\sum_P a_P = m/2$. Writing $x_P = G(u)G(u+v) \geq 0$ gives

$$W_m(G, v) = \sum_{\sum_P a_P = m/2} \frac{m!}{\prod_P (a_P!)^2} \prod_P x_P^{a_P}. \quad (121)$$

Both elements of P contribute $\sqrt{x_P}$ to $\Phi(G, v)$, so $\Phi(G, v) = 2 \sum_P \sqrt{x_P}$ and

$$\Phi(G, v)^m = 2^m \sum_{\sum_P c_P = m} \frac{m!}{\prod_P c_P!} \prod_P x_P^{c_P/2}. \quad (122)$$

Every term of (122) is non-negative. Discard those with some c_P odd and match the remaining terms to (121) by $c_P = 2a_P$. The ratio of the corresponding coefficients is

$$\frac{m! / \prod_P (a_P!)^2}{2^m m! / \prod_P (2a_P)!} = \prod_P \frac{\binom{2a_P}{a_P}}{4^{a_P}} \leq 1.$$

In fact this ratio is strictly less than one, since $m \geq 2$ implies that some $a_P \geq 1$, and $\binom{2a}{a} < 4^a$ for $a \geq 1$. If $\Phi(G, v) > 0$, some $x_P > 0$; assigning $a_P = m/2$ to this pair and zero to the others gives a positive matched term. Summing therefore proves strict inequality in this case. If $\Phi(G, v) = 0$, both sides vanish. $\square$

Lemma 15 (Affinity floor). *Under (A2), $\Phi(G, v) > \varepsilon$ for every $v \in V$. In particular, $\Phi(G, v) \geq \varepsilon$.*

Proof. Hypothesis (A2) gives $W_{m_v}(G, v) \geq c_v \varepsilon^{m_v} > 0$, so $\Phi(G, v) > 0$. The strict part of Lemma 14 yields

$$\Phi(G, v)^{m_v} > W_{m_v}(G, v) \geq c_v \varepsilon^{m_v} \geq \varepsilon^{m_v}.$$

Taking m_v -th roots proves the claim. No limit in m or n is used. $\square$

Lemma 16 (Entropy profile). *Let $p_\varepsilon = \frac{1}{2}(1 - \sqrt{1 - \varepsilon^2})$ and $h(\varepsilon) = H_2(p_\varepsilon)$ for $\varepsilon \in [0, 1]$. Then h is strictly increasing and strictly convex on $(0, 1)$, $h(0) = 0$, $h(1) = 1$, and*

$$h(\varepsilon) = H_2\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right), \quad \delta = \frac{1 - \varepsilon}{2}. \quad (123)$$

Proof. Put $s = \sqrt{1 - \varepsilon^2}$, so $p_\varepsilon = (1 - s)/2$ and $\delta(1 - \delta) = s^2/4$. Thus $\frac{1}{2} - \sqrt{\delta(1 - \delta)} = p_\varepsilon$. For $0 < \varepsilon < 1$, differentiation gives

$$H_2'(p_\varepsilon) = \frac{2 \operatorname{artanh}(s)}{\ln 2}, \quad \frac{dp_\varepsilon}{d\varepsilon} = \frac{\varepsilon}{2s},$$

and hence

$$h'(\varepsilon) = \frac{\varepsilon \operatorname{artanh}(s)}{s \ln 2} > 0, \quad h''(\varepsilon) = \frac{\operatorname{artanh}(s) - s}{s^3 \ln 2} > 0.$$

The second inequality follows from $\operatorname{artanh}(s) = s + s^3/3 + \dots > s$ for $0 < s < 1$; h'' extends continuously to $h''(1) = 1/(3 \ln 2)$. The endpoint values follow from $p_0 = 0$ and $p_1 = 1/2$. $\square$

Lemma 17 (Conditional affinity and Jensen). *Let $X = (X_1, \dots, X_\ell) \sim G$ and, for fixed i , write $Y = X_{-i}$ and $p_Y = \Pr(X_i = 1 \mid Y)$. Then*

$$\begin{aligned} \Phi(G, e_i) &= \mathbb{E}_Y \left[2\sqrt{p_Y(1 - p_Y)} \right], \\ \mathbb{H}(X_i \mid X_{-i}) &= \mathbb{E}_Y \left[h\left(2\sqrt{p_Y(1 - p_Y)}\right) \right]. \end{aligned} \quad (124)$$

Consequently, $\mathbb{H}(X_i \mid X_{-i}) \geq h(\Phi(G, e_i))$.

Proof. Identify u with (y, x_i) . For each y of positive probability, the two points $(y, 0)$ and $(y, 1)$ each contribute

$$\sqrt{G(y, 0)G(y, 1)} = \Pr(Y = y) \sqrt{p_y(1 - p_y)}$$

to $\Phi(G, e_i)$, giving the first identity. Values of p_y on zero-probability events may be chosen arbitrarily. For $b(p) := 2\sqrt{p(1 - p)}$, symmetry of binary entropy gives $H_2(p) = h(b(p))$ for every $p \in [0, 1]$. Averaging over Y gives the second identity. Jensen's inequality and Lemma 16 give the conclusion. $\square$

Theorem 18 (Tensorization obstruction). *Fix $\ell \geq 1$ and $\varepsilon \in (0, 1)$. Let G be a normalized configuration on $\mathbb{F}_2^\ell$, and suppose that a spanning set $V \subseteq \mathbb{F}_2^\ell \setminus \{0\}$ satisfies $\Phi(G, v) \geq \varepsilon$ for every $v \in V$. Then*

$$J_\ell(G) = \frac{\mathbb{H}(G)}{\ell} \geq h(\varepsilon) = H_2\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right), \quad \delta = \frac{1 - \varepsilon}{2}.$$

The right-hand side is the first MRRW expression. This bound is sharp for the relaxed problem with constraints $\Phi(G, e_i) \geq \varepsilon$, $1 \leq i \leq \ell$: $G_\varepsilon = \text{Bernoulli}(p_\varepsilon)^{\otimes \ell}$ attains equality in that problem. Under the stronger hypotheses (A1)–(A2), one has $J_\ell(G) > h(\varepsilon)$.

Proof. Choose a basis $v_1, \dots, v_\ell$ contained in V and an invertible linear map T with $Te_i = v_i$. Define $\tilde{G}(u) = G(Tu)$. Then

$$H(\tilde{G}) = H(G), \quad \Phi(\tilde{G}, e_i) = \Phi(G, v_i) \geq \varepsilon.$$

Let $X \sim \tilde{G}$. The entropy chain rule and conditioning give

$$H(G) = \sum_{i=1}^{\ell} H(X_i \mid X_1, \dots, X_{i-1}) \geq \sum_{i=1}^{\ell} H(X_i \mid X_{-i}) \geq \sum_{i=1}^{\ell} h(\Phi(\tilde{G}, e_i)) \geq \ell h(\varepsilon),$$

where the second inequality uses Lemma 17 and the last uses monotonicity of h . Division by ℓ proves the bound.

For sharpness of the relaxed problem, affinity is multiplicative over tensor products, so

$$\Phi(G_\varepsilon, e_i) = 2\sqrt{p_\varepsilon(1-p_\varepsilon)} = \varepsilon, \quad H(G_\varepsilon) = \ell H_2(p_\varepsilon) = \ell h(\varepsilon).$$

This proves attainment in the stated relaxation, not under (A2).

Finally, under (A1)–(A2), Lemma 15 gives $\Phi(G, v_i) > \varepsilon$ for every basis shift. Strict monotonicity of h makes the last inequality in the entropy chain strict, yielding $J_\ell(G) > h(\varepsilon)$. $\square$

Corollary 19. *Consider the entropy-based rate estimate*

$$R(C) \leq J_\ell(G) + O(\log n/n)$$

for the fixed single-orbit spectral construction (Section 7.4), with ℓ and G fixed and with its feasibility hypotheses satisfied. If G satisfies (A1)–(A2), then its leading term obeys $J_\ell(G) > h(\varepsilon)$. Consequently, minimizing this entropy-based estimate over configurations satisfying (A1)–(A2) cannot give an asymptotic rate bound below the first MRRW expression. The relaxed affinity-constrained minimum equals $h(\varepsilon)$, but this does not assert attainment by a configuration satisfying the finite-walk conditions.

Proof. The rate normalization follows because a level- ℓ dual bounds $|C|^\ell$. Apply Theorem 18 to the leading term $J_\ell(G)$ of the stated estimate. $\square$

7.6.3 Scope and relation to the cited construction

Remark 20 (Scope). The obstruction concerns the entropy objective $J_\ell(G)$ under a spanning set of affinity floors, and therefore under (A1)–(A2). It does not rule out every single-orbit spectral construction. In particular, it does not exclude feasibility mechanisms that bypass (A2), sharper estimates of a dual objective that are not controlled by this entropy bound, alternative sign functions ϕ , spectral functions supported on several configuration orbits, other uses of the partial-Fourier dual families, or general dual feasible solutions. The higher-order hierarchy itself remains complete.

Remark 21 (Sufficiency of the walk condition). Condition (25) of [6] is a sufficient finite- n condition obtained by making the terms of an odd-cardinality expansion of M_m separately non-negative. Hypothesis (A2) is its asymptotic consequence for fixed walk orders, as explained above; it is not an equivalent formulation of finite- n feasibility. Configurations that fail (A2) could in principle still give non-negative Fourier transforms through cancellation in that expansion. The present obstruction does not address such configurations.

Remark 22 (Finite- m admissibility). The equality configuration G_ε is feasible for the affinity relaxation, but not for (A2). Indeed, multiplicativity gives

$$\Phi(G_\varepsilon, v) = \varepsilon^{|v|} \leq \varepsilon \quad (v \neq 0),$$

where $|v|$ is Hamming weight. Lemma 14 therefore gives, for every even $m \geq 2$,

$$W_m(G_\varepsilon, v) < \Phi(G_\varepsilon, v)^m \leq \varepsilon^m.$$

This fails (A2) even with $c_v = 1$. The theorem asserts sharpness only of the affinity relaxation. Any claim that a sequence of feasible finite-walk constructions approaches this value requires a separate approximation argument; no such claim is needed for Corollary 19.

Remark 23 (Equality in the relaxation). Work in coordinates determined by a basis of the tested shifts. If equality holds in the relaxed entropy bound, then every step of the entropy chain in the proof must be an equality. Strict convexity of h and strict monotonicity at the affinity floor imply $2\sqrt{p_Y(1-p_Y)} = \varepsilon$ almost surely for every coordinate. Thus the conditional marginals lie in $\{p_\varepsilon, 1-p_\varepsilon\}$, and equality must also hold in the conditioning step. The tensor-product configuration explicitly satisfies these equalities and explains the value recovered in the relaxed numerical search. We make no uniqueness claim, and these equality statements do not imply finite-walk admissibility.

Remark 24 (Relation to the hypotheses of [6]). The general sufficient criterion in Theorem 6.7 of [6] requires a witnessing shift for each non-zero i with $\langle i, v \rangle = 1$; it does not restrict all witnessing shifts to Hamming weight one. Their explicit configuration examples verify the walk estimates at the standard basis shifts, which already span $\mathbb{F}_2^\ell$. The change of basis above accommodates the general spanning formulation. Their stated finite- n criterion also assumes strictly positive configuration entries, whereas the entropy inequality and walk-affinity comparison hold for all $G \geq 0$. Hence they also apply to configurations with proper support, provided the stated affinity or walk hypotheses hold. This extension is an entropy obstruction, not by itself a proof of feasibility for configurations with zero entries in the cited spectral construction.

7.7 Computational verification of the algebraic reductions

None of the computations in this section is used as part of the analytic proof. They test that the implementation computes the objects Theorem 18 concerns, which is where a hidden factor of $m!$, a 2^m , or an ordered-versus-unordered pair convention would appear.

The literal sum (119) over multiplicity functions was enumerated in exact rational arithmetic and matched the paired form (121) as an exact rational identity, not to a tolerance, for all tested $\ell \in \{1, 2, 3\}$, all $v \neq 0$ and all even $m \leq 8$.

All downstream computation takes place on the configuration lattice, whereas A_v acts on $\mathbb{F}_2^{\ell \times n}$. To confirm that this reduction loses nothing, A_v was constructed literally on the full vertex set and applied as a matrix, rather than assuming that $A_v^m \Lambda(X)$ depends on X only through $\text{config}(X)$. In every tested instance the result was constant on the entire fibre $\text{config}_{n,\ell}^{-1}(g_0)$ and agreed exactly with an independent configuration-lattice walk implementation (Supplementary Table 5). Blocklengths are necessarily small, the largest cases being $4^5 = 1024$ and $8^4 = 4096$ matrices; all $v \neq 0$ were tested and all agree, with $v = e_1$ shown for legibility.

ℓ	n	g_0	v	m	matrix values	config. walk	agree
2	4	(1, 1, 1, 1)	e_1	2	{8}	8	yes
2	4	(1, 1, 1, 1)	e_1	4	{128}	128	yes
2	4	(2, 1, 1, 0)	e_1	2	{8}	8	yes
2	4	(2, 1, 1, 0)	e_1	4	{104}	104	yes
2	5	(2, 1, 1, 1)	e_1	2	{11}	11	yes
2	5	(2, 1, 1, 1)	e_1	4	{245}	245	yes
3	4	(1, 1, 1, 1, 0, 0, 0, 0)	e_1	2	{8}	8	yes
3	4	(1, 1, 1, 1, 0, 0, 0, 0)	e_1	4	{128}	128	yes
3	4	(2, 1, 1, 0, 0, 0, 0, 0)	e_1	2	{8}	8	yes
3	4	(2, 1, 1, 0, 0, 0, 0, 0)	e_1	4	{104}	104	yes

Supplementary Table 5: The operator $A_v^m \Lambda$ computed literally on $\mathbb{F}_2^{\ell \times n}$, against the configuration-lattice walk count. “Matrix values” is the set of values taken by $A_v^m \Lambda(X)$ over the entire fibre $\text{config}_{n,\ell}^{-1}(g_0)$; that this set is a singleton in every row is precisely the assertion that the quantity depends on X only through $\text{config}(X)$. All entries are exact integers.

The finite- n walk count was compared with the leading term $n^m W_m(G, v)$ of (120). The ratios converge to 1, and do so from above, so the finite- n correction is positive:

$\ell = 2, m = 2 :$	$n = 8$	16	32	64	128
	1.6667	1.3333	1.1667	1.0833	1.0417
$\ell = 3, m = 2 :$	$n = 12$	24	48	96	192
	1.7500	1.3750	1.1875	1.0938	1.0469

with the same behaviour at $m = 4$ (for $\ell = 3$, $3.1067 \rightarrow 1.0998$ over the same sequence). The sign of this correction is not used anywhere in the argument, which requires only the $o(n^m)$ bound, but it is recorded because the direction is not obvious a priori.

The inequality of Lemma 14 was checked on 480 exact rational cases, with largest observed ratio 0.4957. The extremal value $\frac{1}{2}$ is attained already at $\ell = 1$, $m = 2$, where there is a single pair and a single admissible multiplicity, so the coefficient comparison reduces exactly to $\binom{2}{1}/4$.

The entropy inequalities carry complete proofs, so the remaining checks are consistency tests rather than evidence. In 60-digit arithmetic, the closed forms of Lemma 16 agreed with numerical differentiation to 10^{-57} ; the entropy chain of Theorem 18 held on 1600 random distributions across $\ell = 1, \dots, 4$ with minimum gap -5.6×10^{-61} , i.e. zero to working precision; and the conditional-affinity identity (124) held on 800 further distributions with maximum deviation 4.6×10^{-61} .

Supplementary Methods: sign-uncertainty discovery and diagnostics

Rigorous tripwires

Two lower bounds are evaluated at run time and used to invalidate the numerics rather than to prove anything. For $d = 1$, $s = +1$,

$$A_+(1) \geq \frac{1}{2(1 + \lambda_{\text{BCK}})} = 0.4107675 \dots,$$

with $\lambda_{\text{BCK}} = -\min_x \sin x/x$ [BCK]. More generally, Theorem 2.1 of [CDG] gives

$$u_0 \geq \lambda(d, n),$$

where $\lambda(d, n)$ is the smallest root of $L_{\lfloor n/2 \rfloor + 1}^{(d/2-1)}$. The latter follows from Gauss quadrature against $e^{-u}u^{d/2-1}du$: positivity of the quadrature weights forces a sign change at or beyond the first node. Any run reporting feasibility below either applicable bound is therefore rejected. During development, an early grid-only relaxation returned $\rho = 0.398942 = 1/\sqrt{2\pi}$ in $d = 1$, immediately falsified by the BCK tripwire.

Discovery campaign and numerical diagnosis

The initial search deliberately explored exact Fourier-eigenfunction mixtures with trainable Gaussian scales and contact locations, using local optimization and Adam-based learned proposals. Low-rank constructions reproduced the expected Bourgain–Clozel–Kahane controls and rapidly improved the objective, but the $d = 1$ search plateaued near $\rho = 0.5785$, above the published Cohn–Gonçalves value.

The failure was diagnostic. When scales crowded or the imposed contact count was incompatible with the local root structure, the coefficient solve became extremely ill-conditioned. Candidate Adam trajectories that appeared to improve the objective entered systems with condition numbers of order 10^{17} – 10^{18} . At this scale, forward accuracy could not be guaranteed in the floating-point formats used here. Multiprecision reevaluation showed that the apparent descent was dominated by numerical error rather than a reproducible improvement. Learned widths and contacts were therefore retained only as proposals, with a hard conditioning guard before verification.

The experiments also showed that the number of double contacts should not be rigidly tied to the number of basis functions. Decoupling the two made comparisons across basis sizes meaningful and showed that additional widths at fixed contact count contributed little. Multistart runs converged to distinct plateaus near 0.5785, 0.5929 and higher values, and verified evaluations could jump when the outermost root pattern changed. This behaviour is consistent with the discontinuity mechanism reported by Cohn and Gonçalves for Newton continuation at $d \leq 2$. We use this only as a diagnostic, not as an optimality theorem.

Implementation details of the convex exchange solver

Three implementation choices proved load-bearing. First, bare LP feasibility with zero objective returns arbitrary vertices that hug zero at many grid points and can dip between them. Maximizing a relative margin instead produces strict positivity between most nodes and sharply reduces the number of exchange rounds. Second, column scaling by the supremum over the full grid is dominated by the far tail and can exclude genuine feasible solutions; rows are therefore equilibrated while columns are left unscaled. Third, multiple far tail anchors are nearly parallel after normalization and degrade the simplex. The tail is imposed instead through a single structural sign constraint on the leading coefficient.

At each bisection value u_0 , the LP is solved on the current grid, the candidate derivative P' is evaluated at high precision, and all real critical points on the active interval are located. Any negative critical value is inserted as a new cutting row. The process stops only after the candidate passes the independent continuous scan to the prescribed numerical tolerance.

Convergence status of the exchange loop

The oracle is Remez-like in mechanics but not in guarantee. The truncated Laguerre space V_N is not a Haar system on $(0, \infty)$: $L_{2N}^{(\alpha)}(2u)$ belongs to V_N and has $2N$ positive roots, exceeding the Haar budget of an $(N + 1)$ -dimensional space. Consequently, uniqueness of the contact configuration and classical equioscillation arguments are not available.

The relevant framework is convex semi-infinite programming. Every finite grid gives an outer relaxation. Under compactness and a Slater point, the exchange sequence drives the maximum constraint violation to zero and the corresponding margins converge to the semi-infinite value. Together with monotonicity and continuity in u_0 , this gives a numerical bracket for the fixed-degree boundary. This concerns optimality within the chosen finite space only. Validity of every quoted upper bound is established by the exact rational certificate and does not depend on convergence of the exchange loop.

Contact degeneracy and leave-one-out reconstruction

The active contact count is discovered at the feasibility boundary. The certified $d = 1$ degree-22, 30 and 38 constructions use $m = 5, 7$ and 9 double contacts, respectively. The non-Haar structure can make the active set non-unique. At degree 38, the exchange oracle reports ten near-active contacts although the square reconstruction uses nine. Collocating all ten produces an admissible but irrelevant function whose outer sign change moves to $u = 274$ ($\rho = 9.34$).

We therefore perform leave-one-out reconstruction: each candidate contact is removed in turn, the remaining contacts are collocated in the square Laguerre system, and only reconstructions whose exact certificate agrees with the convex estimate are retained. At degree 38, four of the ten subsets certify the same 0.5725887 value to nine digits. The same repair is required at $d = 2$. The multiplicity of successful subsets is consistent with a degenerate optimal facet and is not interpreted as uniqueness of the extremizer.

Exact Sturm implementation

After rational reconstruction,

$$P(u) = \prod_i (u - \tau_i)^2 R(u),$$

and only the sign of R on the ray remains to be certified. We build a primitive pseudo-remainder sequence over $\mathbb{Z}$, using only positive rescalings so that Sturm sign-variation counts are preserved. This controls coefficient growth substantially better than naive rational Euclidean remainders, whose bit lengths become impractical by degree 38.

The verifier checks the exact Laguerre-to-monomial expansion, zero remainder after division by every squared contact factor, the sign of the leading coefficient, the exact value $R(\bar{u}) > 0$, and equality of Sturm sign-variation counts at $\bar{u}$ and $+\infty$. No grid, floating-point root finder or interval sampling appears in the final admissibility decision.

Relation to the non-Gaussian search

A numerical A/B analysis explains why the earlier Gaussian-mixture search did not improve the final polynomial construction at small d . Both spaces are evaluated through the same convex machinery, so the comparison concerns function spaces rather than optimizer quality. Projecting individual mixture directions onto the degree-22 polynomial space over the active region gives relative sup-norm residuals of approximately

$$\begin{aligned} 1.3 \times 10^{-15} \ (a = 1.5), \quad 3.0 \times 10^{-10} \ (a = 2.5), \quad 9.2 \times 10^{-7} \ (a = 4), \\ 6.3 \times 10^{-5} \ (a = 6), \quad 1.6 \times 10^{-3} \ (a = 10). \end{aligned}$$

Wider directions are genuinely distinct, but their optimized coefficients are of order 10^{-3} and did not improve the objective beyond the numerical noise floor. This containment analysis is diagnostic only and is not used in the certificate.

Higher-dimensional asymptotic diagnostic

Using the same convex machinery, the scale-free quantity $\rho\sqrt{2\pi/d}$ along one degree–dimension path was

1.435267 ($d=1$), 1.340341 ($d=2$), 1.212474 ($d=4$), 1.079183 ($d=8$), 1.026193 ($d=12$), 1.005480 ($d=16$).

The sequence approaches the sublinear-degree asymptote 1 rather than crossing it. This is only a diagnostic path, not an asymptotic theorem. At $d = 8$, increasing degree from 14 to 38 changes the reported value non-monotonically over a range of approximately 5.6×10^{-4} even though the exact fixed-degree optima are nested. We therefore interpret the spread as a numerical resolution limit of the discovery oracle.

Degree ceiling of the numerical stage

The current floating-point exchange solver is reliable only to polynomial degree approximately 38. Beyond this, the raw polynomial values span an extreme dynamic range: at degree 54, P reaches about 9.7×10^{73} at the far end of its own grid. Row equilibration cannot repair the resulting column-direction scaling, and the LP begins returning points that violate its own constraints.

Near the ceiling, the finite grid can also acquire a recession direction that is non-negative on all nodes while dipping between them, producing an “unbounded” rather than infeasible LP, and the cut pool can accumulate near-duplicate rows across the bisection. We cap the margin variable, refresh the cut pool when necessary, and discard runs whose bisection collides with a rigorous lower tripwire. Multiplication by the positive factor e^{-u} reduces the dynamic range but relocates the failure to underflow in the far tail. Higher-dimensional asymptotic work will therefore require higher precision or iterative/exact refinement inside the optimizer. The certificates reported here are unaffected because all lie at degree at most 38 and are validated by the exact rational stage.

```

NeuralCert repository
|-- neuralcert/                reusable framework
|   |-- core/                  problem and result contracts
|   |-- discovery/             neural models and optimization
|   |-- distill/               structural compression
|   |-- refine/                deterministic local refinement
|   |-- exact/                 reusable exact arithmetic
|   |-- verify/                independent verification primitives
|   |-- data/                  reproducible datasets
|   |-- pipeline.py            typed stage orchestration
|   '-- problems/
|       |-- maynard/           Maynard plugin and adapters
|       |-- delsarte/          coding-theory LP plugin
|       '-- sign_uncertainty/   Gaussian/Laguerre/Sturm plugin
'-- maynard_tools/             specialized compatibility layer
    |-- discovery/             factored, gated, ratio
    |-- certification/         Karatsuba, CRT, FLINT, Arb
    '-- verifier/              independent Maynard checks

```

Supplementary Figure 4: Compact directory view of the NeuralCert package architecture.

8 NeuralCert structure

NeuralCert is organized around explicit interfaces and data boundaries rather than a single shared numerical backend. A problem-specific discovery module produces an explicit candidate representation together with the metadata required to interpret it. A separate certification module reconstructs and evaluates that candidate using rigorous arithmetic and emits a self-contained certificate. An independent verifier consumes only this certificate and recomputes the claimed result without relying on the discovery or certification implementation. Discovery methods can therefore be modified or replaced without changing the trust assumptions underlying certificates that have already been issued. The three applications considered here share this discovery–certification–verification architecture, but not their numerical representations or proof mechanisms. For the Maynard problem, finite- k polynomial candidates are certified through exact rational evaluation of the associated Gram quadratic forms, whereas large- k rational candidates are certified through Fourier-domain evaluation in Arb ball arithmetic. Delsarte bounds are certified by exact rational verification of dual feasibility, while sign-uncertainty candidates are reduced to rational polynomial factorizations and verified using exact Sturm-sequence arguments.

9 Computational performance

The variational problem shards along the pair axis: for a pair (j, ℓ) the entire convolution chain touches only channels j and ℓ and produces two scalars, so the compute-to-communication ratio permits distribution over commodity interconnect. Because the Hellmann–Feynman route detaches the Ritz vector, both quadratic forms are plain sums over pairs, and the exact gradient follows from two scalar all-reductions and one gradient all-reduction per iteration; no autograd-aware collectives are required. Memory is dominated by activations indexed by pair, growing as $m(m+1)/2$, so parameter-sharding schemes (FSDP, ZeRO) address the wrong resource; DDP is also inapplicable, since it averages gradients while the reduction here is a sum inside a nonlinear ratio. Memory is not dominated by the assembled Gram matrices ($m \times m$), but by the activations of the convolution chain, which are indexed by pair and scale as $N \cdot n_q \cdot m(m+1)/2$ per chain step. Because Hellmann–Feynman requires these intermediates for the backward pass, they are retained across all $\sim 2 \log_2 k$ steps of the doubling chain. Supplementary Table 6 shows GPU memory use with different settings for the Maynard–Tao polynomial fitting discovery procedure.

Certification cost: The CRT backend at $k=3000$, degree 60 needs 155,000 primes; ball arithmetic replaces it with one adaptive-precision evaluation. The resolution requirement as a scaling law. n_{rep} grows roughly as $\approx k^{0.78}$ to hold the $g \equiv x$ control at 10^{-3} .

Supplementary Table 6: Overview of memory use and runtime for different settings and options for the Maynard-Tao **Polynomial discovery** regime since other implementations in the package have seconds runtime while using MegaBytes.

k	Channels	N_{grid}	VRAM
100	32	3000	36
100	48	3000	77
100	64	3000	133
100	32	4000	48
100	48	4000	102
100	64	4000	110
500	32	3000	53
500	48	3000	110
500	64	3000	196
500	32	4000	70
500	48	4000	84
500	64	4000	260
1000	32	5000	66
1000	48	5000	218
1000	64	5000	226
1000	64	6000	274
1000	48	4000	171
3000	32	5000	121
3000	48	2500	253
3600	10	5000	22
5000	10	5000	22
5000	20	5000	52
5000	30	5000	102
5000	40	5000	172
5000	50	5000	260

Note. Heavy compute analyses were performed using multiple NVIDIA H100 NVL or NVIDIA H200 SXM GPU cards with respectively 96 or 141 GB VRAM. GPU-consumption = GB VRAM at it's peak. N_{grid} = Representation nodes N ; The Chebyshev–Lobatto nodes per channel which is the resolution of a one-dimensional function, and the k -dimensional integral is handled by the separable structure; memory scales as $N \cdot 3m(m+1)/2$. This is the vanilla setup ($\epsilon = 0$). Runtime in $k = 100, N_{grid} = 3000, \text{Channels} = 32$ is 1 hour and 14 minutes for 4000 Adam iterations and 8 minutes for 20 steps LBFGS.

10 Anatomy of a certificate

A certificate is a compact, exactly specified record from which a claimed inequality can be independently recomputed. It is expensive to produce and cheap to check: verification requires neither the discovery pipeline nor the hardware used to generate the candidate, and a reader who distrusts every stage of the numerical search can nonetheless confirm the stated bound by running a short standalone program. The asymmetry is deliberate: generating a competitive trial function requires the full optimisation and certification stack, whereas checking the resulting inequality requires only the exact trial parameters and a few seconds of interval arithmetic on commodity hardware.

This note makes that concrete by walking through one complete example at $k = 5$ — the smallest case, chosen so that every artefact fits on the page with nothing elided. All files shown are shipped verbatim in `demo/` and are the unmodified output of the two commands in Panels A and C.

Panel A — The claim

The object being certified is one explicit function, fixed in advance:

$$F(t_1, \dots, t_5) = \prod_{i=1}^5 g(t_i) \mathbf{1}_{\mathcal{R}_5}(t), \quad g(t) = \frac{1}{c + 4t}, \quad c = \frac{552637315007}{978860977902}. \quad (125)$$

Because M_k is a supremum over admissible trial functions, *any* such F yields a lower bound, and the certified claim is

$$M_5 \geq 1.9717711764.$$

Three properties of this statement are worth making explicit before the mechanics, because they are what the certificate does and does not assert.

1. It is a statement about F , not about the optimiser. The value of c was found by numerical search; the search is not part of the claim, and a different (or worse) c would give a different (or worse) valid bound.
2. It is a lower bound and is deliberately not sharp. The discovery stage reported $R = 1.9966686079$ for this same F ; directed rounding of the aliasing and truncation budgets costs about 2.5×10^{-2} , and the certified number is the conservative end of that interval.
3. It is weaker than the best published constructions at $k = 5$, and this is the expected behaviour of an honest lower-bound machine. The rank-one rational family is not the optimal family here: Maynard's explicit polynomial trial gives $M_5 \geq 1417255/708216 = 2.00116\dots$, and Bogaert's Krylov computation gives 2.0071451444 . The gap matters mathematically, since under the Elliott–Halberstam conjecture the threshold $M_5 > 2$ implies DHL[5, 2]; the certified rank-one value does not reach it, whereas the polynomial trial does. At $k = 5$ the certificate is therefore a demonstration of the machinery rather than a source of number-theoretic content.

Panel B — The two files

The pipeline writes two records, and the separation between them is the trust boundary.

B1. The discovery record (k5-eps0-ratio.npz).

Produced by the search stage. It carries the trial function in exact rational form and nothing else of mathematical weight; the floating-point fields are provenance, not evidence.

```
k : 5
epsilon : 0/1 (vanilla functional)
mu : [1]
c_num, c_den : ['552637315007'], ['978860977902']
power : [1] g(t) = (c + n t)^{-1}
w_num, w_den : ['1'], ['1'] mixing weight (rank one)
R_discovery : 1.99666861 float, not certified
ceiling : 2.01179739 k/(k-1) log k
canonical : k=5|552637315007/978860977902~-1*1
sha256 : dfbd528eb3900e18771468e642b8cbcde4b2ed9074a593701419a856e2c596e6
```

The `canonical` string is the complete mathematical content of the file: dimension, exact pole, power, weight. Everything the certifier needs is recoverable from that one line, and the `sha256` binds the certificate below to this exact trial function.

B2. The certificate (k5-eps0-ratio.json).

Produced by the certification stage and reproduced here in full.

```
{
  "format": "maynard-Mk-certificate/2",
  "trial_function": {
    "g": "g(t) = 1/(c + n t), n = k-1",
    "form": "F = prod_{i=1..k} g(t_i) on the simplex",
    "c_exact": "552637315007/978860977902",
    "u_exact": "1",
    "note": "R is invariant under scaling of the weight; any fixed weight
            yields a valid lower bound."
  },
  "parameters": {
    "P": 8.0, "prec": 200,
    "nnode": 153, "M0": 153,
    "rmax": 12, "npanels": 80,
    "A_upper": 16.170030887466964,
    "Theta_ball": 32.34006177493393,
    "theta_far": 120.0, "block_ratio": 2.0,
    "tol_bits": 45, "assembly": "directed-arb/2"
  },
  "certified_quantities": {
    "N_enclosure": { "lower": 0.0877189203290238, "upper": 0.0877189203290238 },

```

```

"D_enclosure": { "lower": 0.2196630290925637, "upper": 0.21966302909256372 }
},
"error_terms": {
  "N_band": 0.0, "N_tail": 0.0005786544597255379, "N_alias": 3.51115912146848e-07,
  "D_band": 0.0, "D_tail": 0.0013051009573133488, "D_alias": 4.99060886139429e-07,
  "P_S_gt": 1.2860941636237798e-06, "y": 7.0
},
"far_field_blocks": [],
"final_arithmetic": {
  "N_lower_used": 0.08713991475338612,
  "D_upper_used": 0.22096862911076323,
  "R_lower": 1.9717711763896169,
  "R_midpoint": 1.9966700971800757,
  "ceiling": 2.0117973905426254
},
"claim": {
  "k": 5,
  "statement": "M_5 ≥ 1.971771176",
  "M_k_lower_bound": 1.971771176,
  "known_upper_bound_ceiling": 2.0117973905426254,
  "ceiling_source": "M_k < k/(k-1) log k (Polymath8b)",
  "primes_criterion": "r_k = ceil(theta M_k / 2), theta = 1/2 - eps
    (Bombieri-Vinogradov; Maynard Prop. 4.2)",
  "primes_implied": 1
},
"assembly_note": "All error bounds assembled in Arb ball arithmetic; floats
  exit only through provably outward-rounded conversions
  (nudge-verified against the ball).",
"source": {
  "npz": "./k5-eps0-ratio.npz",
  "sha256": "dfbd528eb3900e18771468e642b8cbcd4b2ed9074a593701419a856e2c596e6",
  "canonical": "k=5|552637315007/978860977902~-1*1",
  "R_discovery": 1.9966686078849576,
  "u_exact": "1"
},
"self_hash_sha256": "7f12647dfe9d7c9e02fc1f636eb8aec4ae05ac359675a0412cddb551a328ee20"
}

```

That the file fits on one page is itself the point: there is nowhere for an unstated assumption to hide. The fields divide into three kinds, and Panel C explains why the distinction matters.

Kind	Fields	Status
Mathematical input	<code>c_exact</code> , <code>u_exact</code> , <code>k</code>	load-bearing; exact rationals
Computational settings	<code>P</code> , <code>nnode</code> , <code>rmax</code> , <code>prec</code> , ...	affect sharpness, not validity
Reported intermediates	<code>*_enclosure</code> , <code>error_terms</code> , <code>final_arithmetic</code>	recomputed, never trusted

Panel C — Verification

C1. The transcript.

The certifier reads the discovery record and writes the certificate:

```

$ neuralcert certify --method ratio --npz k5-eps0-ratio.npz \
  --cert-json k5-eps0-ratio.json

trial function : k=5|552637315007/978860977902~-1*1
sha256 = dfbd528eb3900e18... (verified)
k = 5 R_discovery = 1.9966686079
A = 16.17 Theta_ball = 32.3401 theta_far = 120 nodes = 307 far blocks = 0
aliasing: P(S > 7) <= 1.286e-06 (Markov, r <= 12)
N in [0.08771892032902380 +/- 6.19e-18] err(band,tail,alias) = (0.00e+00, 5.79e-04, 3.51e-07)
D in [0.2196630290925637 +/- 2.42e-17] err = (0.00e+00, 1.31e-03, 4.99e-07)
R midpoint = 1.9966700972 |mid - discovery| = 1.49e-06

CERTIFIED (directed): M_5 ≥ 1.9717711764 [0s]
=> at least 1 primes infinitely often (theta = 1/2 - eps)

```

C2. What is recomputed, and what is ignored.

Every quantity below is regenerated by the verifier from `c_exact` and the computational settings alone. None of the stored values is read as input; they are recomputed and compared.

Quantity	Reconstructed from	Value
$A = 2(c + n)/c$	c	16.170030887466964
$\Theta_{\text{ball}} = 2A$	A	32.34006177493393
$\Delta\theta = 2\pi/P$	$P = 8$	0.7853981634
$\lfloor \Theta_{\text{ball}}/\Delta\theta \rfloor$	A, P	41
$M_0 = \max(41, n_{\text{near}})$	above, $n_{\text{near}} = 153$	153
Band error	$M_0 = n_{\text{near}}$	<i>empty</i>
$N_{\text{tail}}, D_{\text{tail}}$	Lemma (E2) with $A, \Delta\theta, M_0$	$5.7865 \times 10^{-4}, 1.3051 \times 10^{-3}$
$\mathbb{P}(S > 7)$	certified moments, $r \leq 12$	1.286×10^{-6}
$R_{\text{low}} = kN_{\text{low}}/D_{\text{up}}$	above	1.9717711763896

Two features of this table are worth reading off. First, the far-field band is empty because $M_0 = n_{\text{near}} = 153$ exceeds $\lfloor \Theta_{\text{ball}}/\Delta\theta \rfloor = 41$: the analytic tail begins exactly where the computed sum ends, which is the choice discussed in the truncation bound and which at $k = 5$ is worth more than an order of magnitude in the final bound. Second, the entire chain descends from the single rational c ; the verifier never needs the neural stage, a GPU, or any stored enclosure.

Test 1 shows that an inflated claim cannot survive. Test 2 shows the converse and less obvious half: a stored intermediate cannot be used to strengthen a claim either, because the verifier does not read it. Only `c_exact` (guarded by the `sha256` against the discovery record) and the computational settings enter, and altering the latter changes only how sharp the recomputed bound is, never whether the recomputation is valid.

C4. Independent reimplementations.

The strongest form of the check does not use our code at all. Every entry in the Panel C2 table is an elementary formula in c , P , n_{near} and r_{max} , and can be reproduced in a few dozen lines with any interval-arithmetic library. We have verified the present example this way as well as through the shipped verifier.

11 Maynard–Tao Discovery and Certification Tables

Supplementary Table 7: Vanilla discovery and certification results
($\epsilon = 0$).

k	Discovery	Ch_{disc}	Ch_{cert}	Degree	Certification	Fr_{ceiling}
20	3.128345991229	36	33	60	3.127557950496731089199291	0.992
21	3.170106837298	36	30	60	3.169687744135085908710644	0.992
22	3.210448579010	36	29	60	3.209791425249774301029605	0.991
23	3.249812450248	36	30	60	3.249018102873413922834334	0.991
24	3.287477573848	36	32	60	3.286239994066373110168097	0.991
25	3.322508727072	36	28	60	3.322151135960770560379734	0.991
26	3.358033109070	36	30	60	3.356593408977485398051369	0.991
27	3.390797412499	36	30	60	3.389882327187024762678030	0.990
28	3.423550501369	36	29	60	3.421891175367646121306316	0.990
29	3.453912899535	36	33	60	3.452879774954988104889254	0.990
30	3.484519905740	42	35	60	3.482920804908356499173898	0.990
31	3.514154741570	42	35	60	3.512405412198264166419745	0.990
32	3.542948097289	42	34	60	3.540478095710103535569937	0.990
33	3.569408303647	42	35	60	3.568168082140816922057409	0.990
34	3.595856014881	42	36	60	3.594862215205947831119258	0.989
35	3.622637140460	42	36	60	3.620729784609115150023189	0.989
36	3.649049376929	42	36	60	3.645939449133237314781111	0.989
37	3.672649021837	42	36	60	3.670749358191800897953616	0.989
38	3.697450514132	42	38	60	3.694212659617672335430314	0.989
39	3.719968833432	42	37	60	3.718057515927273283488455	0.989
40	3.743412144728	48	45	60	3.740850129189844929384497015046	0.989
41	3.765478132249	48	39	60	3.763326248211363188195226978670	0.989
42	3.786443640815	48	40	60	3.784836398710852090166363045263	0.989
43	3.808393391526	48	44	60	3.806112899332449843513725348999	0.988
44	3.831089249576	48	41	60	3.826734286114500842550587364787	0.988
45	3.850548910793	48	44	60	3.847582396814691842088947101999	0.988
46	3.870708253098	48	45	60	3.867062274460791570055053587204	0.988
47	3.890981548956	48	42	60	3.887050039712155465458257739811	0.988
48	3.911606976491	48	40	60	3.906150910904812934847425559032	0.988
49	3.927617139801	48	42	60	3.925218177946628714063038556246	0.988
50	3.943707217858	40	33	50	3.942445971786197344558734862254	0.988
51	3.962201156342	40	34	50	3.961742909943176882384440577615	0.988
52	3.980085920896	40	33	50	3.979361743352805048779887827229	0.988
53	3.997392090465	40	35	50	3.996569546603520792189153865045	0.988
54	4.014417349448	40	33	50	4.013421392728337468685880863364	0.987
55	4.031158093047	40	33	50	4.029930315883373661648075171587	0.987
56	4.048740196154	40	32	50	4.047211183498640314770985656114	0.987
57	4.064675943291	40	33	50	4.063658911265930366015897090005	0.987
58	4.080897241650	40	33	50	4.079662539803768949658734868762	0.987
59	4.096107675723	40	33	50	4.094929027520673522821234872044	0.987
60	4.111224221271	40	32	50	4.110244679900046307236517232052	0.987
61	4.123180413889	40	37	50	4.122593396892570578487592704836	0.986
62	4.141415806845	40	35	50	4.140421271440397388299091151343	0.987
63	4.151452852615	40	37	50	4.150583732266163623322424042114	0.986
64	4.170843665400	40	33	50	4.169733035246490856721916297730	0.987
65	4.185117958392	40	36	50	4.184200456291009710938969100824	0.987
66	4.199258111085	40	34	50	4.198212352631011423270651543778	0.987
67	4.212801300650	40	32	50	4.211734375259938467462491656228	0.987

Continued on next page

Supplementary Table 7 – continued from previous page

k	Discovery	Ch_{disc}	Ch_{cert}	Degree	Certification	Fr_{ceiling}
68	4.226913707347	40	34	50	4.225671537232203572914298251005	0,987
69	4.240641054970	40	27	50	4.239530843403299471806859841686	0,987
70	4.252043044524	40	34	50	4.250709595458423509274271903750	0,986
71	4.262077283275	40	34	50	4.260781471180062527612810820600	0,985
72	4.280170671493	40	31	50	4.278486659483071740524507466677	0,987
73	4.291652331079	40	36	50	4.290439906377909296284295483681	0,986
74	4.302039567418	40	32	50	4.300766657962825879583571273685	0,986
75	4.315037290653	40	36	50	4.313819804163989594408982684260	0,986
76	4.330839005745	40	33	50	4.328884563460976817456708107930	0,986
77	4.342588581101	40	32	50	4.340903139563253839977695048758	0,986
78	4.353490658660	40	30	50	4.351450026525314703393729301722	0,986
79	4.366332032220	40	25	50	4.363547069467314352935761632810	0,986
80	4.375870655363	50	36	50	4.375305575875195280796809950641	0,986
81	4.388391084261	50	34	50	4.387822836632632171272303086405	0,986
82	4.394774870701	50	40	50	4.394326901844365176252467034760	0,985
83	4.401879201937	50	43	50	4.401472671518373508912545234614	0,984
84	4.417373705040	50	40	50	4.416953094949757411430914506920	0,985
85	4.346441814015	50	41	50	4.346494488379658185500563460563	0,967
86	4.431105148849	50	42	50	4.430626969481280584146214724073	0,983
87	4.452908961195	50	40	50	4.451913396990625293127698559451	0,985
88	4.464302979612	50	41	50	4.463426157257708417355375315939	0,986
89	4.472355081593	50	42	50	4.471988530980810754826227770313	0,985
90	4.487119371775	50	37	50	4.485910734210900551045280545626	0,986
91	4.496698527592	50	37	50	4.495645093532629510509939471985	0,986
92	4.505829693079	50	39	50	4.504547705976628150465431574044	0,985
93	4.518347339954	50	36	50	4.516339208377947949080443322071	0,986
94	4.528059040470	50	29	50	4.526635937140459614406972086093	0,986
95	4.533806772177	50	41	50	4.532436225927372584839614117923	0,985
96	4.545404066027	50	26	50	4.544084221152406068483876743217	0,985
97	4.554320603674	50	31	50	4.553588807459129165253574203320	0,985
98	4.565278947850	50	28	50	4.564004527939718801984156460497	0,985
99	4.509354418861	50	35	50	4.508561572862963088224836065494	0,971
100	4.585389371115	50	39	50	4.583559169120379861185947389471	0,985
100	4.582636529114	50	39	50	4.581993985670381852509553180491	0,985
130	4.822341520115	50	13	50	4.808713824125718836287734368717	0,980
150	4.961245663975	50	6	50	4.955366718182981077061200798091	0,982
200	5.090813348651	50	4	60	5.087280688405278804164639536817	0,955
220	5.258389442417	50	12	60	5.247997786415732591367214256322	0,969
250	5.384166994782	50	5	60	5.349929087705455637995951938908	0,965
350	5.610666510192	50	3	60	4.921779442483077605884943545259	0,838
500	6.080822857000	50	3	60	5.032655240858842848757497207358	0,808

Note. Values $k = 25$ and $k > 30$ have certified values above Bogaert’s Krylov subspace method and the published numbers on the Polymath8b Wiki. Ch_{disc} ; amount of channels in the discovery, Ch_{cert} ; amount of channels in the certification (after pruning). Fr_{ceiling} ; the gap to the Cauchy Schwartz ceiling.

Supplementary Table 8: Discovery and certification results for $\varepsilon = 1/25$.

k	Discovery	Ch_{disc}	Ch_{cert}	Degree	Certification
20	3.210936069786	36	33	60	3.210366921521749737183691731395
21	3.252680335644	36	30	60	3.252298834127242084124289323548

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k	Discovery	Ch_{disc}	Ch_{cert}	Degree	Certification
22	3.292779266722	36	29	60	3.292126369692432458798975243945
23	3.331173442193	36	30	60	3.330322094752801530355410455276
24	3.368036770635	36	30	60	3.366908489372213299919519038109
25	3.402371365754	36	29	60	3.402010619443212328887383163464
26	3.437028098636	36	31	60	3.436157517278351176844746714320
27	3.468827024244	36	29	60	3.468320542253183831988945583237
28	3.500415876452	36	30	60	3.499444954997692030882286887925
29	3.531186516502	36	32	60	3.530010877716944327578014764437
30	3.561700179340	42	39	60	3.559458746545551314730876537135
31	3.590689744972	42	40	60	3.588557568639265632860385653635
32	3.618732963879	42	38	60	3.616127200711337138001580976885
33	3.644253262445	42	36	60	3.642965802069061441160922851345
34	3.671450599407	42	39	60	3.668624924763460294356636286090
35	3.696250084782	42	40	60	3.694175287948157320412197179779
36	3.720908518093	42	39	60	3.718279485810969785557788969075
37	3.744273426646	42	38	60	3.742226163630752703816176588194
38	3.768134969368	42	41	60	3.766021610226650587509688536386
39	3.790213730678	42	39	60	3.788245621497978381906985116415
40	3.813521764234	48	45	60	3.810310054297441303862285913738
41	3.833616627961	48	41	60	3.831637106843201917337701123609
42	3.855906628402	48	45	60	3.852600769154481058291704449249
43	3.876753550031	48	42	60	3.873483797869255723761005318927
44	3.895577913875	48	46	60	3.893301604331862091582323108956
45	3.915649733022	48	45	60	3.913474652772154201185956494902
46	3.936937139139	48	43	60	3.932552485373460445967272733645
47	3.956703033242	48	46	60	3.952079215054289455935133148337
48	3.975823596168	48	44	60	3.970095831528526602285865965250
49	3.991450730929	48	43	60	3.988676716542276674807876834435
50	4.006026386327	40	32	50	4.005637471340251334822223632678
51	4.021337913283	40	21	50	4.020552375066701497376356751144
52	4.040635119278	40	31	50	4.040258681163316069160504782304
53	4.058215551546	40	24	50	4.057203993845967521227902017906
54	4.072969225060	40	28	50	4.072322886063011063853846972776
55	4.084802202882	40	18	50	4.084085711139175414397956372460
56	4.100028783679	40	26	50	4.099206855766737568002165290524
57	4.115585844764	40	23	50	4.115039486353743806900256901164
58	4.135152191312	40	25	50	4.134242300759367593439951066209
59	4.149187895198	40	30	50	4.148472609734504851305763152187
60	4.158966932600	40	19	50	4.157998757293132279032736615835
61	4.175975507879	40	31	50	4.175077906373857205061979626025
62	4.189862734973	40	32	50	4.188904070211809938022792508286
63	4.210516336726	40	33	50	4.208857616045980796408818491228
64	4.219415161437	40	30	50	4.218124437987688545452024305223
65	4.238501800615	40	36	50	4.237384124837738623814213247461
66	4.252120545298	40	29	50	4.250793037094066225539732443814
67	4.261402748608	40	22	50	4.259777456025320664617972564576
68	4.270379618942	40	26	50	4.269124177371744769988813084968
69	4.283984925481	40	5	50	4.277274261946282382619508640328
70	4.301560117306	40	34	50	4.298733418769167808786843497725
71	4.316885083242	40	31	50	4.314413443106201423239218270654
72	4.329503334605	40	35	50	4.327034144580560001136968528718
73	4.334908974772	40	30	50	4.332768096731747934119777486576
74	4.342039995848	40	28	50	4.339799356192207942523849650386

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k	Discovery	Ch_{disc}	Ch_{cert}	Degree	Certification
75	4.358917246793	40	32	50	4.356375192119249086787668845356
76	4.375902255118	40	31	50	4.372974259197422394873126180262
77	4.388092689456	40	35	50	4.385396831879956294549952588437
78	4.399184535712	40	22	50	4.395990292345199428975680930752
79	4.407214416241	40	30	50	4.404183984805812816254054716642
80	4.195941988767	50	40	50	4.197608268366179620141468586531
81	4.075117366264	50	35	50	4.093066546579362186151057435374
82	4.192978689328	50	43	50	4.195259401348774146934234241013
83	4.425644201176	50	35	50	4.424852131550489892382995112584
84	4.096825473475	50	31	50	4.099802658570555784696980423382
85	4.40747525627	50	39	50	4.406760295616284828171584604987
86	4.457072509121	50	31	50	4.458412761836136236841801895336
87	4.453265265989	50	45	50	4.453060114331590189551387134208
88	4.344434571908	50	40	50	4.344422869750479566119971669552
89	4.410538854701	50	39	50	4.417151906430070350388796841891
90	4.470745463723	50	45	50	4.471149582958362588083170625461
91	4.175121841242	50	41	50	4.176550564513205748419816639538
92	4.381872185197	50	45	50	4.389037025811565058867953246127
93	4.312917570269	50	44	50	4.314963190359444288509182754059
94	4.272652151109	50	38	50	4.273951035321736008358360794466
95	4.420217255929	50	38	50	4.419415306457317943090466947241
96	4.311614804044	50	34	50	4.311428453792439589002649755954
97	4.203613002767	50	35	50	4.206802724014883123574465837334
98	4.217114551052	50	39	50	4.225342000024770175903490961201
99	4.548298852384	50	39	50	4.546923190682762183944464394853
100	4.496536005802	50	42	50	4.495512133810554663324778495290
100	4.466983632765	50	40	50	4.466802522812985639158255185770
130	4.847986562151	50	10	50	4.844156480287745811911814959650
150	4.977475423203	50	8	50	4.961947334363613966866801798777
200	5.094346768152	50	7	60	5.087393690044834674577815885368
220	4.978911765299	50	8	60	4.976764489557045481952499319142
220	4.978911765299	50	8	60	4.976764489557045481952499319142
300	4.086884354936	50	1	60	3.689963955854296035616007427184
400	5.381948554900	50	13	60	5.342988749384368640747405260075
500	4.512115207075	50	14	60	4.192789933536702947199563856796

Note. Ch_{disc} ; amount of channels in the discovery, Ch_{cert} ; amount of channels in the certification (after pruning).

Supplementary Table 9: **Certified enlarged-support sweep underlying the ε -crossover analysis.** For each k , the table reports the certified lower bound obtained for the corresponding enlarged-support parameter ε , together with its within- k ranking among the tested values. Rank 1 denotes the strongest certified value at that k .

k	ε	Certified value	Rank
50	0	3.942127742490453604663761985558	5
50	1/5	3.712509239545921204317529071368	8
50	1/6	3.764504991324559702999466932710	7
50	1/8	3.874384180153464414710468398524	6
50	1/12	3.959088983406359921645433483059	4
50	1/16	3.988718438122563563392522885252	3
50	1/25	3.997997740337627844075583884562	1
50	1/50	3.994094475900581244335373138334	2
60	0	4.107867690167866055301464199575	4
60	1/5	3.823138358991218145879633189922	8
60	1/6	3.901108973124642823154420824199	7
60	1/8	4.008638429188118037606214685517	6
60	1/12	4.078919312220589574136195730201	5
60	1/16	4.137429406260009516632280091156	3
60	1/25	4.154030210268053992886263760513	2
60	1/50	4.159671531351647672765832327114	1
70	0	4.250768996013457035617670337865	4
70	1/5	3.932874621775866265776104392099	8
70	1/6	3.992467605762845919987563382701	7
70	1/8	4.097181032238870681623243665487	6
70	1/12	4.203873349967048026646358848602	5
70	1/16	4.264751558230381563655123754834	3
70	1/25	4.284181954725759229768882824961	2
70	1/50	4.298346167280743718826612134536	1
80	0	4.365076286689542864035828077954	4
80	1/5	4.010739510791432389610499056194	8
80	1/6	4.046713573456469114235727637437	7
80	1/8	4.198433911029465756611969501053	6
80	1/12	4.300715218425578908913133664543	5
80	1/16	4.368846840474345630380842204395	3
80	1/25	4.397641633229676353759931557760	2
80	1/50	4.409932699888494327027154402066	1
90	0	4.475394178930606300302930028315	3
90	1/5	4.051193459293446704008504024550	7
90	1/6	4.116620710367156118853873413986	6
90	1/8	4.026133861058892236358880667882	8
90	1/12	4.277640742111484913450494485256	5
90	1/16	4.426300947743744344607385400548	4
90	1/25	4.502108322122445717967006768742	2
90	1/50	4.514812668499281034377537700573	1
100	0	4.571865161766055920977957342516	2
100	1/5	4.131174966202319094573252086929	8
100	1/6	4.228661930221300547990282980216	7
100	1/8	4.376164198382763272282279872439	6
100	1/12	4.455747109584966143321615617427	5

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k	ε	Certified value	Rank
100	1/16	4.483072604316316381523061824105	4
100	1/25	4.520592767325824976955743719607	3
100	1/50	4.600398252159768512024806128765	1

Supplementary References

1. Polymath, D. H. J. *Variants of the Selberg sieve, and bounded intervals containing many primes* 2014. arXiv: [1407.4897](https://arxiv.org/abs/1407.4897) [math.NT]. <https://arxiv.org/abs/1407.4897>.
2. Maynard, J. Small gaps between primes. *Annals of mathematics*, 383–413 (2015).
3. Ghadimi, M. *Heuristic Bounded Prime Gaps via a Chaotic Multidimensional Sieve and Random Matrix Theory* arXiv:2507.17986v1. 2025. arXiv: [2507.17986](https://arxiv.org/abs/2507.17986) [math.NT].
4. Hart, W., Johansson, F. & Pancratz, S. FLINT: Fast Library for Number Theory. *ACM Communications in Computer Algebra* **46**, 88–93 (2012).
5. Probst, D. K. & Alagar, V. S. A Family of Algorithms for Powering Sparse Polynomials. *SIAM Journal on Computing* **8**, 626–644 (1979).
6. Coregliano, L. N., Jeronimo, F. G., Jones, C., Linial, N. & Loyfer, E. Higher-order delarte dual lps: Lifting, constructions and completeness. *arXiv preprint arXiv:2501.04854* (2025).